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Résumé

L’acquisition du vocabulaire expressif et de la grammaire à partir d’une expérience linguistique limitée, où les formes sont inégalement fréquentes, constitue une question centrale en sciences cognitives. Les modèles permettent de tester des hypothèses d’apprentissage ; leur portée développementale dépend de la comparaison des expériences et des comportements. Cette thèse associe une calibration des comparaisons enfants–modèles à quatre études computationnelles sur le vocabulaire, la syntaxe avec récupération d’exemples, la simulation du dialogue et l’apprentissage par récompenses dérivées des réactions des adultes. Avec des modèles et des entraînements distincts, ces études examinent les contributions de l’exposition, de l’accès aux exemples mémorisés et du feedback.

Le protocole CORPUS-CDI ajuste l’exposition estimée, la quantité de productions générées et le vocabulaire évalué pour comparer le développement expressif des enfants à celui de LSTM et de Transformers au niveau des caractères. Les modèles testés ne reproduisent pas conjointement les trajectoires enfantines de diversité lexicale, d’exactitude des formes lexicales et de croissance du vocabulaire ; les mots peu fréquents restent particulièrement difficiles à produire. Une étude distincte augmente des modèles paramétriques par récupération des k plus proches voisins. Cette intervention améliore leur sensibilité aux contrastes syntaxiques contenant des mots peu fréquents et réduit partiellement l’écart lié à la fréquence lexicale, avec une contribution importante de l’information structurelle. Un benchmark d’interactions enfant–adulte montre ensuite que certaines propriétés des énoncés et tendances liées à l’âge peuvent être approchées, tandis que les conversations prolongées révèlent des écarts plus marqués. Les adultes simulés restent trop alignés sur leurs interlocuteurs et insuffisamment diversifiés. Dans une étude indépendante d’apprentissage par renforcement, les récompenses produisent des effets différenciés : l’alignement structurel améliore le plus régulièrement la grammaticalité des énoncés générés, sans amélioration cohérente aux tests de paires minimales BLiMP et Zorro.

Ces bénéfices sélectifs coexistent avec des écarts persistants entre trajectoires et tâches. La thèse apporte des benchmarks de production et d’interaction, des interventions computationnelles contrôlées et un cadre reliant leurs résultats à des conclusions développementales circonscrites. Elle motive des théories distinguant exposition, mémoire et interaction, dont les effets doivent être testés par plusieurs mesures de l’usage du langage.

Mots clés : Acquisition du langage ; modélisation computationnelle ; vocabulaire expressif ; modèles de langage augmentés par récupération ; interaction enfant–adulte ; apprentissage par renforcement

Abstract

How children develop expressive vocabulary and grammatical behavior from limited, unevenly distributed linguistic experience is a central question in cognitive science. Language models make learning assumptions experimentally tractable, but their developmental relevance depends on how experience and behavior are compared. This thesis combines a framework for calibrating child–model comparisons with four computational studies of expressive vocabulary, retrieval-supported syntax, dialogue simulation, and learning from caregiver-derived rewards. The studies use distinct learners and training regimes to examine what predictive exposure, access to stored instances, and feedback contribute to specific linguistic outcomes.

The CORPUS-CDI benchmark aligns estimated training exposure, generated production quantity, and evaluation vocabulary to compare children’s expressive development with character-level LSTMs and Transformers. The tested models do not jointly reproduce children’s trajectories of lexical diversity, word-form accuracy, and vocabulary growth; low-frequency vocabulary remains particularly difficult to produce. A separate study uses k -nearest-neighbor retrieval to supplement parametric language models. Retrieval improves sensitivity to syntactic contrasts involving low-frequency words and partially narrows the lexical frequency gap, with structural information playing an important role. Turning to interaction, a benchmark comparing simulated child–caregiver exchanges with naturalistic dialogue shows that selected utterance-level and age-related patterns can be approximated, while sustained conversations reveal greater divergence. Simulated caregivers remain overly aligned with their interlocutors and insufficiently diverse. In an independent reinforcement-learning study, caregiver-derived rewards have differentiated effects: structural alignment most consistently improves the grammaticality of generated utterances, without consistent improvements on BLiMP and Zorro minimal-pair tests.

Together, these findings identify selective benefits of retrieval and feedback alongside persistent mismatches across developmental trajectories and evaluation tasks. The thesis contributes production and interaction benchmarks, controlled computational interventions, and a framework connecting model outcomes to bounded developmental claims. It motivates theories that distinguish the contributions of exposure, memory, and interaction, and test their effects across multiple measures of language use.

Keywords : Language acquisition; computational modeling; expressive vocabulary; retrieval-augmented language models; child–caregiver interaction; reinforcement learning

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Chapter 1

General Introduction

Objectives

Introduce the language-learning problem addressed by the thesis, explain the role of calibrated child–model comparisons, and connect the empirical studies of expressive vocabulary, retrieval-supported syntactic processing, and caregiver interaction.

1.1 From Linguistic Experience to Language Use

Learning a word involves more than encountering its form. A learner must retain enough of an encounter to recognize the word again, relate it to other knowledge, and eventually use it when a situation calls for it. Learning grammar introduces a related demand: familiar words must become usable in combinations that extend beyond particular previously heard utterances. These changes take place in an uneven environment. Some words and constructions recur throughout everyday conversation, while others occur only occasionally. Children also contribute to this environment through their own productions, which can elicit responses, reformulations, and requests for clarification from caregivers. Understanding how linguistic experience becomes productive language therefore requires an account of the learner, the distribution of experience, and the interaction in which learning occurs [tomasello2003constructing; ambridge2015ubiquity; clark_conversational_2020].

Statistical learning offers one route from experience to knowledge: recurrent patterns can support expectations about which elements occur together and how sequences are organized [saffran1996statistical]. Neural language models make this proposal computationally explicit by learning to predict upcoming linguistic units from preceding context. Their predictions can be evaluated directly, or used to generate language. This makes them useful experimental systems for asking what a specified learning procedure can extract from a specified body of input [dupoux2018cognitive; warstadt2022what]. At the same time, successful prediction leaves open whether the resulting behavior resembles the development of language use in children. A model can assign sensible probabilities to a sentence while rarely producing a particular word, or generate plausible utterances while departing from the dynamics of a conversation.

The cognitive problem motivating this thesis is how learners turn limited and uneven linguistic experience into expressive and grammatical behavior. The thesis approaches this problem through a more specific question: *What do calibrated comparisons between children and language models reveal about the contributions and limits of predictive exposure, access to stored instances, and caregiver-derived feedback?* The studies examine vocabulary production, sensitivity to syntactic contrasts, and child–caregiver interaction. These domains expose different aspects of the relation between what a learner has encountered and what it can subsequently do.

The aim is to use computational models to make parts of the acquisition problem experimentally tractable. Model input can be bounded, access to previously encountered instances can be changed, and particular feedback signals can be isolated. These manipulations provide evidence about the consequences of explicit computational assumptions. Relating that evidence to children requires specifying which aspect of experience or behavior is represented by each experiment. This relation between a substantive learning question and the conditions for interpreting a model comparison provides the organizing thread of the thesis.

1.2 Why the Comparison Requires Calibration

1.2.1 Comparable Experience and Comparable Outcomes

A claim that a model requires more data than a child depends on what counts as input and on which outcome the two learners are expected to reach [frank2023bridging]. Word counts provide a useful description of exposure, but equally sized corpora can differ in register, repetition, conversational organization, and relevance to the learner. Transcribed caregiver speech also differs from the spoken and perceptually grounded events that children encounter. A model trained on a bounded text corpus consequently represents selected properties of a child’s learning situation; the dimensions matched by the design need to be stated explicitly.

The outcome poses a parallel problem. Children’s knowledge is inferred through behavior, including comprehension responses, spontaneous productions, and caregiver reports. Models expose probabilities, representations, or generated sequences. A scoring rule must connect these outputs to a construct such as vocabulary knowledge or grammatical sensitivity. The distinction is concrete in expressive vocabulary. The MacArthur–Bates Communicative Development Inventories (CDI) ask caregivers about words a child produces, while a language model can generate a corpus in which the frequency of those word forms is counted [fenson2007macarthur; frank2017wordbank]. Both measures concern production, but repeated generation of a form does not establish the semantic and contextual knowledge that a caregiver may recognize when reporting that a child says a word.

This thesis therefore treats input alignment and outcome alignment as separate requirements. Input alignment concerns the experience available to the learner; outcome alignment concerns the behavior elicited and the interpretation of its measurement. The comparison also depends on a *task model*: the threshold, decoding rule, prompt, or evaluator that converts a model’s output into an experimental score. Making this component explicit allows an apparent learning difference to be examined rather than attributed immediately to the learner’s architecture. Chapter 2 develops this framework and reviews the relevant literature.

1.2.2 Learning, Simulation, and Intervention

Language models serve several scientific roles in the empirical chapters. A model trained from scratch on a controlled corpus can be studied as a learner. An already pretrained model prompted to speak like a child or caregiver serves as a simulator of observable behavior. A model whose access to memory or learning reward is manipulated provides a system for testing the effect of that intervention. These roles support different conclusions. An age-conditioned prompt tests whether an existing model can express age-associated patterns; an exposure manipulation tests what changes when the training material changes.

The distinction also clarifies the value of success and failure. Under sufficiently aligned conditions, success can support a conditional sufficiency claim: the specified information and learning procedure can produce the measured outcome. It does not establish that children use the same internal process. Failure identifies a mismatch to be explained, but may reflect the learning objective, missing input, or the way knowledge is elicited and scored [dupoux2018cognitive;

warstadt2022what]. Controlled contrasts make these alternatives more precise. They allow the thesis to ask where a change in available information improves a particular behavior and where the mismatch remains.

1.3 From Uneven Input to Memory and Feedback

Three empirical issues motivate the studies. The first is the relation between exposure and expressive vocabulary. Developmental vocabulary data provide a rich human reference, yet measuring lexical knowledge through a model’s prediction probabilities does not show which words are reliably available in its own productions. Generated vocabulary also depends on the amount of output sampled: a word has more opportunities to appear in a large corpus than in a small one. A production-based comparison consequently requires control over both learning opportunities and observation opportunities. It must also examine change across exposure conditions, since a similar final vocabulary score can conceal different growth patterns.

The second issue is the uneven availability of evidence across lexical items. Frequency effects in model output motivate a question about how past experience is retained and accessed. Parametric models encode information through changes to network weights. Retrieval-augmented models can additionally consult stored instances when making a prediction. The distinction has a useful connection to complementary learning systems accounts, which distinguish rapid access to particular experiences from the gradual extraction of statistical regularities [**mcclelland1995there; kumaran2016learning**]. A k -nearest-neighbor language model offers a tractable intervention: its prediction combines the base model’s distribution with evidence retrieved from a datastore [**khandelwal2020knnlm**]. The question tested here is whether this access reduces lexical-frequency effects in syntactic judgments, and what properties of the retrieved instances make it useful. This tests a computational resource motivated by memory theory without requiring the retrieval algorithm to reproduce a complete human memory system.

The third issue is the role of the learner’s own contributions. In child–caregiver interaction, linguistic input is partly responsive to what the child has just said. Caregivers may acknowledge a contribution, request clarification, reuse a structure, continue its topic, or respond with positive affect [**demetras__feedback__1986; clark__conversational__2020; fusaroli__caregiver__2023**]. Such responses can carry information relevant to learning, but their functions differ. A clarification request may concern intelligibility or intended meaning as well as grammar; a response that continues a topic need not endorse every feature of the child’s sentence. Inferring what to change from an utterance-level response therefore presents a credit-assignment problem [**marcus__negative__1993**].

Computational work on interaction must address two questions in this setting. One concerns whether simulated interlocutors reproduce the linguistic and conversational properties of natural dialogue. The other concerns whether particular signals associated with caregiver responses can change a learner’s subsequent language. The thesis investigates both. Separating them makes it possible to evaluate interaction as a behavioral target while also isolating candidate feedback signals as learning interventions.

1.4 Research Questions and Strategy

The empirical work is organized around four questions:

1. **Expressive vocabulary:** Under explicit exposure, production, and lexical-frequency controls, how do the vocabulary productions of predictive language models compare with those of children, and how do the differences change with exposure and decoding?
2. **Retrieval and syntactic sensitivity:** Does access to stored instances reduce the accuracy gap between high- and low-frequency items in syntactic contrast tests, and which retrieved information and configurations support the effect?
3. **Interaction simulation:** How closely do language models reproduce child and caregiver language at word, utterance, and dialogue levels, and does correspondence in single-turn testing persist during extended interactions?
4. **Feedback learning:** How do different caregiver-derived rewards alter generated grammaticality, and do these changes also appear in controlled grammatical preferences?

The studies combine diagnostic comparisons with interventions inside computational systems. The vocabulary and dialogue benchmarks characterize correspondence with child-language references. The retrieval and feedback experiments manipulate a resource or learning signal while retaining an appropriate model baseline. Naturalistic English data from CHILDES provide a recurring source of caregiver language, child production, and interaction records [macwhinney2000chilides]. The experiments also use audiobook text, bounded-data and large-scale pretraining settings, and controlled syntactic test materials where these serve the relevant comparison.

The measurements span free generation and constrained evaluation. Vocabulary coverage, lexical diversity, and word-form validity describe sampled productions. Syntactic minimal pairs test whether a model favors a grammatical sentence over a minimally different ungrammatical alternative. Dialogue metrics describe selected relations among conversational contributions. In the feedback study, automatic measures of generated grammaticality are compared with BLiMP and Zorro performance. This range of outcomes helps distinguish a shift in what the model tends to say from an improvement that transfers to a different test.

The four studies use different models, input regimes, and experimental timescales. Their relationship is a synthesis of complementary questions arising from the completed work. In particular, the retrieval experiment does not test a remedy for the vocabulary benchmark, and the dialogue simulators do not supply the rewards used in feedback learning. These connections instead motivate hypotheses that could be tested in a future integrated learner. The present strategy preserves the interpretability of each comparison before drawing conclusions across them.

1.5 Main Contributions

The first contribution is CORPUS-CDI, a count-based benchmark for expressive vocabulary comparison. It makes training exposure, generated output quantity, and the frequency composi-

tion of the evaluation vocabulary explicit. Applied to character-level LSTMs and Transformers trained on child-directed transcripts or audiobook text, it identifies differences from the child reference in lexical diversity, word-form trajectories, and vocabulary growth. Low-frequency target words are especially difficult for the models to produce often enough to satisfy the scoring criterion. Varying sampling temperature changes the balance among measures without producing a setting that aligns them jointly. The result concerns expressive availability under the benchmark’s generation protocol, and provides a more specific diagnosis than an overall claim about model vocabulary knowledge.

The second contribution is a frequency-stratified test of retrieval augmentation in syntactic processing. Across the tested agreement, *wh*-question, and relative-clause contrasts, *k*NN augmentation improves low-frequency performance and narrows the high–low accuracy gap in both model settings. The gap remains after retrieval. Analyses of retrieved information show that structurally relevant instances are particularly useful, while semantic similarity alone does not reliably reproduce the gains. Retrieval configuration also matters. These findings establish that explicit instance access can selectively compensate for frequency-sensitive predictions in the tested systems, while identifying conditions that constrain the compensation.

The third contribution is a benchmark that evaluates both sides of child–caregiver interaction across linguistic levels and conversational settings. The study compares single-turn responses and multi-turn simulations, with and without conversational examples. Models approximate some word- and utterance-level properties, and the few-shot prompting conditions improve several aspects of their productions. Persistent discrepancies appear in dialogue organization: simulated caregivers display higher semantic alignment and lower semantic diversity than the human reference under the study’s measures. Extended interactions reveal additional differences that are less apparent in isolated responses. The contribution is therefore both an evaluation procedure and evidence that local resemblance alone is insufficient to assess an interaction simulator.

The fourth contribution is a controlled comparison of caregiver-derived feedback signals through reinforcement learning. Small language models pretrained on caregiver input are fine-tuned with rewards predicting structural alignment, communicative feedback, semantic contingency, or affective feedback. Structural alignment yields the clearest improvement in generated grammaticality; communicative feedback has smaller effects, with more consistent benefits from clarification requests than acknowledgments. Semantic and affective rewards generally reduce grammaticality under the evaluations used. The empirical rewards do not consistently improve BLiMP or Zorro performance. The study thus demonstrates differentiated effects on production and leaves the extent of grammatical generalization unresolved.

Together, the findings give empirical substance to the calibration framework. Frequency-sensitive production and syntactic preferences expose limitations under uneven evidence; retrieval provides partial compensation; and feedback changes production in ways that depend on the signal and evaluation. Behavioral resemblance, intervention effects, and developmental explanations remain distinct levels of inference. This separation makes the computational results useful for cognitive science by specifying which learning assumptions they support and which further comparisons are needed.

1.6 Organization of the Thesis

Chapter 2 reviews the conditions under which child–model comparisons support inference about language development. It introduces input and outcome alignment, examines task models and developmental trajectories, and discusses measurement uncertainty and the accounting of linguistic experience. This framework supplies the interpretive basis for the empirical chapters.

Chapter 3 introduces CORPUS-CDI and applies it to expressive vocabulary growth. It establishes the production-based comparison and examines the roles of exposure, decoding, lexical frequency, and word category. Chapter 4 then investigates frequency sensitivity through a different outcome, syntactic minimal-pair judgments, and tests whether retrieval of stored instances can reduce it. The link between these chapters is the availability of low-frequency evidence across distinct language behaviors.

Chapter 5 turns to child–caregiver interaction. Its first study evaluates language models as dialogue simulators; its second isolates the learning effects of caregiver-derived rewards. The chapter’s discussion brings these contributions together while distinguishing the fidelity of an interaction from the effect of a signal on learning.

Chapter 6 synthesizes the results across vocabulary, syntax, and interaction. It considers the complementary roles of predictive learning, instance access, and feedback; examines the limits imposed by input and evaluation choices; and develops directions for longitudinal, multimodal, and interactive modeling. The synthesis returns to the central learning problem with a set of experimentally grounded conclusions about which resources support which measured behaviors.

Chapter 2

Literature review: calibrating comparisons between child and artificial language learners

Objectives

Establish when comparisons between children and artificial language learners support developmental inference by jointly calibrating the learners' input, evaluation outcomes, developmental trajectories, and uncertainty.

2.1 Scope and inferential framework

Understanding how children acquire language from limited experience is a central problem in developmental science, but the relevant learning conditions cannot be manipulated freely in children. Researchers cannot randomly assign infants to radically different linguistic environments, control the timing of every construction they hear, or intervene directly on the learning architecture they bring to the task. Computational models complement observational and experimental work because they instantiate explicit learning hypotheses and can be exposed to controlled input. In this role, a model can address a learnability question: whether a specified learning procedure is sufficient to acquire a target pattern from a specified body of experience [chater2006probabilistic; Perfors2011; dupoux2018cognitive].

This logic predates contemporary large language models. Early connectionist models were deliberately constructed as cognitive simulations and used to test whether apparently rule-governed behavior could emerge from pattern learning [rumelhart_learning_1986; Elman1990]. Recurrent networks were subsequently used to study phenomena such as agreement and auxiliary inversion under distributional learning accounts [Elman1991; Lewis2001; Real2004]. Modern language models differ in an important respect: they are generally optimized for engineering objectives, trained on large collections of adult-written text, and not designed as process models of development [brown2020language; warstadt2022what]. Their strong performance on syntactic, semantic, and reasoning benchmarks is therefore scientifically suggestive, but it does not by itself establish that they learn as children do [Linzen2021; Mahowald2024].

The inference from a successful simulation is asymmetric. If a model learns a phenomenon under conditions that adequately approximate the child’s learning problem, its success provides evidence of *sufficiency*: the represented input and learning biases can, in principle, produce the observed outcome. Failure is less decisive. It may reveal an inadequate learning mechanism, but it may instead arise from missing modalities, an unsuitable objective, an inappropriate evaluation, or input unlike that available to the child. Human learners may also exploit working-memory constraints, perceptual grounding, or social scaffolding that the model does not represent [tomassello2003constructing; warstadt2022what]. A model failure consequently does not establish that a phenomenon is unlearnable from experience, nor does model success establish mechanistic identity.

This chapter develops a calibrated-comparison framework around two necessary dimensions. *Outcome alignment* asks whether the model and the child are evaluated on the same construct, using stimuli, response contrasts, and developmental summaries that support a common interpretation. *Input alignment* asks whether the model’s training experience approximates child-available input in amount, register, modality, interactional structure, and temporal organization. The two dimensions are related but not interchangeable. A model trained on child-directed speech but evaluated only by an engineering benchmark provides little direct evidence about child behavior. Conversely, a model tested with developmental stimuli after training on web-scale adult text answers a different learnability question from the one faced by a child.

Architecture is treated here as an experimental variable, not as a third quantity that can

be matched by inspection. At Marr’s algorithmic level, recurrent, attentional, multimodal, and retrieval-augmented systems embody different hypotheses about memory, representation, and inductive bias [marr_vision_1982]. Yet there is no independently validated specification of the algorithm used by the developing child against which an architecture can simply be checked. Architectural comparisons become interpretable when input and outcomes are held sufficiently constant: a systematic difference can then identify which properties are sufficient or insufficient under those conditions. Claims of architectural plausibility made before these controls remain undetermined.

The review is organized by the methodological decisions that determine those inferences. Section 2.2 examines task affordances, categorical and graded metrics, developmental trajectories, and measurement uncertainty. Section 2.3 examines the quality, quantity, and temporal organization of model input. Section 2.4 then states reporting requirements and identifies the empirical gaps addressed by the remainder of this thesis. The review is concept-driven and selective rather than a systematic or meta-analytic census. The study tables therefore map representative paradigms across domains; they should not be read as exhaustive inventories of every developmental experiment or model benchmark.

2.2 Outcome alignment

2.2.1 Task affordances across linguistic domains

Outcome alignment is stronger than topical similarity. Giving a child and a model stimuli about “syntax” is not sufficient if the child’s dependent variable indexes preference after familiarization while the model’s score is the probability of an isolated sentence. A calibrated comparison should specify at least four correspondences: the linguistic construct being probed, the information available in the stimulus, the decision or response that produces the dependent variable, and the point in development or training at which it is observed. Where any one of these changes, the interpretation must be qualified.

Child competence is usually inferred through an overt behavior—sucking rate, head turning, visual fixation, pointing, or production—after sensory encoding, attention, decision making, and motor execution. A model usually exposes a probability or hidden representation, which is converted into an experimental outcome by an additional *task model*: a distance function, probe, classifier, prompting protocol, or scoring rule. The task model is part of the hypothesis. Layer choice, pooling, prompt wording, and decision threshold can change the apparent competence even when the underlying trained model is fixed. Table 2.1 uses four deliberately conservative alignment categories: *close* when matched stimuli and decisions permit trial-level comparison; *partial* when construct and contrast are shared but response generation differs; *indirect* when only a broader theoretical construct is shared; and *none established* when the literature lacks a defensible counterpart.

Table 2.1 – Alignment of representative developmental and model evaluation paradigms. Ratings describe the current measurement correspondence, not the importance of the linguistic domain.

Domain	Child measure	Model measure	Alignment	Representative evidence
Phonology	Habituation, high-amplitude sucking, or conditioned head turning to an acoustic contrast	Distance-based ABX discrimination over selected model representations	Partial	[Eimas1971; Werker1984; Schatz2013; matusevych2023infant]
Morphology	Contextualized elicitation of an inflected or derived nonce form	Generation from a nonce lemma and sentence frame, sometimes with explicit feature labels	Partial to close	[BerkoGleason1958; weissweiler2023counting; anh2024morphology]
Word form	Familiarization followed by preference for words versus part-words; spontaneous or reported vocabulary production	Probability-based word/nonword discrimination or generation of forms	Partial	[saffran1996statistical; Aslin1998; nguyen2022word]
Word meaning	Spoken word–referent matching by looking or pointing	Forced-choice image–text matching	Close for matched trials	[bergelson_at_2012; Lany2011; Wang2025]
Semantic categories	Sorting or response-time effects for category membership and typicality	Geometry or similarity of learned embeddings	Indirect	[Lenci2022]
Syntax	Artificial-grammar preference or child production; later explicit judgments	Probability comparison for grammatical minimal pairs	Partial; potentially close for matched discrimination	[Marcus1999; warstadt2020blimp; Hu2020]
Sentence meaning	Looking-time, violation-of-expectation, or neural response to semantic anomaly	Sentence probability, surprisal, or semantic minimal-pair accuracy	Partial	[Margoni2023; friedrich2006early; kauf2023event; michaelov2024strong]
Pragmatics	Action prediction or response in socially grounded belief and communicative tasks	Prompted textual answers to narrated scenarios	Indirect; no grounded production analogue	[WimmerPerner1983; Wellman2001; kosinski2024evaluating; Strachan2024]

Phonological knowledge. Early phonological perception is commonly studied with habituation and dishabituation, high-amplitude sucking, or conditioned head-turn procedures. A change in sucking, looking, or head orientation after a stimulus change is interpreted as discrimination [Eimas1971; Best1988; Kuhl1992; Sundara2018]. These paradigms revealed the reorganization of sensitivity to native and non-native contrasts during the first year [Werker1984; tsuji2014perceptual]. The dependent variable, however, reflects more than auditory separation: the infant must encode the signal, judge it sufficiently novel, and produce a measurable response.

Familiarity, fatigue, and task parameters affect whether discrimination is expressed.

Speech models are often evaluated with an ABX task. Given tokens A , B , and X , the task asks whether the representation of X is closer to A or B [Schatz2013; Schatz2014; dunbar2021zero; lavechin2023babyslm]. This avoids training a downstream phoneme classifier, but it does not eliminate modeling choices. The selected layer and distance metric determine which acoustic or phonological information contributes to the score. ABX is also structurally closer to a three-item psychophysical comparison than to an infant’s novelty response. Comparisons of ABX trajectories with infant contrast sensitivity are therefore informative when languages, contrasts, exposure, and testing intervals are matched, but current correspondence remains partial [schatz2021early; matusevych2023infant; poli2024modeling].

Production offers a possible closer bridge because acoustic properties can be extracted from both child and generated speech. In practice, child vocalizations involve articulatory development, phonological approximation, and transcription decisions, whereas most computational production work has been evaluated qualitatively or against adult targets. A calibrated production comparison would need common acoustic measures, age-appropriate targets, and an explicit model of how representational knowledge becomes an articulatory output.

Morphological knowledge. Children’s productive morphology is classically probed by eliciting inflected or derived forms of nonce words in meaningful sentence and picture contexts [BerkoGleason1958; engelmann2019input; michaly2025development]. Perceptual work can reveal earlier sensitivity to morphological structure—for example, 6- and 8-month-olds distinguish relations between suffixed nonce forms and stems [kim2021morphology]—but that outcome is not the same as productive generalization. The point of the Wug paradigm is that a response cannot be a stored form: it must extend a pattern to a novel stem. Computational evaluations range from supervised paradigm completion, where a model receives a source form and a derivational slot [Cotterell2017], to Wug-style generation in which a nonce form appears in a sentence frame [weissweiler2023counting; anh2024morphology].

Sentence-framed generation is the closest counterpart because the populations can receive parallel contexts and their produced forms can be coded by the same rule. Alignment weakens when the model is given an explicit feature bundle or list of candidate affixes that the child does not see. Discriminative well-formedness judgments are useful diagnostics, but they are not substitutes for an elicited-production comparison unless a corresponding child judgment task is administered. The strongest design therefore matches the context, nonce inventory, response opportunity, and scoring procedure rather than treating every morphology benchmark as equivalent to a Wug test.

Word forms and lexical production. Infant word segmentation is often tested by familiarizing infants with a continuous speech stream and then contrasting words with part-words [saffran1996statistical; Aslin1998]. Evidence that infants exploit sequential statistics extends to natural-language materials [Pelucchi2009]; infant-directed prosody can also facilitate mapping a novel word form to a referent [grafestes2013prosody]. Yet the direction of listening prefer-

ence is not a transparent scale of knowledge. Brief exposure can produce a familiarity preference, whereas longer or easier exposure can produce a novelty preference [hunter1988multifactor; houston2004distinguishing]. Treating either direction as success requires a theory of how familiarization, complexity, and processing stage determine behavior.

Model benchmarks commonly compare the probability of a word with that of a matched non-word, or evaluate segmentation from speech representations [nguyen2022word; dunbar2021zero]. The target contrast is related to the infant task, but an explicit probability ranking is not the same response as a listening preference. Trial-level comparison is most credible when the same speech materials and foils are used and when the analysis predicts the magnitude and direction of infant behavior rather than coding both possible preferences as binary success.

Expressive vocabulary introduces a different alignment problem. Naturalistic transcripts record child productions after adult interpretation and normalization, while caregiver inventories summarize reported comprehension or production over a fixed list [macwhinney__chilides__2000; fenson1994variability]. Model generation provides directly observable token output, but it depends strongly on the prompt, decoding policy, and sampling opportunities. Comparing vocabulary size alone can therefore conflate lexical knowledge with observation and elicitation. A defensible comparison aligns the opportunity to produce a word, the unit of counting, the lexical normalization policy, and the cumulative exposure available at each measurement point.

Word meaning. Early comprehension is commonly assessed with looking-while-listening or forced-choice procedures: the child hears a word while viewing a target and distractor, and gaze or pointing toward the named referent indexes comprehension [bergelson__at__2012; Lany2011; Bergelson2017; barbir2023rapid]. Multimodal benchmarks can preserve the same basic decision by requiring a model to match a word or short phrase to one of two images [Wang2025]. When the visual alternatives, linguistic prompt, and scoring rule are matched, this is among the closest available child-model comparisons.

Important differences remain. Children’s referential decisions unfold in time and are supported by prosody, joint attention, and expectations about the speaker. Static image-text pairs remove those cues. Conversely, distributional models are often evaluated through lexical-relation classification or similarity among learned representations [Lenci2022]; these evaluations do not directly reproduce an infant category-sorting or referent-selection task. Such representations can be explanatory predictors, but they should not be labeled behaviorally aligned without showing that they predict trial-level child responses.

Syntactic structure. Artificial grammar learning exposes infants or children to sequences generated by a hidden regularity and tests their response to novel conforming and violating sequences [Marcus1999; gervain2012binding]. Child production additionally reveals generalization and characteristic errors that an endpoint judgment may conceal. Model evaluations such as BLiMP and SyntaxGym contrast grammatical and minimally different ungrammatical sentences using probability or surprisal [warstadt2020blimp; Hu2020; marvin2018targeted; salazar__masked__2020].

The binary contrast creates a useful structural analogy, but equivalence depends on more than the number of choices. Many model benchmarks reflect adult grammatical norms, contain vocabulary or constructions inappropriate for infants, and provide written strings rather than spoken sequences. Alignment improves when stimuli are developmentally licensed, are presented in a comparable modality, and support the same generalization claim. Production trajectories remain a major gap: probability differences at isolated checkpoints do not yet reproduce the sequence and distribution of children’s spontaneous or elicited syntactic errors.

Sentence meaning. Infant sentence comprehension can be inferred from looking or violation-of-expectation responses when a described event conflicts with a visual or causal outcome [aslin2007s; Margoni2023]. Event-related potentials, including N400 responses to semantic violations, provide a complementary processing measure [friedrich2006early; lozano2025development]. Models are evaluated through plausibility minimal pairs or word-by-word surprisal,

$$I(w_i) = -\log P_{\text{LM}}(w_i \mid w_{<i}),$$

which has been related to adult reading time and N400 amplitude [hale2001probabilistic; SmithLevy2013; frank2015erp; michaelov2024strong]. A common anomalous stimulus can support a calibrated contrast, but response magnitude is not automatically commensurable: surprisal is a model-internal quantity, whereas looking and neural signals reflect a larger processing chain. The comparison should test an explicit linking function rather than infer equality from a shared anomaly label.

Pragmatic inference. False-belief and related tasks ask whether children predict an agent’s behavior from that agent’s informational state rather than from reality [WimmerPerner1983; Wellman2001]. Language models can answer textual versions of these scenarios and sometimes reach high accuracy [kosinski2024evaluating; Strachan2024]. Surface-level task identity is deceptive here. Children act or answer within a socially interpretable event; models usually generate text after a decontextualized narrative, and the answer may be recoverable from recurring linguistic patterns. Existing tasks establish benchmark performance, not equivalence between grounded mental-state reasoning and textual continuation. Pragmatic production—adapting an utterance to a particular interlocutor’s knowledge and communicative needs—has no well-established developmental model analogue.

Across domains, the best current alignment occurs when a common forced-choice structure can be applied to matched perceptual materials, as in word–referent matching. Preference tasks and minimal-pair scores share contrasts but differ in how evidence is expressed. Similarity measures are generally explanatory proxies rather than behavioral equivalents. Production can in principle be scored jointly, but only if elicitation and opportunity are aligned. These distinctions motivate the representative evidence maps in Tables 2.2 and 2.3.

2.2.2 Selected empirical coverage

The following tables replace numerical cross-indexing with direct citations. They are intentionally selective: their purpose is to show which combinations of domain, input, and outcome are represented, and where the human and model literatures still fail to meet at the same task.

Table 2.2 – Selected computational studies spanning the outcome domains reviewed in this chapter.

Study	Domain	Model or task	Training or evaluation input	Primary outcome
[Schatz2013]	Phonology	ABX evaluation	Speech tokens and learned acoustic representations	Contrast discrimination error
[Schatz2014]	Phonology	Noise-sensitive ABX	Controlled speech and acoustic degradation	Robustness of discrimination
[Dunbar2017]	Phonology	Zero-resource speech systems	English, French, and Mandarin speech	ABX and representation quality
[dunbar2021zero]	Phonology / lexical / syntax	ZeroSpeech evaluation suite	Untranscribed speech with lexical and syntactic probes	ABX, spoken-word, and grammar scores
[lavechin2023babyspeech]	Speech development	BabySLM	Developmentally motivated speech input	Phonetic, lexical, and syntactic scores
[matusevych2023infant]	Phonology	Alternative acoustic learners	English, French, and Japanese speech	Contrast-specific developmental trajectories
[poli2024modeling]	Phonology	Contrast-learning model	Speech embedded in broader ambient sound	Trajectory of discrimination
[Cotterell2017]	Morphology	Sequence-to-sequence derivation	Source forms with derivational paradigm slots	Derived-form accuracy
[weissweiler2023counting]	Morphology	Wug-style LM evaluation	Nonce roots in linguistic contexts	Productive inflection
[anh2024morphology]	Morphology	Generative language models	Multilingual Wug-style items	Human-rated generalization accuracy
[nguyen2022word]	Word form	Spoken language model probes	Continuous or segmented speech	Lexical discrimination and segmentation
[vong2025robustness]	Word meaning	Multimodal neural learners	Child-specific SAYCam audiovisual streams	Robustness of word–referent learning across children
[Wang2025]	Word meaning	Developmental vision–language model	Synthetic child-directed captions and visual scenes	Visual two-word matching
[Lenci2022]	Lexical semantics	Semantic-relation classifier	Distributional representations and lexical-relation datasets	Relation-classification performance
[warstadt2020blimp]	Syntax	BLiMP minimal pairs	Controlled grammatical contrasts	Forced-choice accuracy
[Hu2020]	Syntax	SyntaxGym-style targeted evaluation	Sentences instantiating syntactic dependencies	Surprisal interactions and accuracy
[marvin2018targeted]	Syntax	Targeted LM evaluation	Grammatical and ungrammatical sentence pairs	Relative sentence probability
[salazar_masked_2020]	Syntax / semantics	Pseudo-log-likelihood scoring	Masked language models and minimal pairs	Sentence-level acceptability
[hale2001probabilistic]	Sentence processing	Probabilistic parser	Incrementally parsed sentences	Word-by-word surprisal
[SmithLevy2013]	Sentence processing	Probabilistic language model	Adult reading materials	Relation between surprisal and reading time
[frank2015erp]	Sentence meaning	Recurrent language model	Sentence stimuli paired with EEG evidence	Relation between surprisal and N400

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Table 2.2 continued

Study	Domain	Model or task	Training or evaluation input	Primary outcome
[kauf2023event]	Sentence meaning	Autoregressive language models	Plausible and implausible event descriptions	Semantic minimal-pair accuracy
[michaelov2024strong]	Sentence meaning	Large language models	ERP stimulus materials	Surprisal–N400 correspondence
[kosinski2024evaluating]	Pragmatics	Prompted language models	Textual false-belief scenarios	Mental-state answer accuracy
[Strachan2024]	Pragmatics	Comparative LLM evaluation	Theory-of-mind tasks used with humans	Cross-task accuracy and error profile

Table 2.3 – Selected infant and child studies illustrating the behavioral and neural measures discussed in this chapter.

Study	Domain / age	Paradigm	Stimulus or setting	Observed response
[Eimas1971]	Phonology / young infants	High-amplitude sucking	Synthetic voice-onset-time contrast	Recovery of sucking response
[Werker1984]	Phonology / 6–12 months	Conditioned head turn	Native and non-native consonant contrasts	Detection of sound change
[Best1988]	Phonology / infancy	Habituation	Non-native click consonants	Dishabituation or looking recovery
[Kuhl1992]	Phonology / 6 months	Conditioned response	Vowel prototypes and variants	Category-sensitive discrimination
[Sundara2018]	Phonology / 6–10 months	Infant-controlled habituation	Language-specific coronal contrasts	Looking-time recovery
[BerkoGleason1958]	Morphology / preschool and school age	Wug elicitation	Pictures and nonce words	Produced plural and past-tense forms
[engelmann2019input]	Morphology / preschool children	Elicited production	Real and novel verbs in Finnish and Polish	Inflectional generalization
[kim2021morphology]	Morphology / 6–8 months	Segmentation and recognition	Suffixed nonce forms and bare stems	Sensitivity to morphological relatedness
[michaly2025development]	Morphology / school-age children	Comprehension and production	Real- and pseudoword derivational analogies	Accuracy and use of morphological elements
[safran1996statistical]	Word segmentation / 8 months	Familiarization preference	Artificial continuous speech	Word versus part-word listening
[Aslin1998]	Word segmentation / 8 months	Head-turn preference	Artificial speech stream	Statistical segmentation preference
[Pelucchi2009]	Word segmentation / 8 months	Familiarization preference	Natural Italian child-directed speech	Word versus part-word listening
[grafestes2013prosody]	Word learning / 17 months	Habituation word learning	Novel labels in infant- or adult-directed prosody	Learning of label–object mappings
[bergelson__at__2012]	Word meaning / 6–9 months	Looking while listening	Familiar nouns and paired images	Target-directed gaze
[Lany2011]	Word meaning / second year	Forced-choice comprehension	Novel word with two referents	Gaze to selected target
[Bergelson2017]	Word meaning / infancy	Looking while listening	Named objects in visual scenes	Time course of target gaze

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Table 2.3 continued

Study	Domain / age	Paradigm	Stimulus or setting	Observed response
[barbir2023rapid]	Word meaning / toddlers	Forced-choice comprehension	Spoken word and paired pictures	Looking or pointing choice
[Marcus1999]	Syntax / 7 months	Artificial grammar learning	Repetition patterns over syllables	Listening preference at test
[gervain2012binding]	Sequence structure / newborns	Functional near-infrared spectroscopy	Speech sequences varying identity relations and position	Cortical response to identity and position
[Margoni2023]	Event and sentence meaning / infancy	Violation of expectation	Possible and impossible event outcomes	Differential looking time
[friedrich2006early]	Word / sentence meaning / toddlers	Picture–word semantic violation	Congruent and incongruent spoken items	N400 response
[lozano2025developmental]	Semantic processing / bilingual toddlers	ERP experiment	Semantically congruent and incongruent materials	Development of N400 effects
[WimmerPerner1983]	Pragmatics / preschool onward	False-belief task	Character, object transfer, and search story	Predicted search location
[Wellman2001]	Pragmatics / childhood	Meta-analysis of false-belief tasks	Diverse belief scenarios	Age-related proportion correct

2.2.3 Thresholded evidence and graded performance

The same task can support different claims depending on how competence is operationalized. A *thresholded* or milestone-based analysis asks whether evidence for a specified contrast exceeds a criterion. A *graded* analysis asks how strongly, accurately, or efficiently the learner expresses that contrast. These questions are complementary, but neither metric can stand in for the other.

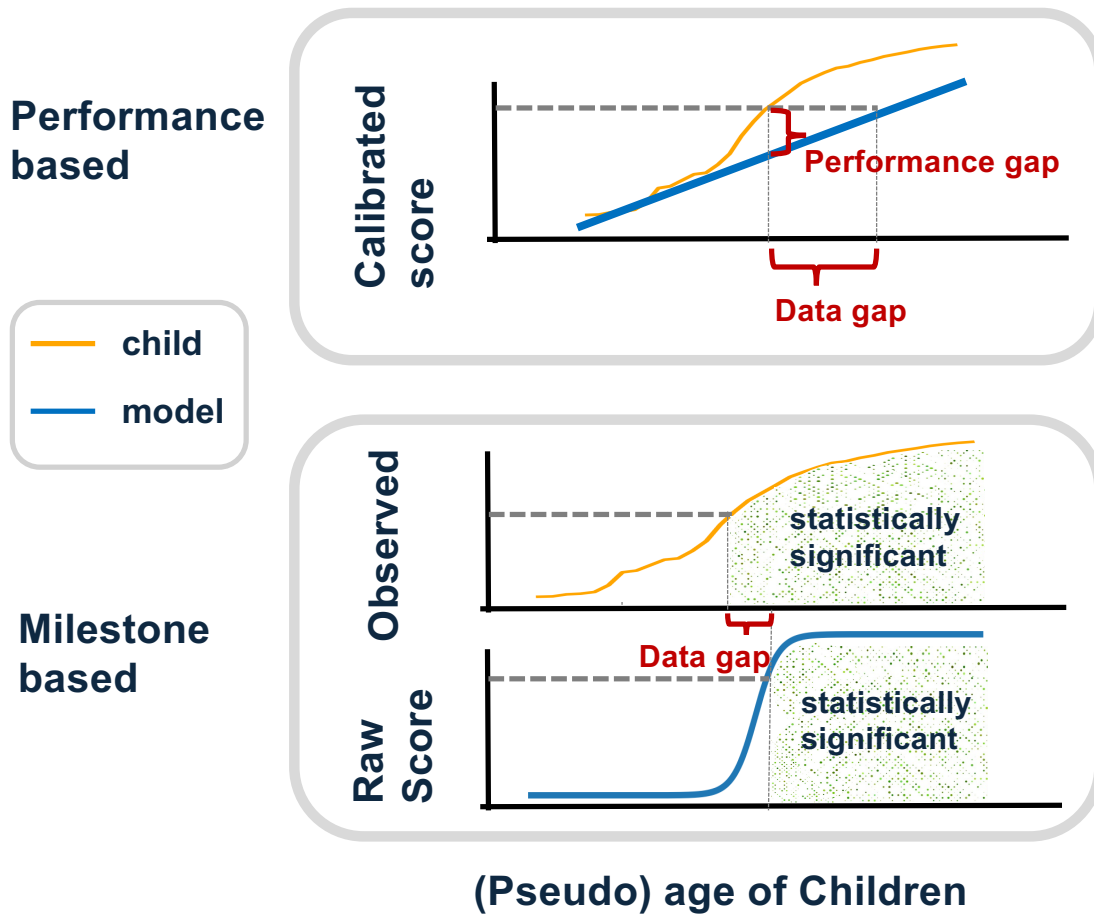


Figure 2.1 – Schematic contrast between thresholded and graded evaluation. A milestone analysis treats evidence for a specified competence as present once an explicit statistical criterion is met. A graded analysis retains continuous performance and supports comparison of strength and trajectory. Both depend on the validity of the underlying task.

Thresholded evaluation is common in targeted syntactic work. A model is said to exhibit a dependency when it assigns higher probability to the predicted member of a controlled contrast reliably above chance, or when a theoretically specified surprisal interaction is present [linzen2016assessing; marvin2018targeted; Hu2020; PearlSprouse2013]. This formulation clarifies what has been demonstrated: the model has crossed a criterion for one operationally defined construction. It does not imply mastery of syntax more generally, and it does not show that the mechanism producing the effect is the same as a child’s. Results can be sensitive to item construction, probability estimation, and the statistical rule used to declare the milestone [warstadt2020blimp; wilcox2023testing].

Graded evaluation reports accuracy, surprisal, perplexity, correlation, or another continuous score [warstadt2023findings; hu2024findings]. Continuous measures preserve differences among checkpoints and architectures, but global improvement can obscure a missing generalization. Perplexity may decrease as frequent sequences become more predictable while a particular long-distance dependency remains absent [Hu2020]. Likewise, reaching or exceeding an adult aggregate score does not establish that every component competence has been acquired.

For child–model comparison, thresholded analyses are most informative when the criterion corresponds to an independently established developmental effect, while graded analyses are most informative when both populations contribute observations on a shared scale or through an explicit linking model. Reporting both is often preferable: a threshold locates the emergence of a theoretically defined behavior, and the continuous effect size describes how it strengthens. Figure 2.1 illustrates that the two summaries compress different features of the same evidence.

2.2.4 Developmental trajectories

Endpoint similarity is weak evidence for developmental similarity because different learning routes can converge on the same final score. Trajectory analysis instead evaluates performance across exposure and asks about acquisition order, curve shape, error distribution, and sensitivity to changes in the learning environment [chang2022word; evanson2023language; ma2025babysit]. It links milestone and graded analyses: milestones identify transitions, while continuous measurements reveal whether change is gradual, abrupt, monotonic, or temporarily regressive.

Lexical work illustrates this approach through neural age of acquisition: the point in training at which the model’s evidence for an individual word meets a predeclared criterion. Model acquisition order can then be compared with child age-of-acquisition norms and related to frequency, contextual diversity, and interaction [chang2022word; ma2025babysit]. The comparison requires care. A checkpoint is an amount of optimization, not a child’s chronological age; the mapping becomes meaningful only after accounting for the amount and order of input processed and for repeated exposure across epochs.

Curve shape provides additional evidence. Children sometimes show U-shaped development, producing stored irregular forms correctly, then overgeneralizing a productive rule, and finally recovering the exceptions [marcus_overregularization_1992]. If analogous non-monotonic patterns appear in neural learners, they are more informative than endpoint accuracy because they constrain the interaction among item frequency, generalization, and memory. Still, a visually similar curve is not sufficient. The error types, item-level ordering, exposure scale, and conditions producing the reversal must also agree.

Trajectory analysis also tests whether grammatical phenomena emerge in an order resembling child development. Some model capabilities improve smoothly, whereas others appear more abruptly, and their relative ordering may change with architecture or corpus composition [evanson2023language]. Distributional analyses of word-learning trajectories can test whether acquisition order reflects interpretable lexical properties or merely the statistics of a particular corpus [ficarra2025distributional]. Such comparisons are most useful when age norms are precise and the evaluation remains constant across checkpoints; post-hoc selection of thresholds or checkpoints can manufacture apparent stages.

Finally, a developmental trajectory includes the sequence of experience, not only the cumulative token count. Curriculum learning and interaction may change what is learned at a given budget. Evidence that preserving local discourse continuity can matter more than globally sorting child-directed utterances by child age cautions against equating any “simple-to-

complex” ordering with the child’s curriculum [feng2024child]. Synthetic variation can improve some morphosyntactic outcomes without producing equivalent gains in semantics or pragmatics [haga2024babylm]. Trajectory alignment must therefore be reported per domain rather than reduced to one global developmental score.

2.2.5 Measurement uncertainty

Both developmental experiments and model evaluations vary, but their uncertainty enters at different stages. The relevant question is not simply which learner is noisier. It is which components of the observed variance reflect the target knowledge, which reflect the measurement procedure, and which can legitimately be compared.

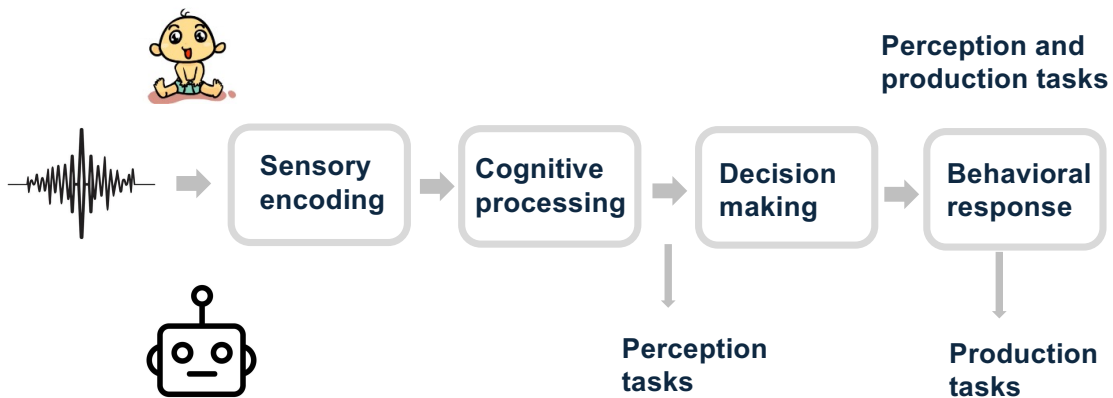


Figure 2.2 – Sources of variation across a simplified evaluation pipeline. Infant and child behavior is observed after sensory encoding, cognitive processing, decision or response selection, and motor execution. Many model perception scores are read from representations through a task model before an overt response stage; generated model output is observed later in the pipeline. The comparison must identify, rather than assume away, these distinct measurement stages.

For children, sensory encoding depends on signal quality and hearing; cognitive processing includes the developing representation of interest; decision and response selection depend on arousal, familiarity, memory, and task interpretation; execution adds oculomotor, articulatory, equipment, and coding error. Some variation is nuisance, but some may be developmental signal: inconsistent responding can reflect a transitional representation rather than random failure [Mitchell2022; kucharsky2024habituation]. Large collaborative studies document substantial heterogeneity across laboratories, procedures, ages, and individual infants [ManyBabies2020]. Preferential-looking effects can be robust at the group level while showing limited individual test–retest reliability [Monroy2021; DeBolt2020; byers2022six; schreiner2024limited].

For models, variability arises from initialization, data order, stochastic optimization, dropout, preprocessing, hardware and numerical nondeterminism, and sampling at inference [srivastava_dropout_2014; dodge2020fine; summers2021nondeterminism]. Conditional on a fixed trained model and a deterministic scoring procedure, repeated evaluation of the same input can be identical. That narrow repeatability should not be confused with eliminating model uncertainty. Different seeds may produce materially different learning trajectories, and even a fixed model’s apparent

competence depends on task-model decisions such as layer, probe, prompt, tokenization, and decoding. These sources are often controllable and estimable, but they are not absent.

The central asymmetry is thus structural. A child’s discrimination score is observed after several embodied response stages, whereas an ABX or cosine score may be calculated directly from a selected representation. Production moves model evaluation closer to an overt response, but child articulation and transcription still have no simple analogue in model decoding. Adding arbitrary noise to model outputs does not solve this problem: it can mimic the width of a behavioral distribution without reproducing its causal sources. Likewise, collecting more infant trials reduces sampling error but cannot by itself separate unstable knowledge from state-dependent behavior.

Calibrated analysis should instead propagate uncertainty on both sides. Human estimates should report exclusions, trial counts, coding reliability, individual and laboratory effects, and uncertainty around age-specific means. Model estimates should report independent training runs, checkpoint selection, task-model choices, and item-level confidence intervals. Where possible, the same hierarchical model should estimate stimulus-level and learner-level effects, while preserving the distinction between biological participants and random model instances. This does not make the noise structures identical; it prevents differences in precision from being mistaken for differences in competence.

2.3 Input alignment

Outcome alignment determines whether the two learners are measured comparably; input alignment determines whether they have been asked to solve comparable learning problems. “Child-scale” is not a property of token count alone. Children encounter spoken language in recurring activities, accompanied by visible events, interlocutors, and feedback. The distribution changes with age and with the child’s own behavior. A corpus can match one part of this experience while diverging sharply on others. For that reason, input claims should be decomposed into register, modality, interaction, quantity, and temporal organization rather than summarized by a single realism label.

2.3.1 Quality, register, modality, and interaction

Language-only text. Web corpora such as Common Crawl, C4, and Wikipedia provide extensive coverage of lexical, syntactic, and factual patterns [buck2014n; raffel2020exploring; Kaplan2020]. They are nevertheless dominated by written, adult-directed genres: news, reference material, online discussion, and technical prose. Compared with child-directed speech, these sources differ in vocabulary, utterance length, discourse function, repetition, and topic distribution [huebner2021babyberta]. Shuffling documents also removes the contingent sequence in which one conversational turn responds to another. A model trained on this material may answer whether a pattern is learnable from broad written evidence, but not whether it is learnable from the experience available to a young child.

The BabyLM Challenge constrains training to 10 or 100 million words and combines child-directed transcripts with educational and simplified materials [warstadt2023findings]. It therefore improves control over both scale and register and makes comparison across architectures possible within a common budget. Its input remains largely textual, however. Transcribed speech omits prosody, voice, timing, overlap, and much of the situational context that makes an utterance interpretable. A developmentally bounded text corpus is thus a valuable experimental resource, but it is not a complete reconstruction of infant input.

Naturalistic child-directed transcripts, especially CHILDES, more closely represent the vocabulary and conversational structure addressed to children [macwhinney_childes_2000]. Child-directed speech tends to contain shorter utterances, concrete and frequent vocabulary, repetition, reformulation, and prosodic adaptation; its composition predicts developmental outcomes [WeislederFernald2013]. The main limitation is not only corpus size. Recordings sample particular families and activities, transcription maps a continuous multimodal exchange into discrete words, and the resulting database does not preserve everything that was perceptually available. “Training on CHILDES” should consequently be accompanied by the languages, participants, age ranges, speaker roles, and preprocessing decisions used.

Synthetic augmentation attempts to expand such corpora without returning to adult web register. Paraphrases and variation sets generated from child-directed utterances can increase lexical or syntactic diversity and improve some formal evaluations [haga2024babylm]. But register imitation is not equivalent to interaction. Models generating caregiver-like dialogue can reproduce local lexical and utterance statistics while overstating alignment and failing to reproduce turn-by-turn contingency [liu2024benchmarking]. Synthetic data are best described by the property they add—paraphrastic variety, simplified syntax, or a target vocabulary—rather than as globally child-like.

Language-only audio. Speech supplies information removed by transcription: segmental variation, prosody, speaker identity, timing, and cues to turn and utterance boundaries. These signals are central to phonetic category formation and segmentation [Kuhl2006; saffran1996statistical]. Read-speech and audiobook corpora offer clean acoustic conditions and relatively reliable annotation, but their adult-directed register and performance style differ from home interaction. Spontaneous adult speech improves discourse naturalness without necessarily matching what is addressed to a child.

Long-form child-centered recordings preserve more of the natural acoustic ecology, including background sound, overlapping voices, changing distance from speakers, and extended daily routines [cristia2020accuracy; bergelson2023everyday]. That realism creates a difficult learning signal: target speech is sparse, noisy, and only partly directed to the child. Systems evaluated on clean audiobooks and on long-form recordings are therefore changing acoustic quality and register together [lavechin2024modeling]. This trade-off is scientifically useful if it is reported explicitly; otherwise, a performance gap can be misattributed to the amount of data or to architecture.

Language with vision. Visual context makes referential learning possible in a way that text or audio alone does not. Large image–caption collections such as LAION and COCO provide millions of paired examples [schuhmann2022laion; lin_microsoft_2014]. Their captions describe images for adult readers rather than participating in a child’s unfolding activity. The pairing is curated and static, often identifying a salient object more cleanly than a natural scene would. These resources establish what vision–language architectures can learn from large aligned datasets, but they provide limited evidence about early cross-situational learning.

Synthetic child-directed captions occupy an intermediate position. Developmental vision–language work can pair child-perspective images with simplified descriptions and evaluate grounded combinations on child-oriented tests [Wang2025]. Such data improve register correspondence while retaining a clean supervised pairing. They do not recreate the ambiguity of a caregiver utterance whose referent depends on joint attention, current action, and conversational history.

Egocentric video more directly samples the visual stream available to individual children. SAY-Cam follows a small number of infants longitudinally and has supported demonstrations of word–referent learning from a single child’s visual and linguistic experience [sullivan_saycam_2022; vong2024grounded]. BabyView broadens the recording effort and improves visual quality and participant diversity [Long2024]. Yet a head-mounted camera is not an infant’s perceptual system: camera field of view differs from gaze and attention, and usable paired speech is far scarcer than web-scale image–text examples. High ecological validity therefore comes with limited coverage and measurement uncertainty of its own.

Language, vision, and social interaction. Children learn in exchanges structured by turn taking, joint attention, communicative intention, and contingent caregiver response [Tomasello1995; Csibra2009]. Dyadic corpora retain some of this structure. Shared book-reading recordings combine caregiver and infant perspectives in a constrained but genuinely interactive setting [ahamat2025shared]; ECOLANG annotates speech, gesture, and gaze in interactions with older children [gu2025ecolang]; and game-based corpora extend multimodal dialogue to middle childhood [bodur_chico_2021; goumri_chica_2024]. Their value lies in temporal coordination across channels, not merely in the coexistence of audio and video.

These corpora remain too small and specialized to serve as drop-in replacements for large language-model training sets. They can instead support focused learning experiments, evaluation of contingent behavior, or estimation of the distributions that a larger simulation should preserve. Treating every audiovisual recording as “social” would erase the key distinction: co-occurrence presents signals together, whereas interaction makes one participant’s behavior conditional on the other’s.

Table 2.4 – Representative input resources organized by modality, register, interactional structure, and principal limitation. The entries characterize resource types and do not imply that every study uses identical preprocessing.

Resource	Modality and register	Interactional structure	Developmental value	Principal limitation
Common Crawl [buck2014n]	Written, mostly adult-directed web text	Documents rather than interaction	Broad lexical and domain coverage	Register, scale, and temporal mismatch
C4 [raffel2020exploring]	Filtered written web	Documents rather than interaction	Reproducible large-scale LM training	Adult-written genres and weak grounding
BabyLM [warstadt2023findings]	Mixed text with child-directed and simplified components	Some dialogue after transcription	Controlled 10M/100M-word budgets	Predominantly textual; heterogeneous sources
CHILDES [macwhinney_childes_2000]	Transcribed child speech	Natural conversational turns	High register correspondence across languages	Sparse sampling and loss of acoustic/visual context
Synthetic CDS variation sets [haga2024babylm; liu2024benchmarking]	Generated child-directed-style text	Simulated turns or paraphrases	Scalable control of surface register	Limited contingency and inherited generator bias
Read-speech collections [lavechin2023babyslm]	Clean adult-directed audio	Narration rather than dialogue	Acoustic and prosodic learning at scale	Low interactional and register correspondence
Child-centered long-form audio [cristia2020accuracy; bergelson2023everyday]	Naturalistic home audio	Extended daily interaction	Realistic acoustic ecology and exposure estimates	Noise, overlap, privacy, and sparse annotation
LAION [schuhmann2022laion]	Web image–text pairs	No contingent interaction	Large-scale visual grounding	Adult web captions and curated pairing
COCO [lin_microsoft_2014]	Curated images and descriptive captions	No contingent interaction	Controlled object and scene descriptions	Captioning task unlike child-directed talk
Developmental synthetic captions [Wang2025]	Child-oriented text with visual scenes	Static pairing	Register-controlled grounded evaluation	Generated language and absent joint attention
SAYCam [sullivan_saycam_2022; vong2024grounded]	Infant-perspective with speech	Naturalistic home activity	Longitudinal, child-specific grounding	Few children and camera–attention mismatch
BabyView [Long2024]	Egocentric developmental video and audio	Naturalistic activity	Broader and higher-quality child-perspective sampling	Modest scale relative to web corpora
Shared book reading [ahamat2025shared]	Dyadic audiovisual child-directed speech	Turn taking and joint attention	Aligned caregiver and infant perspectives	Narrow activity and small sample
ECOLANG [gu2025ecolang]	Speech, gesture, and gaze	Naturalistic dyadic interaction	Explicit cross-channel coordination	Older age group and limited training scale
CHICO/CHICA [bodur_chico_2021; goudmri_chica_2024]	Multimodal task dialogue	Collaborative word games	Naturalistic communicative adaptation	Task- and age-specific interaction

2.3.2 Quantity and temporal organization

Claims about data efficiency require a defined denominator. Studies variously report words, subword tokens, utterances, audio hours, image–text pairs, gradient updates, or epochs. These

units do not convert transparently. One hour of long-form audio contains silence, distant speech, overlapping speakers, and utterances not directed to the child; one hour of an audiobook is densely linguistic. Likewise, a word processed during a hundredth epoch is not equivalent to a newly encountered word in a new situation. Reporting a nominal corpus size without total presentations and sampling procedure can therefore make a heavily recycled training regime appear child-scale.

Estimates of children’s cumulative linguistic exposure vary with recording method, socioeconomic and cultural context, the definition of speech addressed to or merely audible to the child, and extrapolation from sampled days [hart1992american; WeislederFernald2013; frank2023baby]. A defensible comparison should state a range and sensitivity analysis rather than a single universal number. It should also distinguish raw exposure from *effective exposure*: what the learner can encode given signal quality, attention, vocabulary, and multimodal context.

Fixed-corpus trajectories. In a fixed-corpus design, one dataset is repeated while the model is evaluated across checkpoints [chang2022word; evanson2023language]. Holding composition constant helps isolate learning dynamics and makes frequent measurement inexpensive. It is useful for studying item acquisition order and the emergence of particular generalizations. But the developmental analogy is limited. Children do not receive the same closed corpus in a new random order on every pass; their vocabulary and activities expand, interlocutors respond to them, and old events cannot be replayed exactly.

Repeated epochs also change the effective quantity. A model trained for 100 epochs over ten million words has processed one billion word presentations, even though its unique corpus is described as ten million words. Memorization, overtraining, and declining lexical novelty may shape the resulting trajectory. At minimum, studies should report unique tokens, total token presentations, number of epochs, sampling with or without replacement, and the exposure count at every evaluated checkpoint.

Successive scaling. Scaling designs train otherwise comparable models on increasingly large samples and ask how much input is needed for a target behavior [warstadt2020learning]. BabyLM makes this comparison explicit within bounded text budgets [warstadt2023findings; hu2024findings]; STELA and BabySLM extend related logic to quantities and registers of speech [lavechin2023babyslm; lavechin2024modeling]. The input level at which a model crosses a milestone can be compared with a plausible exposure range for children.

This design better represents cumulative amount, but independent models do not form a longitudinal learner. Each scale may begin from a new initialization, and the larger training set is not necessarily an ordered extension of the smaller one. Apparent movement along a learning curve can therefore combine data quantity, sample composition, and between-run variation. Nested corpora, multiple seeds per scale, and identical evaluation make the interpretation stronger.

Curriculum and continual learning. Children’s input is incrementally arriving and contextually clustered. A caregiver repeats a topic within an exchange, adapts utterances to a response,

and introduces new vocabulary through changing activities. Curriculum experiments attempt to preserve selected parts of that structure. Globally ordering child-directed speech by the child’s age has not consistently improved model outcomes, while keeping local conversational context can matter [feng2024child]. The relevant curriculum may therefore lie in coherent interaction rather than a simple global ordering from short to long sentences.

Continual learning offers the closest formal analogy to an accumulating stream: the model updates on new data while retaining what it learned earlier. Standard neural systems are vulnerable to catastrophic forgetting, in which new learning disrupts earlier representations [mccloskey1989catastrophic]. That failure is not merely an inconvenience to be hidden by replay. It identifies a property—stable integration of new evidence with long-term knowledge—that a developmental model must explain. Replay buffers, retrieval, and external memory can be tested as hypotheses, but their stored amount should count toward the model’s effective exposure.

No current design aligns quantity, lexical novelty, temporal order, interaction, and learning continuity simultaneously. Fixed corpora maximize experimental control but recycle a static distribution; successive scaling improves amount calibration but often substitutes different learners for developmental time; continual learning represents sequential arrival but introduces retention problems. The appropriate choice depends on the claim. Studies should state which dimension is controlled and which remains an abstraction.

2.4 Implications, reporting standards, and thesis research gaps

2.4.1 What a calibrated comparison licenses

Input and outcome alignment are distinct requirements. Strong outcome alignment with weak input alignment shows that a model can reproduce a behavioral contrast under a different learning regime. Strong input alignment with weak outcome alignment shows what representations or engineering scores arise from child-oriented data, but not whether child behavior has been reproduced. When both are reasonably aligned, model success supports a bounded sufficiency claim: the specified learner can acquire the specified outcome from the specified experience. Even then, success does not establish that children use the same internal mechanism.

This framework separates two legitimate research orientations. Developmentally oriented modeling uses a model to test hypotheses about learning and therefore requires calibrated input and outcomes. Engineering-oriented evaluation uses human performance to diagnose or improve artificial systems; it can be scientifically valuable without recreating development, but its results should not be presented as evidence about how children learn. Problems arise when the orientation changes between method and conclusion—for example, when a web-trained model’s benchmark success is interpreted as a developmental proof, or when failure under a mismatched task is treated as evidence for an innate human constraint.

Model failure becomes informative through controlled contrasts. If input and evaluation are fixed while memory architecture changes, a performance difference can bear on the role of memory. If architecture and task are fixed while register changes, the comparison can isolate

properties of the input distribution. If all dimensions vary at once, attribution is impossible. Calibration is therefore not a demand that every model imitate a child in every respect; it is a discipline for identifying which difference is the experimental variable.

2.4.2 Minimum reporting for interpretable evidence

Transparent reporting should make the inferential chain reconstructible. At a minimum, a developmental model comparison should specify:

1. **The orientation and claim.** State whether the study tests developmental sufficiency, mechanism, prediction of human behavior, or engineering performance, and formulate the conclusion at the same level.
2. **The input population.** Report language, register, speaker roles, child ages, modality, activity or genre, collection procedure, filtering, deduplication, and synthetic-data provenance.
3. **The exposure budget.** Give unique and total tokens or hours, epochs, sampling and curriculum, amount processed at each checkpoint, and a justified range for the human comparison.
4. **The outcome and task model.** Identify the construct, stimuli, response alternatives, prompt or probe, selected layer, scoring rule, threshold, and the linking assumption between child and model measures.
5. **The developmental comparison.** Predefine the human age range, checkpoints, acquisition criterion, and whether evidence concerns endpoint, order, curve shape, or item-level errors.
6. **Uncertainty and robustness.** Report participant exclusions and measurement reliability for humans; seeds, runs, numerical settings, checkpoint selection, and decoding for models; provide item- and learner-level uncertainty where possible.
7. **Residual mismatches.** Name the missing modalities, interaction, cognitive resources, or response stages and state how they limit the conclusion.

The checklist does not require every study to solve the entire calibration problem. It requires enough detail to know which problem was solved. Standardized benchmarks remain useful, but their scores should be accompanied by developmental task analyses. Naturalistic corpora remain useful despite incomplete coverage, but their sampling and preprocessing must be visible. Cumulative evidence emerges when later work can reproduce both the learner and the comparison, not only the final score.

2.4.3 Research gaps addressed in this thesis

The review identifies three connected gaps that motivate the empirical chapters that follow. First, lexical development lacks a production-oriented measure that aligns model generation with the cumulative growth recorded for children while making input budgets explicit. Chapter 3 addresses this gap with CORPUS-CDI, comparing the expressive vocabularies of children and

language models across developmental stages and controlled exposure. This operationalizes the present chapter’s argument that a vocabulary count is meaningful only when observation opportunities, lexical units, and developmental time are defined jointly.

Second, differences between child and model syntax may reflect memory and lexical exposure rather than grammatical representation alone. Chapter 4 tests whether episodic retrieval changes language models’ sensitivity to syntactic minimal pairs and whether it reduces a lexical-frequency gap. Retrieval is thereby treated as a controlled architectural hypothesis under a common outcome measure, rather than invoked after the fact to explain a mismatch.

Third, existing child-oriented corpora preserve linguistic register more readily than interactional contingency. Chapter 5 evaluates how well large language models reproduce child–caregiver dialogue and then isolates which forms of caregiver feedback support grammar learning. The benchmark distinguishes surface resemblance from turn-level behavior, while the learning experiment treats feedback as an input variable. Together, these studies address the interactional component of input alignment that static corpora leave unresolved.

These contributions do not attempt to build a globally child-equivalent learner. They make narrower comparisons in which the inputs, outcomes, or hypothesized mechanisms can be varied and measured. Their conclusions should therefore be read as evidence about lexical growth, episodic retrieval, and caregiver feedback under the stated conditions, not as a general equivalence between children and contemporary language models.

2.4.4 Limits of the framework

Calibration is necessary for developmental interpretation but not sufficient for mechanistic identity. Two systems can receive similar observable input and produce similar responses while relying on different internal representations. Stronger mechanistic evidence requires intervention, process-level prediction, and generalization to new tasks, not only behavioral correlation [marr_vision_1982; Perfors2011].

The framework also abstracts over individual and cultural variation. Children’s environments differ in quantity, interlocutors, activity structure, multilingual exposure, and the social organization of learning [WeislederFernald2013; bergelson2023everyday]. Average developmental curves can conceal distinct routes and can make one narrowly sampled corpus appear universal. Model ensembles are not replicas of this variation because their differences arise from initialization and training rather than lived environments. Future comparisons should link individual input histories to individual outcomes where the data permit.

Finally, the evidence base is uneven. Phonological and lexical perception have established infant paradigms, syntactic model evaluation has extensive minimal-pair resources, and word–referent matching provides a promising shared task. Naturalistic production, multimodal sentence meaning, and socially grounded pragmatics remain less well aligned. The selective study maps in this chapter represent that imbalance rather than a complete quantitative review, and claims about gaps should be updated as new corpora and matched paradigms become available.

2.5 Chapter summary

Computational models can inform language acquisition when the comparison is calibrated to the inference being made. Outcome alignment requires more than a shared linguistic label: child and model tasks must target the same construct through stimuli, response alternatives, and measures whose relationship is explicit. Thresholded tests, graded scores, developmental trajectories, and uncertainty each reveal different aspects of learning. They should be combined where possible rather than treated as interchangeable summaries.

Input alignment is similarly multidimensional. A child-oriented token budget does not compensate for adult-written register, and naturalistic child-directed language does not automatically preserve perceptual modality or social contingency. Repeated epochs, shuffled corpora, successive scaling, and continual learning embody different assumptions about developmental time. Consequently, every success and failure must be indexed to the quantity, quality, ordering, and interactional structure of the experience that produced it.

The strongest conclusion licensed by aligned success is a conditional sufficiency claim: a specified learner can acquire a specified outcome from specified input. Mechanistic conclusions require controlled variation and process-level evidence. This logic guides the following chapters on lexical growth under defined exposure, episodic retrieval with shared syntactic outcomes, and caregiver interaction as a component of the learning environment.

Chapter 3

CORPUS-CDI: comparing expressive vocabulary growth in children and language models

Objectives

Develop CORPUS-CDI as a calibrated production benchmark and test whether character-level language models reproduce child vocabulary trajectories across input scales, decoding conditions, and lexical frequency bands.

3.1 From lexical knowledge to observed production

The first empirical study in this thesis asks how a model’s vocabulary growth can be compared with children’s when the comparison is based on what each learner produces. Chapter 2 established that such a comparison requires a specified relation between input, learning, and measured behavior. Expressive vocabulary makes this requirement concrete. A learner may encounter a word many times without producing it in a short observation, and a model may generate a plausible word form without establishing that it understands its meaning. The present chapter develops an explicit measurement framework for this problem and uses it to examine the lexical trajectories of predictive language learners.

Vocabulary is informative because it supports comparisons at several levels simultaneously: how many target words appear in production, how their availability changes with experience, and whether frequent and infrequent words follow different trajectories. The MacArthur–Bates Communicative Development Inventories (CDI) and the developmental data assembled in Wordbank provide a substantial basis for studying early expressive vocabulary through caregiver reports [fenson2007macarthur; frank2017wordbank]. Naturalistic child transcripts provide a complementary record of observed productions. Together, these sources make lexical development a useful setting in which to ask whether learning from bounded quantities of linguistic input yields patterns resembling those found in children.

The measures nevertheless capture different aspects of vocabulary. A caregiver inventory summarizes whether a child is reported to say particular words across everyday situations. A transcript records the forms observed during sampled interactions. Model generation supplies another sample of forms, whose composition depends on the trained distribution and the decoding procedure. Throughout this chapter, *lexical knowledge* refers to the broader capacity to represent and use words, *expressive vocabulary* to the words available for production, and *observed production* to the output actually recorded under a specified procedure. The benchmark operationalizes the last of these as evidence about the second; it does not independently measure the full lexical knowledge of either learner.

CORPUS-CDI makes this link explicit through three coordinated choices. It scales training corpus sizes to estimated quantities of childhood input, sets generation budgets using estimates of monthly child production, and constructs content-word evaluation sets aligned in lexical category and frequency. A count threshold calibrated on child productions links corpus observations to the CDI reference trajectory. These choices address a central ambiguity in vocabulary comparison: a larger vocabulary count can result from more learning, more opportunities to produce words, or a test list containing easier items. The benchmark brings these factors into the experimental design rather than leaving them implicit.

The study compares character-level LSTMs and Transformers trained on child-directed speech or audiobook transcripts. It asks whether increasing the available training data reproduces the child production trajectory, whether the answer converges across vocabulary coverage, lexical diversity, and non-word rate, and whether decoding temperature or lexical frequency accounts for the observed differences. Age labels index estimated input and output budgets; the experiment

does not simulate biological maturation or one child’s continuous learning history.

The clearest result is that the evaluated models do not jointly reproduce the child reference patterns. Target-word coverage increases with input size, but its trajectory differs from the accelerated growth in the child corpus; low-frequency words are particularly sensitive to model and sampling conditions. Lexical diversity and non-word rates also depend on architecture and corpus, making them complementary constraints rather than interchangeable measures of progress. The following study presents the benchmark and these comparisons in detail. The chapter discussion then considers what they establish about data efficiency and how they motivate the subsequent test of retrieval on syntactic contrasts.

3.2 Study rationale

Large language models (LLMs) have shown that a general-purpose architecture trained on large text corpora can acquire rich syntactic and semantic knowledge [brown2020language; devlin2019bert; touvron2023llama]. Beyond their impact on NLP applications [bommasani2021opportunities], LLMs provide scalable tools for testing theories of early language acquisition [dupoux2018cognitive; evanson2023language; lavechin2022reverse; apidianaki2024language; hu2025language]. In particular, they offer a controlled way to investigate the long-standing hypothesis that children bootstrap into language learning by accumulating statistics from linguistic input [saffran1996statistical].

A central challenge when applying LLMs as models of the early language learner is establishing valid *quantitative* comparisons between the learning trajectories of real children and those of LLMs fed with comparable data. For instance, it has been claimed that LLMs require significantly more input data than children to reach the same level of performance [frank2023bridging]. But measuring this ‘data gap’ depends critically on developing sound metrics that can be applied consistently to both human learners and computational models.

We focus on measures of vocabulary, as there is a wealth of developmental data that provides human side of the comparison. Previous work has primarily explored two approaches for evaluating vocabulary in models: (i) **familiarity metrics**, which exploit LLMs’ probabilistic outputs (e.g., surprisal or word–pseudoword preference) to probe lexical knowledge [chang2022word; ma2025babysit; lavechin2025simulating]; and (ii) **cross-modal matching**, which assesses word–referent grounding using multimodal training and behavioral benchmarks [vong2024grounded; tan2024devbench; nikolaus2021modeling], analogous to looking-while-listening paradigms in children [swingley2011looking]. While informative, familiarity metrics often rely on limited behavioral data, and cross-modal methods require multimodal input unavailable to unimodal language models. Also, neither of them has focused on expressive vocabulary.

To overcome these limitations and enable large-scale, quantitative comparison with child vocabulary, we introduce CORPUS-CDI, a benchmark that aligns model generations with children’s expressive vocabulary. CORPUS-CDI calibrates three key dimensions: (1) training exposure, scaled from 6 to 36 months based on daylong recordings [cristia2019child]; (2) cumulative

generated word count, matched to children’s monthly word production rates; and (3) evaluation vocabulary, matched in part-of-speech categories and frequency distribution to the CDI word list [frank2017wordbank].

Using CORPUS-CDI, we evaluate character-level LSTMs and Transformers trained on child-directed speech (CDS) and audiobook transcripts. We assess vocabulary acquisition across three complementary dimensions: lexical diversity (type-token ratio), word-form validity (non-word rate), and target-word coverage (CORPUS-CDI score). The evaluated models do not jointly reproduce the child reference patterns. Coverage increases with input size but does not match the accelerated middle portion of the child corpus trajectory. Lexical diversity and non-word rates vary with architecture and corpus: audiobook-trained models have especially high TTR, whereas some CDS-trained conditions approach children’s TTR or show substantial reductions in non-word rate. Frequency analyses identify a further discrepancy in the availability of lower-frequency target words under the production criterion.

3.3 Related Work

3.3.1 Developmental Trajectories

Prior work has examined the alignment between language model learning and child language development across multiple linguistic levels. A key challenge is mapping model training onto corresponding developmental milestones. Early studies used training checkpoints as proxies for age, tracking when models acquire words [chang2022word] or syntactic structures [evanson2023language]. However, these approaches repeatedly expose models to identical input batches, unlike children who encounter increasingly varied linguistic input. To address this, lavecchin2023babyslm varied input size to model developmental trajectories, improving zero-shot lexical accuracy with increased exposure. Beyond vocabulary, studies have investigated phonological development via word segmentation [goriely2024babble], morphological acquisition through overregularization patterns [haga2024modeling], and syntactic learning [evanson2023language]. Most prior work, however, calibrates only input quantity, leaving output production and evaluation frequency uncontrolled. In contrast, our framework aligns training exposure, generation quantity, and evaluation timing to enable direct comparison between models’ and children’s expressive vocabulary.

3.3.2 Vocabulary Evaluation

Evaluating vocabulary acquisition in models has followed two main strategies: developing developmentally inspired metrics or applying existing metrics to child-relevant tasks [kosoy2023comparing; nikolaus2021evaluating; jiang2023mewl]. Notably, chang2022word first introduced the MacArthur-Bates Communicative Development Inventory (CDI) to evaluate when models acquire individual words during training. Their findings highlight that models rely disproportionately on input frequency, with more variance in acquisition timing than child age-of-acquisition norms. While children also benefit from Zipfian distributions, the challenge for models is pri-

marily learning words in the *long tail* of the distribution [wolters2024zipfian; lavi2023zipfian; lavi2022learnability]. This partially explains the gap between model and human acquisition patterns, particularly for low-frequency words.

Cross-linguistic analyses show that contextual surprisal predicts age of acquisition for function words and predicates, but not nouns [portelance2023predicting], suggesting distributional learning mechanisms affect lexical categories differently. To address this, our evaluation sets are frequency-matched *and* controlled for part-of-speech categories to ensure that differences are not confounded by lexical class or frequency effects.

3.3.3 Calibrated Evaluation Frameworks

Direct comparisons between models and children require alignment across training data, learning objectives, and evaluation tasks [warstadt2022what]. The BabyLM Challenge [warstadt2023findings; hu2024findings; charpentier2025findings] established shared tasks by restricting model training to 10M or 100M words to approximate child exposure at ages 2 and 13. Despite these constraints, findings remain mixed: some studies report benefits from child-directed speech [feng2024child], whereas others observe that Wikipedia training outperforms CHILDES-based input [padovani2025child].

Input representation is another consideration: most models use BPE tokens [huebner2021babyberta; evanson2023language], whereas infants perceive continuous acoustic speech [dupoux2018cognitive]. We use character-level input, requiring models to generate word forms from character sequences rather than a pretrained subword vocabulary [bunzeck2024graphemes; goriely2024babble]. Spaces and punctuation remain available as orthographic boundary cues. This simplifies aspects of the input relative to speech, but does not establish a quantitative upper bound for speech-based learning.

Finally, production metrics such as lexical diversity often use the type-token ratio (TTR), which is sample-size dependent [richards1987type] and requires fixed-size windows for comparability [mckee2022measurement]. Sampling temperature introduces tradeoffs between diversity and quality, with no single setting optimizing both [verine2025improving].

Our framework addresses these challenges through three calibrations: (1) age-appropriate training exposure derived from daylong recordings [cristia2019child], (2) production quantity matched to children’s monthly output, and (3) frequency- and POS-matched evaluation sets reflecting child-directed speech patterns. Together, these enable a developmentally grounded evaluation of expressive vocabulary.

3.4 Methods

3.4.1 corpus-CDI Benchmark Overview

The CORPUS-CDI benchmark enables direct comparison between language model generation and child production through three methodological calibrations to address common mismatches in prior work (Figure 3.1):

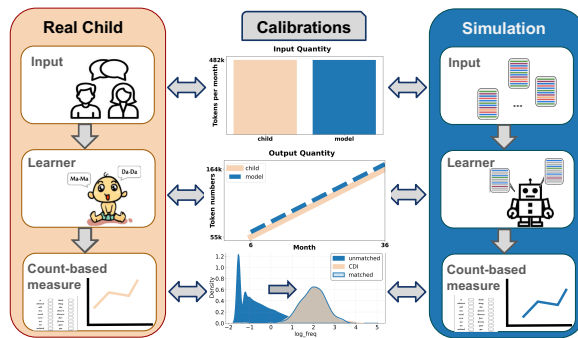


Figure 3.1 – **Overview of the corpus-CDI benchmark framework.** (a) Training and generation sizes are aligned with age-appropriate estimates of linguistic exposure and production from 6–36 months [cristia2019child]. (b) Evaluation sets match the frequency distribution of MacArthur–Bates CDI words in child-directed speech (English-CHILDES).

- (1) **Age-appropriate training exposure:** Training sets are scaled to match cumulative linguistic input from 6–36 months of age estimated from daylong recordings [cristia2019child].
- (2) **Production quantity matching:** Generation sizes are calibrated to children’s monthly word production rates estimated from longitudinal corpora [cristia2019child] to ensure comparable sample sizes for vocabulary assessment.
- (3) **Frequency-matched evaluation words:** The evaluation vocabulary is drawn from the MacArthur–Bates Communicative Development Inventory [fenson2007macarthur] and frequency matched to words in English CHILDES corpora.

We evaluate models on three dimensions: *lexical diversity* (type–token ratio), *word form accuracy* (non-word rate), and *expressive vocabulary size* (CORPUS-CDI score).

3.4.2 Training Data

We construct two training corpora representing different linguistic environments:

Audiobook corpus: English audiobook transcripts from LibriVox [kearns2014librivox], representing narrative-rich, syntactically complex adult-directed language. **Child-directed speech (CDS) corpus:** English CHILDES transcripts [macwhinney2000childes], representing the naturalistic register of early child input.

Age-appropriate scaling Following cristia2019child, we estimate exposure from 250 hours at 6 months to 1,500 hours at 36 months of age. This corresponds to approximately 500 hours of speech input per year, representing a mid-range estimate of children’s linguistic environments. Converting speech hours to words using a standard rate of 3 words per second yields training sets ranging from 2.89M words (6 months) to 17.35M words (36 months). Developmental datasets are constructed every 6 months by randomly sampling the target quantity from each corpus (see Table 3.1). Given English CHILDES contains a total of 15.6M words of caregiver inputs, the available data is equivalent to those at the 30-month level, which thus serves as the maximum

training input. We emphasize that this scaling controls for *total input quantity* rather than assuming uniform structure of children’s linguistic environments across months.

Preprocessing All text is lowercased, with spaces and basic punctuation (periods, commas, apostrophes) preserved as word boundary markers. No subword tokenization is applied: models learn to generate word forms from character sequences while retaining these explicit orthographic boundary cues.

3.4.3 Model Architectures and Training

We train character-level autoregressive models under two architectures: **LSTM** Three layers with 200-dimensional character embeddings, 1024-dimensional hidden states, and 200-dimensional output projections, following [lavechin2023babyslm](#). **Transformer** GPT-2-style decoder-only architecture with 12 transformer layers, 512-dimensional hidden states, 12 attention heads per layer, following [goriely2024babble](#).

Training procedure Models are trained to minimize cross-entropy loss using the Adam optimizer (learning rate 0.0001, $\beta_1 = 0.9$, $\beta_2 = 0.98$) with inverse square root learning rate decay and 10,000 warmup steps. We train until validation perplexity converges. For each age-corpus pair, two models are trained with same random seeds and non-overlapping data subsets. Training details are provided in Table 3.2.

Generation procedure At inference time, we generate text using temperature sampling with $T \in \{0.3, 0.6, 1.0, 1.5\}$. Generation begins with a beginning-of-sequence (BOS) token and continues until reaching the target word count corresponding to children’s estimated monthly production at that developmental stage (see Table 3.3 in Appendix). Character sequences are segmented into words using space and punctuation boundaries. For months 31–36, outputs are generated from the same model checkpoints. Because the CDS corpus contains a total of 15.6M words of caregiver input, this limits the age-appropriate training to the 30-month equivalent. For months 31–36, we generate outputs using the same 30-month checkpoint, scaling generation lengths according to children’s estimated production, while noting that the training exposure does not increase beyond 30 months. This ensures a consistent comparison across developmental stages while respecting the corpus limitations.

3.4.4 Evaluation Set Construction

Base vocabulary and POS filtering We start with the MacArthur-Bates Communicative Development Inventory (CDI) [[fenson2007macarthur](#)], a standardized checklist of 680 English words with normative age-of-production data. To control for lexical class effects, we apply POS tagging using a RoBERTa-based SpaCy pipeline. For each word, POS labels are aggregated across contexts and majority voting is applied separately in each corpus. Function words (articles, pronouns, prepositions, auxiliaries) are excluded, retaining nouns, verbs, adjectives, and adverbs.

Frequency stratification Filtered CDI words are divided into six frequency bands based on their occurrence in child-directed speech (adult input in English CHILDES), with an equal number of words per band.

Audiobook subset selection For a comparable evaluation, we construct a matched audiobook subset using the same POS filtering and frequency banding. Words are selected such that their frequency distribution mirrors the filtered CDI evaluation set. This process yields two aligned datasets: i.) Filtered CDI evaluation set with 420 words in six frequency bands; and ii.) frequency-matched audiobook subset with the same size per frequency band. This enables a developmentally grounded comparison between model generations and children’s expressive vocabulary.

3.4.5 Evaluation Metrics

We assess model performance using three complementary metrics capturing lexical diversity, form accuracy, and expressive vocabulary.

Type-Token Ratio (TTR) Measures lexical diversity as the ratio of unique word types to total word tokens. Following [rasanen2024age](#), we compute TTR over fixed-size chunks of 3,500 words to control for sample size effects, then average across all chunks.

Non-word Rate Quantifies word-form validity as the proportion of generated word types not found in the reference lexical resources: CELEX, the Enchant library,¹ and Wiktionary.² A word is classified as valid if it appears in any of these resources. This is an orthographic dictionary-membership measure, not a direct measure of speech accuracy or contextual appropriateness.

corpus-CDI Score Adapts the human CDI vocabulary assessment to observed productions through a count-based criterion. A word meets the production criterion if it appears at least τ times in the evaluated output. To determine τ , we compare vocabulary growth curves derived from CHILDES child productions with the empirical CDI trajectory. We use $\tau = 60$, selected on the basis of the reported slope comparison. This is a calibration rule linking corpus counts to caregiver reports, rather than an independent test of lexical knowledge. The CORPUS-CDI score is computed as:

$$\text{CDI} = \frac{|\{w \in \text{eval set} : \text{count}(w) \geq \tau\}|}{|\text{eval set}|}$$

where the evaluation set contains 420 target words. The value represents the proportion of that set reaching the count criterion under the specified production budget. Sensitivity analyses for alternative thresholds ($\tau \in \{1, 500\}$) are shown in Figure 3.6.

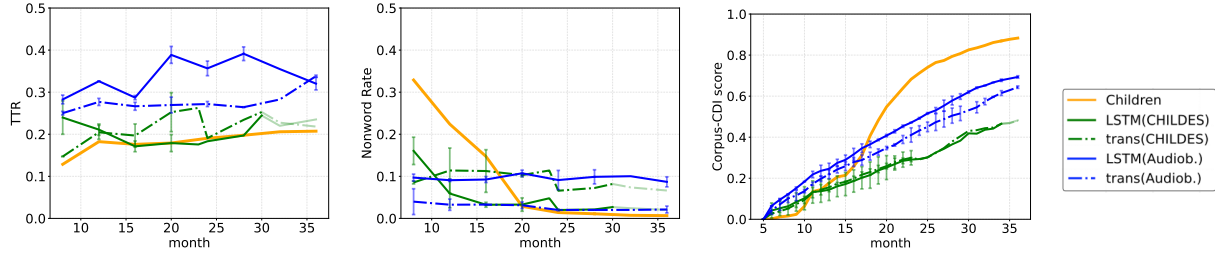


Figure 3.2 – **Core metrics comparing child and model vocabulary development across age groups.** Left: type-token ratio (TTR). Middle: non-word type rate under dictionary matching. Right: CORPUS-CDI coverage, with words counted when produced at least 60 times. Results are shown for LSTM and Transformer models trained on audiobooks and child-directed speech (CDS), compared with CHILDES child production data. Error bars are reported as 95% confidence intervals across training subsets. The lighter CDS points after the 30-month-equivalent checkpoint use increasing output budgets without additional training input.

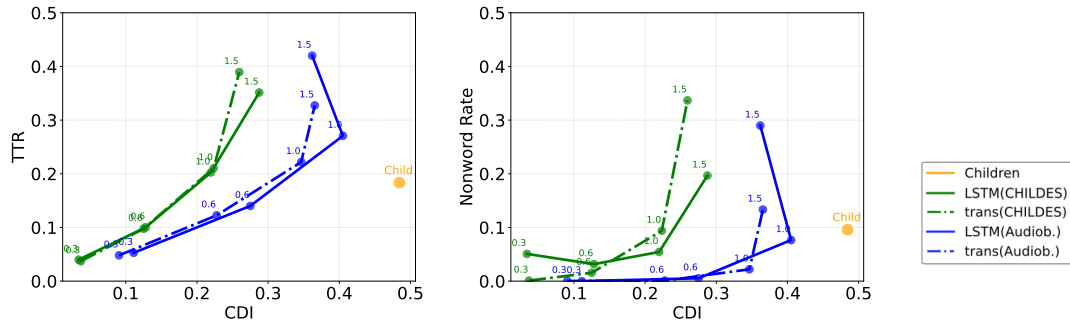


Figure 3.3 – **Temperature trade-off across different metrics; temperature settings (0.3-1.5)** Left: Trade-off between type-token ratio and CDI score. Right: Relationship between non-word rate and CDI score. Points show mean scores across models, with numbers illustrating temperature values.

3.5 Results

3.5.1 Developmental Trajectories

Figure 3.2 presents three production measures against developmental labels derived from estimated input and output quantities, alongside the CHILDES child reference. All results use temperature $T = 1.0$ as the default setting. The later CDS points retain the same training checkpoint and vary the production budget, as described in the generation procedure.

Lexical diversity Audiobook-trained models exhibit higher type-token ratios than the child reference across the plotted stages. CDS-trained models are closer to the child values at several stages, with differences depending on architecture and input scale. The child TTR rises overall, whereas model curves show condition-specific fluctuations rather than a common developmental trend. Thus lexical diversity distinguishes input registers and model conditions, but does not imply that every model consistently exceeds the child reference.

1. <https://pypi.org/project/pyenchant/>
2. <https://www.wiktionary.org>

Word form accuracy The proportion of child word types absent from the reference dictionaries decreases markedly with age. Model trajectories vary: several are relatively flat, whereas the CDS-trained LSTM shows a substantial decline. No single model curve reproduces the full child pattern across the plotted range. Because this measure is computed from written forms, it cannot identify the contribution of children’s articulatory or phonological development separately from transcription and dictionary coverage.

Vocabulary growth The child CORPUS-CDI trajectory contains a pronounced period of accelerated growth, followed by a slower rise. Model coverage also increases, but follows a more nearly linear pattern over the observed range. Audiobook-trained models achieve greater coverage than CDS-trained models in the figure, although their evaluation sets are frequency-matched rather than lexically identical. The main comparison concerns the differing trajectory shapes; a universal ranking of the architectures is not supported by these curves.

Taken together, the measures show that the tested models do not jointly reproduce the child production profile. The vocabulary-coverage discrepancy appears across architectures and corpora, while the magnitude and direction of TTR and non-word differences depend on condition. The next analyses test whether the sampled decoding temperatures improve this joint alignment (Section 3.5.2) and how the coverage discrepancy varies with lexical frequency (Section 3.5.3).

3.5.2 Temperature Tradeoff

The developmental gaps observed in Section 3.5.1 raise an important question: could these differences arise from suboptimal generation parameters rather than inherent learning limitations? Temperature sampling controls generation stochasticity, with lower temperatures favoring high-probability outputs and higher temperatures increasing diversity. We evaluate whether any temperature setting can align models with children across all three metrics.

Figure 3.3 shows the relationships between metrics as temperature varies from 0.3 to 1.5. Each point represents the mean performance across all developmental stages for a given architecture, corpus, and temperature combination.

TTR-CDI trade-off Temperature changes the relation between lexical diversity and target-word coverage. Lower temperatures concentrate output on fewer forms, whereas higher temperatures increase diversity. Coverage does not increase monotonically in every condition: in the audiobook models it declines between $T = 1.0$ and $T = 1.5$, even as TTR increases. None of the tested settings jointly matches the child reference on TTR and CORPUS-CDI coverage. Thus this temperature manipulation does not resolve the observed discrepancy between lexical variety and repeated production of target words.

Non-word-CDI trade-off High temperatures generally increase non-word rates, with the pattern depending on architecture and corpus. Some conditions approach the child’s mean non-word rate without matching its vocabulary coverage. Moreover, similar means can summarize

different developmental trajectories, as shown in Figure 3.2. Alignment on this single aggregate measure therefore does not establish alignment of the overall production profile.

Within the tested temperature range, decoding changes do not remove the joint discrepancy between child and model production measures. This rules out a particular adjustment as a sufficient explanation of the result, while leaving open other decoding, elicitation, and learning procedures. The next section examines the relation between the coverage discrepancy and input frequency.

3.5.3 Frequency effect

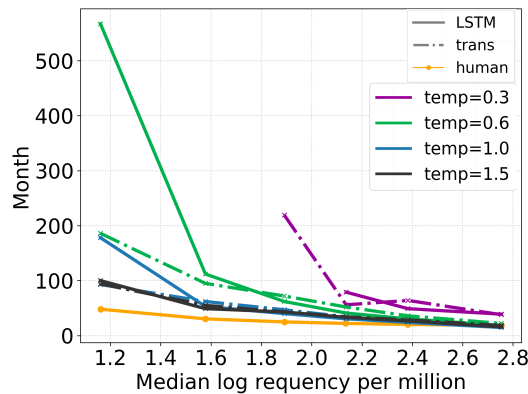


Figure 3.4 – **Fitted criterion-crossing months by frequency band for audiobook-trained models and the child reference.** The horizontal axis gives median log frequency per million; the vertical axis gives the month at which the fitted trajectory reaches 80% coverage within the band. Values beyond the observed age or exposure range are extrapolations, not observed training requirements. Missing model points indicate that no criterion-crossing estimate is reported for that condition.

We next examine whether the vocabulary-coverage discrepancy is uniform across frequency. The evaluation sets are divided into six bands, and sigmoid curves are fitted to the proportion of words meeting the count criterion in each band. A criterion-crossing month is extracted where a fitted curve reaches 80% coverage. This summarizes the fitted relation between frequency and production, conditional on the exposure and output scaling. Estimates outside the observed range are extrapolations, including any child-reference estimates beyond the ages represented in the data.

Divergent frequency sensitivity Figure 3.4 shows a steeper relation between frequency and fitted criterion-crossing time for the models than for the child reference, especially in the lowest-frequency band. Model estimates also vary substantially with temperature. At high frequencies, several model estimates approach the child estimate. The result identifies frequency-dependent difficulty in meeting this production criterion; it does not directly measure how much input is necessary for a word to be represented or understood.

Temperature modulation Temperature modulates the frequency pattern. At $T = 0.3$, sampling is concentrated on high-probability continuations, and some low-frequency bands have no reported criterion-crossing estimate. Higher temperatures make additional forms available in generation, although lower-frequency model estimates remain especially sensitive to condition. Because the count criterion depends on generated occurrences, this analysis reflects both what the trained model assigns probability to and how that probability is sampled.

Long-tail truncation The frequency pattern is consistent with underproduction of forms in the lexical long tail. In empirical cross-entropy training, frequent events contribute repeatedly to the aggregate training signal. During finite generation, low-probability forms also have fewer opportunities to accumulate enough occurrences to pass a fixed count threshold. Together, these factors provide a plausible account of the observed production pattern. The present design does not isolate the objective from representation, corpus composition, optimization, or decoding, and does not establish that representational capacity is allocated in direct proportion to word frequency.

Evidence from child word learning motivates possible additional sources of support, including memory for particular learning events [markson1997evidence], cross-situational information [smith2008infants], and social-pragmatic cues [tomasello2000social]. These possibilities suggest interventions for subsequent modeling work. They are not manipulated here, so the current results cannot determine which resource explains the child-model production discrepancy or exclude a contribution from predictive learning in children.

3.5.4 Production of Content Words

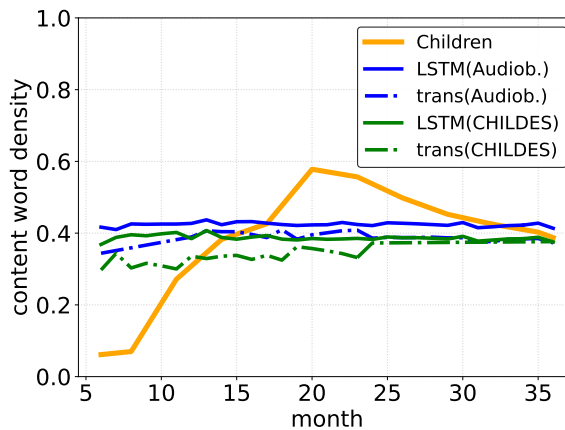


Figure 3.5 – **Proportion of content words in generated text across development.** Children show an inverse U-shaped trajectory, while all model configurations maintain stable proportions around 40% across training stages.

The frequency-dependent pattern in Section 3.5.3 motivates examining the allocation of output to content and function words. The CORPUS-CDI evaluation targets content words (Section 3.4.4), whereas TTR is calculated over the sampled vocabulary more broadly. A model

can therefore have a high overall TTR without repeatedly generating enough target content words to attain high CORPUS-CDI coverage.

Children’s inverse U-shaped trajectory Figure 3.5 shows a pronounced rise and subsequent fall in the proportion of content words in child productions. This is consistent with a changing lexical composition as vocabulary expands and grammatical function words become more prevalent [bates1994developmental; brown1973first]. Coverage of the target vocabulary continues to increase while its share of all produced tokens declines, illustrating that vocabulary coverage and token allocation measure different properties.

Models’ narrower range Model content-word proportions vary within a narrower range, generally around 40%, although some conditions change with training scale. The models do not show the pronounced peak present in the child curve. This establishes a distributional difference without requiring the assumption that model output is identical to an adult corpus distribution.

The joint analysis shows that target-word coverage, lexical diversity, and lexical-category proportions need not move together. The content-word pattern is relevant to interpreting coverage on a content-word evaluation set, but it does not by itself explain the cause of elevated TTR or establish that function-word production displaces particular target words.

These observations extend the trajectory comparison from the number of target words produced to the distribution of output across lexical categories. Testing why those distributions differ would require controlled changes to the learning or production procedure.

3.6 Chapter discussion

3.6.1 What the comparison establishes

This chapter makes expressive production an explicit target for developmental comparison. The main empirical finding is a discrepancy in the joint pattern of vocabulary coverage, lexical diversity, and word-form validity. With larger training corpora, models produce more of the evaluation vocabulary often enough to cross the CORPUS-CDI threshold. They nevertheless fail to reproduce the accelerated middle portion of the child corpus trajectory. This mismatch is visible across the tested architectures and input registers, although individual measures vary across conditions: audiobook-trained models have especially high lexical diversity, whereas some child-directed-speech conditions approach children’s TTR or show substantial declines in non-word production (Figure 3.2). The result is therefore a constraint on the overall production profile, rather than evidence that every model is equally distant from children on every measure.

The temperature experiment strengthens this conclusion by testing one concrete alternative explanation. Among the sampled temperatures, moving toward more diverse output does not yield a configuration that jointly matches the child reference on vocabulary coverage and TTR (Figure 3.3). It can also increase the rate of forms absent from the reference dictionaries. This is a useful robustness result: the discrepancy is not removed by choosing another value within the

tested temperature range. It leaves open whether other elicitation contexts, decoding policies, or training procedures would produce a different relation among the measures.

3.6.2 Measurement, benchmark, and model contributions

The measurement contribution is to specify the observation process through which a vocabulary estimate is obtained. Fixed-size windows make TTR less dependent on the amount of sampled text; production budgets define opportunities to observe words; and the count threshold defines how much observed evidence is required for an item to contribute to the vocabulary score. The benchmark contribution is to combine these decisions with explicit input scales and frequency-matched evaluation sets, allowing a common protocol to be applied to two architectures and two registers. The model comparison then provides empirical constraints on those particular learners under that protocol. These contributions remain useful even when a numerical score cannot be read as an inventory of everything a learner knows.

The distinction matters especially for CORPUS-CDI. Its threshold was chosen by comparing child corpus counts with a caregiver-report trajectory. This calibration supplies a linking rule between measurements; agreement on the calibration material is not independent validation of lexical knowledge. A target form may remain below threshold because it is weakly represented, because relevant contexts are rarely sampled, or because more probable alternatives occupy the generated output. Conversely, frequent production alone does not establish appropriate reference or contextual use. A stronger validation would apply the fixed calibration to held-out children or corpora and test whether the resulting estimates agree with additional vocabulary measures. The present score is most directly interpretable as the proportion of target forms reaching a production criterion under the specified sampling conditions.

3.6.3 Frequency and the meaning of data efficiency

Frequency analysis identifies where the production discrepancy is largest. The fitted trajectories are more frequency-sensitive for the evaluated models than for the child reference, particularly in the lower-frequency bands (Figure 3.4). The associated content-word analysis adds an informative distributional difference: the child corpus shows a pronounced rise and subsequent fall in content-word proportion, whereas the model proportions vary within a much narrower range (Figure 3.5). Thus vocabulary coverage and the distribution of output across lexical categories need to be considered together. A model can generate many distinct forms while repeatedly producing too few of the particular content words scored by the benchmark.

This pattern is compatible with an interaction between frequency-weighted predictive training and finite sampling. Under an empirical cross-entropy objective, frequent events contribute repeatedly to the training signal. During generation, low-probability forms also have fewer opportunities to accumulate enough occurrences to pass a fixed count threshold. These considerations motivate the hypothesis that the tested learning and production procedures provide insufficient support for parts of the lexical long tail. The experiment does not isolate the objective as the unique cause: it does not compare cross-entropy with another objective while holding the

remaining conditions fixed. Likewise, the content-word proportions do not by themselves identify a causal explanation for TTR. The results locate a frequency-related discrepancy and define hypotheses that can be tested through interventions on memory, objectives, input, or production.

The time axis further limits a numerical claim about data efficiency. Model conditions vary in available corpus size and are trained to convergence, so the amount of distinct text is not the total number of word presentations during optimization. The age-equivalent scale also combines estimated input with increasing output budgets. In the CHILDES conditions, the available training corpus limits further input growth after the 30-month-equivalent checkpoint, while later generation budgets continue to change. Those later observations cannot therefore be attributed wholly to additional learning. Finally, frequency-band estimates that cross the acquisition criterion far beyond the observed range are curve extrapolations, not results from models actually trained on hundreds of months of input. Their magnitude depends on the fitted function and threshold. Within these boundaries, the study shows that increasing corpus size under the tested protocol does not automatically recover the child production profile; it does not determine a universal child–model data multiplier.

3.6.4 From a diagnostic discrepancy to a controlled intervention

The strongest next step is to preserve the production measure while manipulating a candidate source of the discrepancy. An episodic store could test whether access to particular encounters improves low-frequency target coverage; alternative training objectives could test whether changing the weighting of rare events changes that coverage; and grounded or interactive input could test whether referential and conversational context changes which words are produced. Such comparisons should count any additional information and evaluate effects across multiple outcomes, since an improvement in thresholded coverage could coexist with deteriorating lexical diversity or word-form validity.

Chapter 4 takes up one related intervention by asking whether retrieval of stored linguistic instances improves sensitivity to low-frequency syntactic contrasts. It tests the potential contribution of instance-based access using a different outcome and experimental setting. It consequently extends the inquiry into sparse evidence without directly repairing or explaining the expressive-vocabulary curves reported here. Across the two studies, the common question is what information remains available to a predictive learner when relevant evidence is infrequent, and how that availability depends on the task used to measure it.

3.7 Limitations

Our evaluation framework has several limitations. First, we evaluate character-level models rather than speech-based systems. Character input supplies stable orthographic forms and explicit boundaries, bypassing acoustic variability and aspects of segmentation [dupoux2018cognitive]. Speech-based language models [lakhoria2021generative; nguyen2024spirit] face a different representation problem. The direction and magnitude of a child–model gap under speech input cannot be established from the present comparison. Extending CORPUS-CDI to speech would

require a validated mapping from acoustic output to the forms used for vocabulary scoring. The models also lack the multimodal and interactive information available in children’s environments, whose contribution is not isolated here.

Second, the CHILDES transcripts used as reference data introduce possible biases. Adult transcription and normalization can change which child forms are recorded and whether they match reference dictionaries. A dictionary-based non-word score therefore combines properties of the child’s production, transcription choices, and lexical-resource coverage. Character-level model output can be assessed directly as written forms, but the model can still generate non-words; this does not supply an equivalent measure of children’s spoken production accuracy.

Third, caregiver CDI reports and counts in generated text are related but distinct measurements. A report that a child says a word summarizes observations across contexts; CORPUS-CDI applies a frequency criterion to a sampled output. Neither one occurrence nor repeated occurrences alone establishes the full meaning or appropriate use of a word. The threshold ($\tau = 60$) calibrates the corpus measure against the reference trajectory, but its construct validity requires further testing on independent materials and outcomes.

Finally, our study focuses on English and a single vocabulary assessment tool (CDI). Cross-linguistic validation is essential for establishing universality: languages with richer morphology, less transparent orthography, or different child-directed registers may yield different alignments between model and child learning. Character-level models in particular may be sensitive to orthographic properties, limiting generalizability beyond English.

3.8 Ethics Statement

The study uses existing CHILDES child-language data, LibriVox audiobook transcripts, and CDI vocabulary reference materials. Its purpose is model benchmarking and research on language acquisition. These evaluations do not establish suitability for real-world deployment. Any reuse of the source materials should follow the applicable access and data-use conditions.

3.9 Appendix

3.9.1 Training and generation details

Table 3.1 summarizes the age-appropriate scaling of training data across three exposure scenarios. Linguistic exposure estimates are based on daylong recording studies [cristia2019child], which document substantial variability in the amount of speech heard by children across environments. Annual exposure ranges from approximately 100 hours in low-input households to 1000 hours in high-input ones, with 500 hours per year representing a mid-range estimate.

Cumulative exposure thus increases from 50–500 hours at 6 months to 300–3000 hours by 36 months. Assuming a standard speech rate of three words per second, this corresponds to training sets ranging from 0.58M words (6 months, low exposure) to 34.71M words (36 months, high exposure).

Age	Hours	Words	Characters	Utterances
<i>Low exposure scenario (100 hours/year)</i>				
6	50	0.58	2.95	58
12	100	1.16	5.89	116
18	150	1.73	8.84	173
24	200	2.31	11.78	231
30	250	2.89	14.73	289
36	300	3.47	17.67	347
<i>Medium exposure scenario (500 hours/year)</i>				
6	250	2.89	14.73	289
12	500	5.78	29.45	578
18	750	8.68	44.18	868
24	1000	11.57	58.90	1157
30	1250	14.46	73.63	1446
36	1500	17.35	88.35	1735
<i>High exposure scenario (1000 hours/year)</i>				
6	500	5.78	29.45	578
12	1000	11.57	58.90	1157
18	1500	17.35	88.35	1735
24	2000	23.14	117.80	2314
30	2500	28.92	147.25	2892
36	3000	34.71	176.70	3471

Table 3.1 – **Age-appropriate scaling of training datasets across three exposure scenarios.** Estimates are derived from three exposure scenarios (100h, 500h, 1000h per year), reflecting observed variability in children’s linguistic environments [cristia2019child]. Word counts assume 3 words/second; character counts assume 5.1 characters/word; utterance counts assume 10 words/utterance.

Model training employed carefully tuned hyperparameters (Table 3.2), selected through systematic validation experiments to ensure training stability and computational efficiency across different data scales.

Table 3.4 provides illustrative child productions and model generations at a sampling tem-

Hyperparameter	Value
Max sequence length	1024
Batch size	64
Learning rate	0.0001
Learning rate scheduler	inverse sqrt
Warmup steps	10000
Optimizer	Adam
Adam-beta1	0.9
Adam-beta2	0.98
Dropout	0.1
LSTM hyperparameter	Value
decoder layers	3
hidden size	1024
embedding dimension	200
Transformer hyperparameter	Value
Transformer layers	12
Intermediate hidden size	512
Attention heads	12
Attention dropout	0.05

Table 3.2 – **Language model training hyperparameters.** Configurations balance model capacity, stability, and efficiency across training scales.

perature of $T = 1.0$. The examples show variation in output length and word forms, including sequences containing non-words. They are not a controlled assessment of grounding, pragmatic coherence, or syntactic dependency length and should be read alongside the quantitative analyses.

3.9.2 Result details

To establish reliable vocabulary acquisition metrics, we conducted systematic analyses of different word count thresholds (Figure 3.6). This calibration process aimed to align our CORPUS-CDI scores with empirical observations of child vocabulary development.

Notably, the frequency effects observed in the Machine CDI are evaluated on a selected subset matched with the human-CDI set. It is unclear whether this effect is influenced by the randomness of sampling. Also, the selected words are typically in the median frequency bands, making it unknown whether such an effect stems from the limited generation size. To address this concern, we expanded our analysis of the model generations using the same and larger sampling sets. Figure 3.7 shows the word frequency distributions of the training and generated datasets by the LSTM model trained on 3.4M words, with different generation sizes. Expanding the generation set mitigates some truncation effects, as more high-frequency words are generated, but low-frequency words remain underrepresented. This pattern reflects the model’s limited ability to reproduce realistic word distributions, especially for rare words, even with larger sampling sizes.

The exposure-scaling analysis in Figure 3.8 compares CORPUS-CDI trajectories under three estimated annual input budgets. Larger input budgets improve coverage in some conditions but do not reproduce the child reference trajectory within the tested protocol. This is evidence

month	#sec/hour	estimation	production
6	61.25	55,125	782
7	65.30	58,769	1,323
8	69.35	62,412	1,278
9	73.40	66,056	3,964
10	77.44	69,699	627
11	81.49	73,343	1,148
12	85.54	76,986	1,361
13	89.59	80,630	568
14	93.64	84,273	2,720
15	97.69	87,917	13,765
16	101.73	91,560	3,620
17	105.78	95,204	7,589
18	109.83	98,847	15,197
19	113.88	102,491	17,090
20	117.93	106,134	19,912
21	121.98	109,778	33,458
22	126.02	113,421	33,880
23	130.07	117,065	63,738
24	134.12	120,708	191,062
25	138.17	124,352	130,525
26	142.22	127,995	136,195
27	146.27	131,639	148,741
28	150.31	135,282	158,910
29	154.36	138,926	190,263
30	158.41	142,569	195,549
31	162.46	146,213	196,677
32	166.51	149,856	184,024
33	170.56	153,500	185,534
34	174.60	157,143	185,691
35	178.65	160,787	153,515
36	182.70	164,430	299,537

Table 3.3 – **Monthly progression of speech metrics.** Estimated exposure and production rates over developmental time.

about the effect of the tested corpus-size manipulation, not identification of a unique learning mechanism responsible for the discrepancy.

Setting	Generated Sequences
Child	dad dada mommy looked it missed its wheel i want my plate for it
Audiobook (LSTM)	wounded for woman says no i don't know what i fuller over his week scully a
Audiobook (Trans)	to o he had been and the work of the lord ascend mass of softly rounded head
CHILDES (LSTM)	can bump he said I was wondering what yeah detfive wroggy on drh aulf
CHILDES (Trans)	on helsewhere he doesn't like him does he hafta dragged on doh aunty quite

Table 3.4 – **Examples of generated sequences.** Samples are drawn from models trained on different datasets and architectures at a sampling temperature of 1.0. The boundary marker is replaced with blank space for readability.

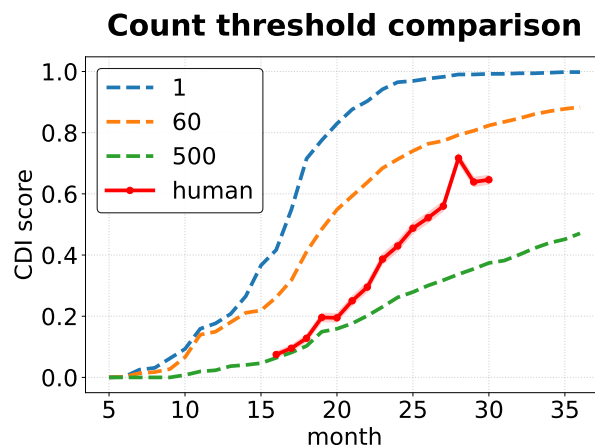


Figure 3.6 – **Count-based calibration.** Child corpus vocabulary trajectories obtained with count thresholds of 1, 60, and 500, alongside the caregiver-report reference (red). The selected threshold of 60 is the calibration used in the main analysis. The curves illustrate the sensitivity of coverage and trajectory to the count criterion; their comparison does not by itself validate equivalence to caregiver reports.

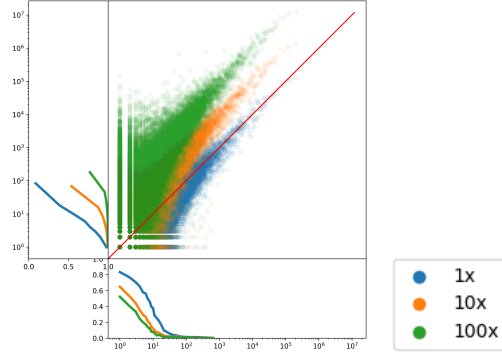


Figure 3.7 – **Word count distribution(temperature = 1.0)**. Word frequency relationships between training and generation sets under different sampling conditions (1x, 10x, 100x generation volume). x-axis: word frequency in the training set; y-axis: word frequency in the generation set. Side plot on the bottom show Missing rates as a function of word count in input corpus.

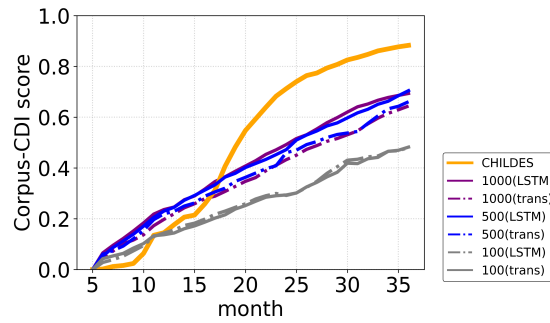


Figure 3.8 – **Model vocabulary growth across estimated annual exposure levels**. Comparison of CORPUS-CDI trajectories for models trained on 100-, 500-, and 1000-hour exposure estimates, alongside CHILDES child production data. Models show roughly linear vocabulary growth that scales modestly with total exposure but remains far below the human developmental curve.

Chapter 4

Does episodic memory help close the lexical frequency gap in sensitivity to syntactic contrasts? A test using retrieval-augmented language models

Objectives

Test whether k -nearest-neighbour language models reduce lexical-frequency gaps on syntactic minimal pairs, and examine how retrieved content, representations, and retrieval configuration affect the gains.

4.1 From sparse lexical experience to retrievable evidence

Learning a word involves more than making its form available for production. A learner must also use it in combinations that respect grammatical dependencies, including when the word has been encountered infrequently. Chapter 3 examined the first problem through expressive vocabulary: under the specified exposure and generation budgets, low-frequency words were less reliably represented in model output. The present chapter turns to a related question about syntactic use. When a predictive model assigns unreliable probabilities to sentences containing infrequent content words, can access to particular previous instances improve its sensitivity to grammatical contrasts?

The link between these questions is the availability of evidence from sparse experience. It is not a direct continuation of the preceding experiment. The vocabulary study used character-level learners and a production-based criterion, whereas this study uses GPT-2 models and probability comparisons between grammatical and ungrammatical sentences. Their corpora and training histories also differ. Retrieval is therefore tested here as a possible resource for frequency-sensitive linguistic performance, without assuming that it resolves the expressive-vocabulary discrepancy established in the previous chapter.

The distinction between learning from experience and retaining access to experience motivates the intervention. Parametric learning changes shared weights across many examples; retrieval allows a particular stored context to contribute directly to a later prediction. Complementary Learning Systems theory provides a cognitive motivation for investigating such a division of labour between gradually acquired regularities and access to specific experiences [mcclelland1995there; kumaran2016learning]. The hypothesis examined here is functional: when the parametric prediction is unreliable, a relevant stored instance may supply useful evidence. The computational implementation does not reproduce hippocampal encoding or establish the neural basis of human grammatical processing. It makes one proposed memory function experimentally manipulable.

The study implements retrieval augmentation through a k -nearest-neighbour language model (k NN-LM; [khandelwal2020knnlm]). A datastore associates representations of previous contexts with their following tokens. At inference time, nearest-neighbour search supplies a token distribution that is interpolated with the base model’s distribution, with retrieval weight $\lambda = 0.25$ in the reported experiments. This form of retrieval-augmented language modelling changes the evidence used to score a sentence. Its immediate effect is measured without retraining the base model, so improved performance can reflect better access to stored evidence without demonstrating that new grammatical knowledge has been consolidated into the model’s parameters.

Two settings test the intervention: GPT-2 Small trained from scratch on a 50.03M-word mixture of child-directed speech, conversational speech, stories, and subtitles, and the publicly pretrained GPT-2 XL model. The smaller setting constrains the base learner’s training corpus, while the larger setting asks whether a similar intervention remains useful under broad pretraining. They are separate tests within different systems, rather than a controlled manipulation of data scale alone. The datastore and any externally trained representations must also be distinguished

from the base learner’s training input when interpreting the amount and source of available experience.

The experiments compare each base model with its retrieval-augmented counterpart on frequency-stratified subject–verb agreement, wh-question, and relative-clause contrasts selected from BLiMP, Zorro, and BIG-bench. The central comparison is whether retrieval improves low-frequency items more than high-frequency items within the same model and corpus. Further analyses vary the representations used for retrieval, the presence of examples containing the target syntactic phenomenon, and retrieval granularity, context size, and neighbour count. These comparisons ask both whether access helps and what makes a stored instance useful.

The reported results show selective but incomplete compensation. Retrieval narrows the high–low accuracy gap in all three phenomenon groups and both model settings, with particularly large absolute gains for relative clauses. The gain depends on representation and retrieval configuration; simply making more neighbours available does not produce uniformly better performance. The article below presents these results and their technical analyses. The chapter-level discussion then considers their contribution to the thesis: a controlled demonstration that access to stored instances can improve specific linguistic decisions, together with a more precise account of the evidence needed to connect that effect to language learning and memory.

4.2 Study rationale

Language models are known to exhibit systematic lexical frequency biases: their performance on linguistic evaluations correlates with how often the lexical items in the test examples appear in training data, leading to low-frequency items posing a persistent challenge [wei-etal-2021-frequency; kim2022uncontrolled; diehl-martinez-etal-2024-mitigating]. While frequency does influence human performance in empirical tests of grammatical knowledge such as acceptability judgments [kempen2005relationship; bader2010toward; lau_grammaticality_2020], recent work [christensen2024complexity] has shown evidence for robustness to lexical frequency in delineating the critical contrasts. What accounts for this gap between human robustness and model fragility?

One possible answer comes from Complementary Learning Systems (CLS) theory [mcclelland1995there; kumaran2016learning], which proposes that the brain employs two distinct memory systems: a hippocampal system that rapidly encodes and retrieves specific experiences, and a neocortical system that gradually extracts generalizable statistical patterns. When parametric knowledge is weak, as it is for rare items, episodic memory can compensate by directly retrieving relevant past encounters [davis2009complementary]. This division of labor has also been proposed to be relevant to linguistic processing [borensztajn2011episodic; duff2012hippocampus], motivating the functional hypothesis that access to specific stored instances may compensate for unreliable predictions involving low-frequency words.

We test this functional hypothesis in language models by asking whether an explicit retrieval mechanism narrows the lexical frequency gap in syntactic contrast evaluations. We use k -nearest-neighbour language models (k NN-LM; [khandelwal2020knnlm]) to combine

parametric prediction with access to stored context–target pairs. This is a computational intervention on access to instances, motivated by episodic-memory accounts; it is not a model of hippocampal encoding or a direct test of the neural mechanism supporting human performance. Prior work motivates examining both the benefits and limits of instance access: retrieval can support rare knowledge [**mallen2023trust**; **kandpal2023large**], while the generalization biases associated with information in context and in weights depend on the task and training regime [**lampinen2025latent**; **chan2022transformers**]. Recent evidence that k NN-LMs do not consistently improve low-frequency token prediction in standard language modeling [**nishida2025long**] leaves the effect on syntactic contrasts empirically open.

We investigate this question through syntactic contrast tests across three different phenomena (subject-verb agreement, wh-questions, relative clauses) using frequency-stratified test items, under both child-realistic and large-scale pretraining. We further analyze what drives retrieval effectiveness by contrasting full versus semantics-only retrieval, by examining the role of structural similarity in retrieved instances, and by varying the configurations of retrieval (retrieval granularity, context window size, and number of retrieved neighbors). Our results show that retrieval augmentation consistently narrows the lexical frequency gap across all phenomena and both model scales, with particularly large absolute gains for relative-clause items. Nevertheless, the gap remains not fully closed. To this end, we analyze the limitations of the current modeling approach and posit preferential reweighting, improved representations and retrieval for structural information, and flexible configurations of storage and retrieval as promising future directions.

4.3 Related Work

Frequency effects on linguistic evaluation in language models. Language models exhibit systematic frequency biases, where performance on linguistic evaluation often degrades when the test items contain low-frequency lexical items [**wei-etal-2021-frequency**; **kim2022uncontrolled**]. In the context of syntactic contrasts, models perform worse on minimal pairs that contain low-frequency words, even when the underlying grammatical contrast is identical to high-frequency cases. **diehl-martinez-etal-2024-mitigating** directly quantify this frequency bias in BLiMP (one of our syntactic test datasets) and propose a training-time intervention (Syntactic Smoothing) to reduce it, finding that substantial gaps persist even after mitigation.

wettig-etal-2023-mask examine how the masking rate during training determines which co-occurring token pairs (measured via pointwise mutual information) are exposed to the model, shaping both convergence speed and final performance. Since pointwise mutual information is itself derived from corpus-level token frequencies, these findings are consistent with frequency distributions in pretraining data influencing model behavior, though this connection is inferential. More directly, **kandpal2023large** show that language models struggle to acquire long-tail knowledge whose supporting documents are rare in the pretraining corpus, establishing a clear link between corpus frequency and what is learned.

Together, these findings establish that (lexical) frequency creates differential strength in

parametric representations, with low-frequency items learned far less robustly. Our work asks a related question from a different methodological angle: rather than reducing frequency bias through better training, can it be compensated for at inference time through retrieval?

Complementary Learning Systems theory. Complementary Learning Systems (CLS) theory [mcclelland1995there; kumaran2016learning] proposes that the human brain employs two specialized memory systems: a hippocampal system that rapidly encodes specific episodes, and a neocortical system that gradually extracts statistical regularities across experiences. This allows new information to be acquired quickly without disrupting slower cortical learning, and computational implementations of CLS show that it enables both rapid memorization and gradual generalization [mcclelland1995there]. Critically for our purposes, eichenbaum2001hippocampus argues that the hippocampus does not merely store episodes but actively supports flexible use of past experiences through retrieval, enabling inferences that parametric learning alone may fail to achieve. Applied to language models, this predicts that when parametric representations are weak (as for low-frequency items), episodic retrieval should provide compensatory benefits by making specific relevant experiences directly accessible. lampinen2025latent in particular demonstrates that language models fail to generalize certain types of knowledge learned parametrically (e.g., relation reversals) but succeed when the same information is available in context, suggesting that episodic retrieval enables more flexible use of past experiences than parametric consolidation. Similarly, studies of in-context learning show that generalization from information presented in context can differ from generalization from information stored in weights [chan2022transformers]. Our work extends these insights to syntactic processing by testing whether retrieval enables more robust generalization across frequency bands than parametric-only inferences.

Retrieval-augmented language models. Retrieval-augmented methods enhance language models by supplementing parametric knowledge with information drawn from an explicit memory at inference time. k NN-LMs [khandelwal2020knnlm], which we adopt in this work, interpolate the base model’s next-token distribution with a distribution computed from nearest-neighbor contexts retrieved from a datastore of training instances. While prior work shows benefits for rare patterns and knowledge-intensive tasks [borgeaud2022improving; kandpal2023large; mallen2023trust], recent work finds that standard k NN-LMs do not improve prediction performance for low-frequency tokens in language modeling evaluation [nishida2025long], leaving open questions about retrieval benefits for linguistic processing.

4.4 Method

We first show that frequency gaps in tests of grammatical knowledge exist in vanilla language models (“base models”), and then test whether k NN-LMs help close these gaps. To this end, we use targeted syntactic contrast datasets where we divide test instances based on the frequency of the content words they contain. We describe the details of the models, datasets, evaluation tasks and the retrieval procedure below.

4.4.1 Models and Training

Base models. We evaluate two decoder-only Transformer models: (1) GPT-2 XL (1.5B parameters), using the publicly released checkpoint pretrained on WebText [radford2019language], and (2) GPT-2 Small (124M parameters), trained from scratch for ten epochs on a 50.03M-word mixture of child-directed speech, conversational speech, children’s stories, and subtitles. The four sources and their proportions are reported in Appendix 4.8.1. Both models use an autoregressive objective and serve as parametric-only baselines. Model size, corpus composition, and training history differ between settings, so retrieval effects are assessed against each model’s own baseline.

k NN language models. Following khandelwal2020knnlm, we augment base models with a datastore $\mathcal{D}$ containing context–target pairs. For each token x_t , we store the key-value pair: $(\mathbf{f}_\theta(\mathbf{x}_{<t}), x_t)$, where $\mathbf{f}_\theta(\mathbf{x}_{<t})$ is the layer-normalized contextual representation fed as input to the feed-forward sublayer of the final Transformer block. For GPT-2, this corresponds to the output of the final block’s second layer-normalization module and the input to its MLP. At inference time, we:

1. Compute the query: $\mathbf{q}_t = \mathbf{f}_\theta(\mathbf{x}_{<t})$
2. Retrieve k nearest neighbors via squared L2 distance
3. Compute the k NN probability: $p_{k\text{NN}}(x_t|\mathbf{x}_{<t}) \propto \sum_{(\mathbf{k}, v) \in \mathcal{N}} \mathbb{1}_{x_t=v} \exp(-d(\mathbf{k}, \mathbf{q}_t)/\tau)$
4. Interpolate: $p(x_t|\mathbf{x}_{<t}) = \lambda p_{k\text{NN}} + (1 - \lambda)p_{\text{LM}}$

We fix $\lambda = 0.25$ following khandelwal2020knnlm and vary $k \in \{1, 16, 1024\}$, using FAISS [johnson2019billion] for retrieval.

4.4.2 Evaluation Task

Syntactic contrasts. We evaluate sensitivity to three syntactic phenomena: Subject-Verb agreement (SV), Wh-questions (Wh), and Relative Clauses (RC). We collect examples in these categories from BLiMP [warstadt2020blimp], Zorro [huebner2021babyberta], and BIG-bench [srivastava2023beyond], filtering examples containing out-of-vocabulary items for each model. Each test item is a minimal pair with one grammatical and one ungrammatical sentence, where models must assign a higher probability to the grammatical sentence.

Frequency stratification. To test our hypothesis regarding lexical frequency, we further divided the evaluation data based on mean frequency across all the content words in the sentences (*high-frequency* ($f > 10^4$) versus *low-frequency* ($f < 10^3$); everything in-between *mid-frequency*). Our main hypothesis concerns the contrast between the high- and low-frequency bands; we report the mid-frequency band throughout to give a complete picture of how retrieval benefits scale with frequency.

We emphasize that all frequency gap claims are scoped *within* a single model/corpus: each gap and its narrowing under retrieval is measured against the same model’s own baseline. Because the two pretraining corpora differ substantially in size and composition, the absolute frequency thresholds do not render items in the same band directly comparable across pretraining settings, and we therefore do not compare numerical trends across the two settings.

Table 4.1 – Syntactic contrast accuracy and perplexity (PPL) for baseline and kNN-LM models ($k = 16$, $\tau = 3$, sequence-level retrieval). kNN augmentation improves accuracy across all phenomena and both pretraining scales, with relative clauses showing the largest gains. Perplexity reductions are larger for the large-scale model. Bold indicates improvement over baseline.

	Child-realistic		Large-scale	
	baseline	kNN	baseline	kNN
PPL. ↓	14.88	14.30	13.68	11.60
SV ↑	0.76	0.82	0.84	0.89
Wh ↑	0.77	0.81	0.81	0.86
RC ↑	0.68	0.78	0.65	0.84

4.4.3 Retrieval Procedure

We construct three datastore variants to disentangle the contributions of structural and semantic information: (1) *Full Retrieval*: standard contextualized embeddings encoding both semantic and syntactic information; (2) *Semantics-only (averaged)*: position-averaged contextualized embeddings from (1);¹ and (3) *Semantics-only (uncontextualized)*: summed GloVe [pennington2014glove] and fastText [bojanowski2017enriching] embeddings, providing non-contextual lexical representations without an explicit encoding of word order. We additionally vary three retrieval configuration parameters: granularity of retrieved instances (token, phrase, or sequence), context window size (τ), and number of retrieved neighbors (k), to characterize which retrieval configurations are most effective across syntactic phenomena.

4.5 Results

4.5.1 Retrieval Benefits Syntactic Contrasts

We begin by investigating the general effectiveness of retrieval augmentation on syntactic contrasts. Table 4.1² shows that k NN augmentation improves performance across all three syntactic phenomena and both pretraining scales, as well as replicating previously reported perplexity improvements [khandelwal2020knnlm]. Relative clauses benefit the most from retrieval augmentation across both model scales. This pattern holds consistently regardless of pretraining scale, though perplexity reductions are larger for the large-scale model. The disproportionate benefit for relative clauses suggests that retrieval is most helpful where baseline decisions are least accurate, which motivates us to analyze the performance gains stratified by frequency.

1. In practice, averaging contextualized embeddings may not fully scrub syntactic information; we return to this issue later.

2. Note for all tables in this section: Baseline models use parametric knowledge only; kNN models are retrieval-augmented. Abbreviations: SV (subject-verb agreement), Wh (wh-questions), RC (relative clauses), PPL (perplexity). We use $k = 16$, $\tau = 3$, and sequence-level retrieval (see Section 4.4.3) as the representative configuration, since this setting yields general gains in low-frequency conditions. However, as discussed in Section 4.6.2, there may exist alternative configurations that outperform this representative configuration for individual phenomena.

4.5.2 Frequency Gap is Narrowed but not Fully Closed

Table 4.2 – Frequency-stratified syntactic contrast performance ($k = 16$, $\tau = 3$; sequence-level retrieval). ER denotes error reduction, $\Delta\text{acc}/(1 - \text{acc}_{\text{base}})$. Bold indicates improvement over baseline.

		Child-realistic			Large-scale		
		base	kNN	ER (%)	base	kNN	ER (%)
SV	high	0.91	0.90	−11.1	0.96	0.95	−25.0
	mid	0.82	0.83	5.6	0.90	0.89	−10.0
	low	0.60	0.76	40.0	0.71	0.83	41.4
Wh	high	0.92	0.88	−50.0	0.93	0.93	0.0
	mid	0.83	0.86	17.6	0.90	0.91	10.0
	low	0.61	0.70	23.1	0.69	0.74	16.1
RC	high	0.88	0.89	8.3	0.89	0.95	54.5
	mid	0.73	0.81	29.6	0.82	0.89	38.9
	low	0.48	0.65	32.7	0.41	0.70	49.2

Next, we examine whether retrieval helps close the lexical frequency gap. Table 4.2 confirms the existence of frequency gaps in both baseline models across three syntactic phenomena, with performance on low-frequency items consistently lower than on high-frequency items, and mid-frequency items falling in between. With retrieval augmentation, the high–low performance gap narrows across all phenomena and both model scales. In terms of error reduction ($\Delta\text{acc}/(1 - \text{acc}_{\text{base}})$), low-frequency items show substantial relative gains (e.g., 49.2% for RCs in the large-scale model), though we note that this metric can be misleading near ceiling, where small absolute changes in already-high accuracy produce large negative percentages (e.g., child-realistic high-frequency Wh, $0.92 \rightarrow 0.88$, yields -50%). Raw accuracies should be used to adjust the interpretations in such cases.

Statistical analysis supports the interpretation that narrowing is driven primarily by gains on low-frequency items. Bootstrapped 95% confidence intervals on the per-frequency band performance delta (kNN – baseline) confirm a robust separation: for the child-realistic model, low-frequency items show $\Delta = 0.140$ (95% CI $[0.115, 0.166]$) versus $\Delta = -0.013$ (95% CI $[-0.030, 0.004]$) for high-frequency items, with non-overlapping intervals (large-scale model shows the same pattern; see Appendix 4.8.3).

A mixed-effects logistic regression used treatment coding with high-frequency items and the baseline model as reference levels, with the specification

$$\text{correct} \sim \text{Frequency} * \text{ModelType} \\ + (1 \mid \text{item}) + (1 \mid \text{phenomenon}).$$

The retrieval effect is significantly larger for low-frequency than for high-frequency items (Frequency [low] $\times$ ModelType [kNN] interaction: $\beta = 0.52$, $p < 0.001$). The effect of retrieval within the high-frequency band is not significant (ModelType [kNN]: $\beta = -0.04$, $p = 0.62$; Appendix 4.8.3).

Nevertheless, this compensation is not sufficient to eliminate the frequency gap: low-frequency items still yield lower accuracy than high-frequency items across all tested phenomena and both model scales, even after retrieval augmentation.

4.6 Analysis

The analyses in this section fix the retrieval configuration to $k = 16$, $\tau = 3$ with sequence-level retrieval, which isolates the contribution of the factors studied from the effects of configuration.

4.6.1 Importance of Structural vs. Semantic Information

We proceed to investigate which aspects of the retrieved information drive the observed benefits by examining (1) the effect of semantics-only retrieval, and (2) the effect of retrieving structurally similar instances.

Semantics-only retrieval is mostly ineffective. Figure 4.1 shows that semantics-only retrieval (FastText, GloVe, G-Avg; see Section 4.4.3) performs at or substantially *below* the baseline without retrieval across most conditions. This shows that semantic similarity of the retrieved examples alone does not reliably improve syntactic judgments. One partial exception is GPT2-Averaged (G-Avg), where retrieval does lead to substantive gains over the baseline for low-frequency items for RCs, although performance on high-frequency items degrades by a large margin. FastText and GloVe also yield gains for low-frequency RCs in the large-scale setting, although relatively minor. There are two possible, non-mutually exclusive interpretations for this: (1) semantic information does sometimes yield benefits, as discussed in the next paragraph; or (2) that averaging of contextualized embeddings does not totally remove structural information.

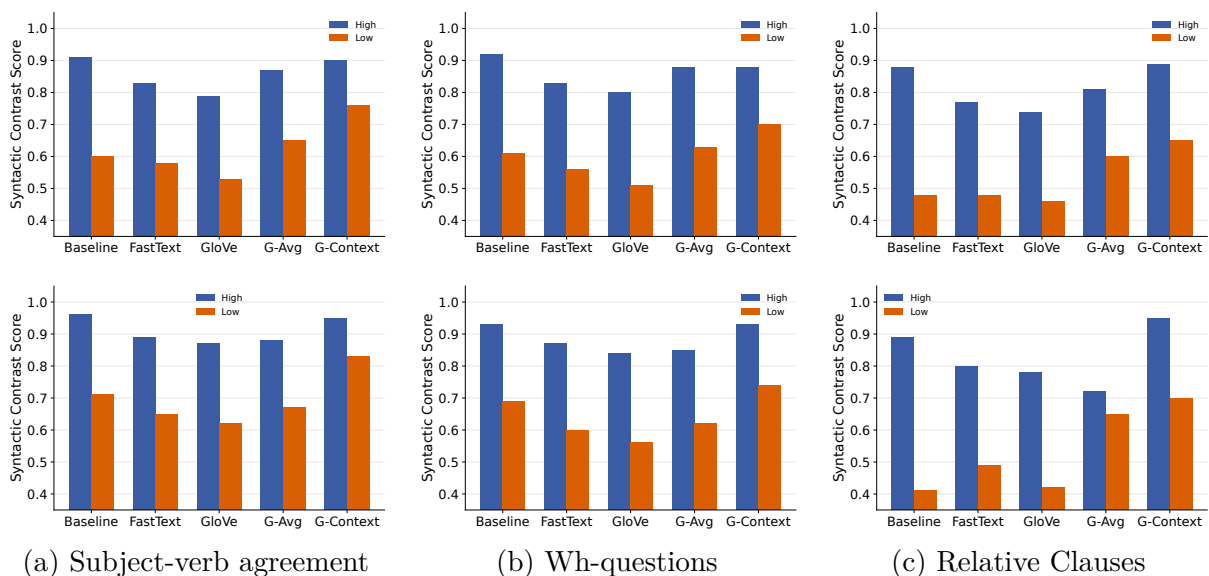


Figure 4.1 – Performance on syntactic contrasts using semantics-only retrieval strategies compared to full retrieval ($k = 16$, $\tau = 3$) of the child-realistic (top row) and large-scale (bottom row) models. Blue bars indicate high-frequency items and orange bars low-frequency items.

Structural information is critical, and similarity-based example selection does not compensate for missing structure. Figure 4.2 examines retrieval benefits from two angles: whether the set of retrieved items contains structurally matching examples (*with* (w) vs. *without* (w/o)),³ and whether retrieval ranks them by embedding similarity or at random ($+r$). Across all phenomena and both model scales, the w bars consistently exceed w/o for low-frequency items (orange bars) with the strongest effects for relative clauses. This illustrates the importance of structure-matched examples in the retrieved set. For low-frequency items, both structural matches and similarity-based selection contribute. However, similarity does not compensate for missing structural information. While it does add smaller, consistent gains overall, similarity-based selection without structural matches (w/o : middle orange bars with hatched gains) consistently fails to outperform random selection with structural matches ($w/+r$: right orange bars without hatched gains) for any model or phenomenon. Thus, examples containing the target syntactic phenomenon are the main contributor under the study’s operational definition of frequency-selective retrieval benefits, with secondary contributions from similarity-based selection. Given the ineffectiveness of semantics-only retrieval discussed above, the embedding similarity gains likely reflect partial structural similarity between the retrieved examples and the target, in addition to semantic similarity.

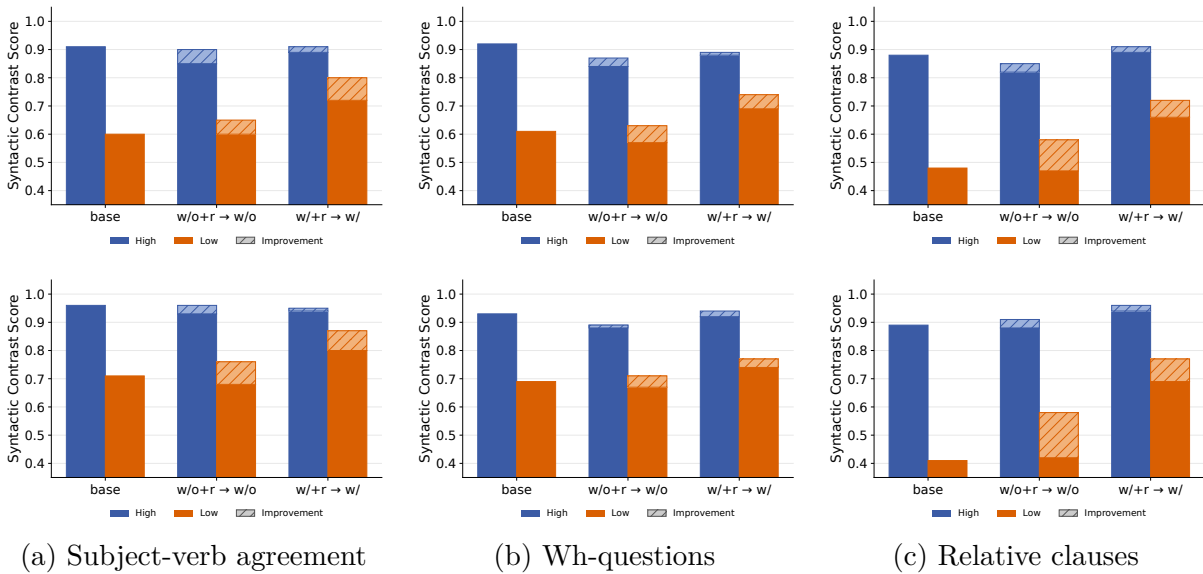


Figure 4.2 – Performance when retrieved sequences contain target syntactic structures (with: w /) versus when they do not (without: w/o). Top row: child-realistic model; bottom row: large-scale model ($k = 16$, $\tau = 3$, sequence-level). Blue and orange bars correspond to high- and low-frequency items, respectively. Within each panel, the x -axis shows three conditions: **base** (no retrieval), **$w/o+r \rightarrow w/o$** (retrieved set does not contain structurally matched examples, random selection $\rightarrow$ embedding-similarity), and **$w/+r \rightarrow w/$** (retrieved set contains structurally matched examples, random selection $\rightarrow$ embedding-similarity). Solid bars denote random selection ($+r$); hatched segments denote the additional gain from embedding-similarity ranking over random selection.

3. See Appendix 4.8.4 for our structural match metric.

4.6.2 Effects of Retrieval Configuration

We next analyze whether and how retrieval effectiveness is modulated by three configuration parameters: retrieval granularity (token, phrase, or sequence), number of neighbors (k), and context window size (τ).

Effects of retrieval granularity. Figure 4.3 shows that, at $k = 16$ and $\tau = 3$, sequence-level retrieval numerically provides the strongest or tied-strongest performance for low-frequency items across all phenomena and both model settings, but the effect of retrieval granularity overall is small.

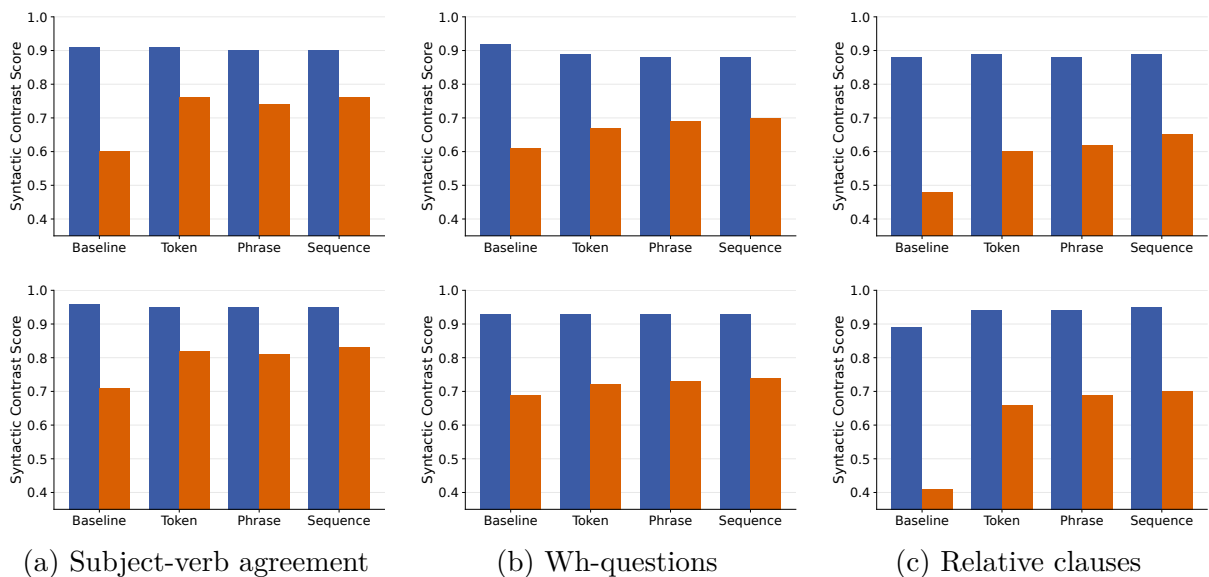


Figure 4.3 – Performance across retrieval granularities (token, phrase, sequence) in child-realistic (top row) and large-scale (bottom row) models ($k = 16$, $\tau = 3$). Blue bars indicate high-frequency items and orange bars low-frequency items. Sequence-level retrieval is numerically the overall strongest, but the effect of retrieval granularity is overall small.

Effects of context window size. Figure 4.4 shows that the effect of context window size is idiosyncratic. For RC, $\tau = 10$ yields the highest accuracy across both model scales and both frequency bands, whereas for SV, $\tau = 3$ yields the strongest performance for low-frequency. For Wh, the effect of τ is minimal, except $\tau = 3$ leading to degradation in high-frequency items in the child-realistic model. Thus, increasing context window size does not uniformly improve performance, and the preferred window size depends on the phenomenon, frequency band, and model.

Effects of number of retrieved instances. Figure 4.5 shows that the number of retrieved instances has a non-monotonic effect on performance in the low-frequency regime. Moving from $k = 1$ to a moderate number ($k = 16$) consistently improves performance across most conditions, indicating the benefit of aggregating evidence from multiple exemplars. However, further increasing the number of neighbors ($k = 1024$) leads to performance degradation, consistent with interference from less relevant neighbours, although this comparison does not isolate interference from the changed neighbour distribution and its interaction with the fixed interpolation weight.

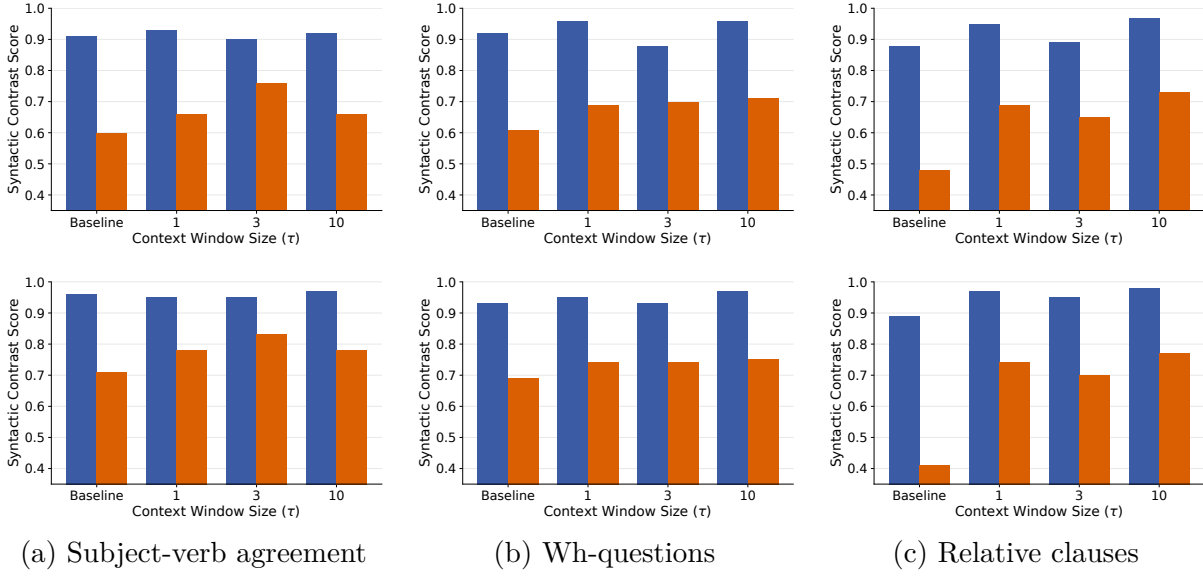


Figure 4.4 – Performance across context-window sizes for the child-realistic (top row) and large-scale (bottom row) settings, using sequence-level retrieval with $k = 16$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items. The effects of context size are non-monotonic and phenomenon-, frequency band- and model-dependent.

This pattern holds across models, granularities, and syntactic phenomena, suggesting the optimal choice lies at intermediate values of k . The effect for high-frequency items is smaller and mixed. **No cross-phenomena optimal configuration.** Across the three parameters (retrieval granularity, context window size, and number of retrieved instances), no single configuration consistently yields optimal performance across all syntactic phenomena, model, and frequency strata. While sequence-level retrieval, larger context windows, and a moderate number of neighbors ($k = 16$) provide the most robust overall performance (and therefore serves as our representative configuration), it does not uniformly dominate across all settings, particularly for simpler or high-frequency dependencies where more localized representations or smaller contexts can be competitive. These results suggest that effective retrieval is jointly modulated by dependency complexity, data frequency, and model capacity, and that optimal retrieval configurations are likely to be phenomenon-dependent rather than universally fixed.

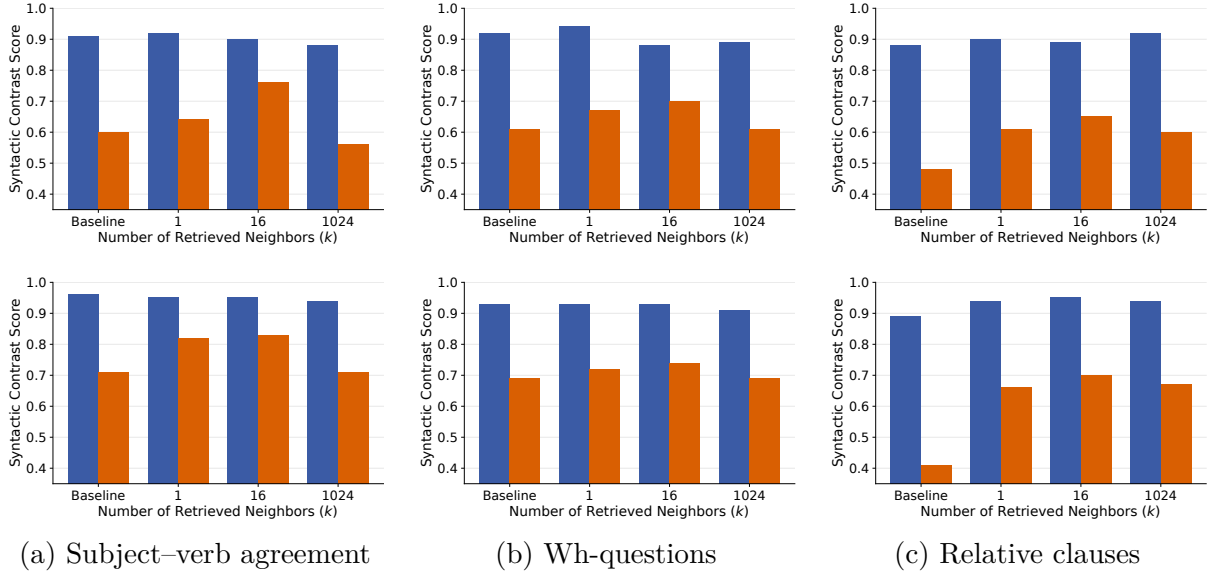


Figure 4.5 – Performance across numbers of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic (top row) and large-scale (bottom row) settings, using $\tau = 3$ and sequence-level retrieval. Blue bars indicate high-frequency items and orange bars indicate low-frequency items. For all six low-frequency conditions, $k = 16$ outperforms both $k = 1$ and $k = 1024$. Effects on high-frequency items are smaller and mixed.

4.7 What retrieval contributes to learning from sparse evidence

4.7.1 Selective compensation at inference time

The main contribution of this chapter is a controlled change in how linguistic evidence is used. For a given trained base model and evaluation set, adding k NN retrieval changes the probabilities assigned to the two members of a syntactic minimal pair. Low-frequency items improve in all three phenomenon groups and both model settings in the representative configuration. For example, subject-verb agreement accuracy in the smaller model rises from 0.60 to 0.76 for low-frequency items, while high-frequency accuracy changes from 0.91 to 0.90 (Table 4.2). The frequency gap therefore narrows primarily through improved low-frequency performance, although small declines in some high-frequency conditions also contribute to gap reduction.

Relative clauses show the largest absolute low-frequency gains in the reported table: from 0.48 to 0.65 in the smaller model and from 0.41 to 0.70 in GPT-2 XL. This pattern identifies relative-clause items as especially responsive to the tested intervention. It does not by itself establish a general law relating syntactic complexity to retrieval benefit. The phenomenon groups differ in baseline accuracy, lexical composition, and benchmark sources, and lower baselines leave more room for improvement. The stronger conclusion is that retrieval can supply useful evidence in conditions where the base model’s decisions are least reliable. Low-frequency accuracy nevertheless remains below high-frequency accuracy after augmentation.

This result concerns the contribution of retrieval at inference time. Interpolation can improve a decision even when the base model’s parameters remain unchanged. The experiment therefore distinguishes access to useful evidence from its prior incorporation into the parametric

predictor, but does not measure subsequent consolidation or durable learning after the datastore is removed. Likewise, lower perplexity and higher minimal-pair accuracy are complementary outcomes: improved average prediction does not automatically explain improved discrimination of a particular grammatical contrast.

4.7.2 The usefulness of memory depends on its contents and representation

The ablations show that the effect depends on how stored experience is represented and selected. Averaged contextual representations and the GloVe and fastText alternatives generally fail to reproduce the benefits of full contextual retrieval (Figure 4.1). In the structural-content analysis, retrieval sets containing the target phenomenon are more useful for low-frequency items than sets without it, and similarity ranking does not fully compensate for the absence of those examples (Figure 4.2). A useful memory must therefore make task-relevant information accessible, rather than merely return lexically related material.

These comparisons do not cleanly separate syntax from semantics. Averaging contextual representations removes positional distinctions in the pooling operation but may retain information about structure encoded before pooling. Static lexical embeddings differ from contextual embeddings in several properties, including their training sources and similarity geometry. Their poorer performance cannot be attributed uniquely to the removal of syntax. Moreover, the structural-match indicator in Appendix 4.8.4 records whether a retrieved sequence contains the same broad phenomenon, using parser-based criteria. It is not a measure of identity between complete syntactic analyses or between the particular dependencies in two sentences. The results support the usefulness of these operationally defined structural cues; a stronger decomposition would require closer control of lexical content, construction, and representational quality.

Configuration matters as well. In the main sequence-level comparison, 16 neighbours outperform both 1 and 1,024 neighbours on the six low-frequency conditions. The additional analyses support an intermediate setting as a useful choice among the tested values, rather than a universal optimum. Context-size effects differ across phenomena, and the advantage of sequence over smaller retrieval units is modest. Deterioration with more neighbours is compatible with interference from less relevant instances, but the experiment does not isolate interference as the mechanism. Changes in the resulting token distribution and in its interaction with the fixed interpolation weight are alternative contributors. Adaptive retrieval is consequently a motivated future intervention, not a mechanism already demonstrated by this study.

4.7.3 Cognitive and methodological implications

For cognitive modelling, the study supplies a functional demonstration of complementarity. A system that combines parametric prediction with direct access to instances can make more accurate low-frequency syntactic decisions than its parametric component alone. This is consistent with the broad motivation for complementary learning systems, while leaving open whether children use a comparable retrieval process during the same tasks. The experiment contains no matched child retrieval manipulation, no neural measurement, and no developmental trajectory of

memory formation. Hippocampal storage, replay, selective encoding, forgetting, and consolidation are therefore candidate mechanisms for further modelling, rather than findings of the present comparison.

For AI methods, the contribution is more specific than an overall gain from retrieval augmentation. Frequency-stratified evaluation shows where an apparently successful configuration helps, where it leaves a residual gap, and where it can reduce already strong performance. The representation and retrieval-content ablations then identify conditions under which that compensation is lost. This combination turns a general retrieval method into a diagnostic instrument for examining the reliability of syntactic predictions across lexical frequencies. It also motivates evaluating retrieval strategies on both aggregate accuracy and the distribution of gains across items.

The developmental interpretation depends on a complete account of the system’s information sources. The smaller base model was trained for ten epochs over a 50.03M-word corpus whose largest component is subtitles; this is bounded textual training, not a record of one child’s successive encounters. The full-retrieval method is described using the base GPT-2 model’s contextual states, while GloVe and fastText introduce different learned representations in the ablations. The datastore’s corpus, construction split, size, and overlap with training and evaluation need to be reported explicitly. Only a datastore restricted to previously available material would isolate renewed access to that material; a store containing additional text would also change the evidence available to the system. This distinction is especially consequential for claims about learning from limited experience.

Finally, the evaluation measures probabilistic preference on selected written minimal pairs. BLiMP and the other sources provide controlled contrasts, but the present subsets do not establish mastery of all English syntax, comprehension in context, spontaneous grammatical production, or a child’s developmental course. Frequency bands are defined by the mean frequency of content words in an item, so they should not be equated with an experimentally manipulated number of encounters with a single target word. Comparisons across the two model settings are further limited by simultaneous differences in model size, corpus composition, and training history.

4.7.4 Targeted next steps

Three interventions follow from these results. First, selective storage or weighting could prioritize rare or informative instances rather than treating similarity as the only retrieval criterion. Error-driven learning and selective data weighting motivate this comparison [chang2006dual; rescorla1972theory; toneva2018empirical; xie2023doremi]. A useful test would keep the datastore’s total size fixed while varying which instances are retained, then assess both low-frequency improvement and any loss on frequent items.

Second, retrieval could be constrained by the target construction or use alternative contextual representations. Comparing the present layer-normalized keys with other points in the residual stream, while controlling the datastore and distance scaling, would test whether a persistent gap reflects selection of the wrong instances or inadequate information in the selected representation. Structural filtering would need matched lexical controls so that an apparent structural gain is

not explained by a different distribution of words.

Third, an adaptive rule could choose context scope and neighbour count for the current prediction. The present configuration effects motivate such a rule, but do not establish its benefit. Comparing a rule selected on held-out development material with fixed configurations would test whether adaptation improves the distribution of gains without selecting settings on the final syntactic test.

Together with Chapter 3, these results make frequency a recurring question about the availability and use of linguistic evidence. The preceding chapter examined whether words appear reliably in generated output; this chapter shows that stored instances can selectively improve another expression of linguistic knowledge. A direct bridge would test retrieval on the same expressive-production measure, or evaluate one learner on both production and minimal pairs while controlling its information sources. The following chapter examines a different resource, caregiver interaction and feedback. These studies extend the thesis by testing distinct ways in which experience can influence model behaviour, without treating them as successive stages of one developing system.

4.8 Appendix

4.8.1 Pretrained models and datasets

Child-realistic pretraining data. We constructed a dataset to approximate the language input a child receives during early development, drawing on child-directed speech (CDS) from CHILDES [macwhinney2000childes], transcribed conversational speech from the Switchboard Dialog Act Corpus [godfrey1992switchboard; stolcke_dialogue_2000], children’s stories from the Children’s Stories Text Corpus [bensaid2021fairytailor], and movie and television subtitles from OpenSubtitles [lison2016opensubtitles]. These four sources also appear in the 2023 BabyLM corpus release [warstadt2023findings]. The complete dataset comprises 50.03M words, consistent with estimates of cumulative linguistic input by early childhood [gilkerson2017mapping; frank2023bridging]. Table 4.3 shows the distribution of constituent sub-datasets.

Table 4.3 – Composition of child-realistic pretraining corpus combining child-directed speech, conversational dialogue, and children’s books to approximate naturalistic language exposure during early childhood. CDS = child-directed speech; SWBD = Switchboard; CSTC = Children’s Stories Text Corpus; OSub = OpenSubtitles.

Dataset	Domain	# words	Prop. (%)
CHILDES	CDS	15M	30.0
SWBD	Dialogue	1.18M	2.4
CSTC	Books	3.22M	6.4
OSub	Dialogue	30.63M	61.2
Total		50.03M	100.0

Large-scale pretraining data. For the large-scale condition, we use GPT-2 XL (1.5B parameters) pretrained on the WebText corpus [radford2019language], which contains approximately 40GB of text scraped from internet sources.

Model architectures and training. Table 4.4 summarizes the architectural specifications for both models. The child-realistic model uses the GPT-2 Small architecture with 124M parameters, trained from scratch for 10 epochs, following the BabyLM Challenge setup [hu2024findings]. The GPT-2 XL model uses the publicly available pretrained checkpoint from radford2019language. The training details for the child-realistic model are provided in Table 4.5.

Table 4.4 – Model architecture specifications. GPT-2 Small was trained from scratch on child-realistic data (50M words); GPT-2 XL uses the publicly released pretrained checkpoint from radford2019language trained on WebText (~ 40 GB).

Parameter	GPT-2 Small	GPT-2 XL
# parameters	124M	1.5B
Transformer layers	12	48
Hidden size	768	1600
FFN size	3072	6400
Attention heads	12	25
Attention dropout	0.1	0.1
Embedding dimension	768	1600
Vocabulary size	50257	50257

Table 4.5 – Training hyperparameters for GPT-2 Small on the child-realistic corpus, trained for 10 epochs using a standard autoregressive language modeling objective with cross-entropy loss.

Hyperparameter	Value
Max sequence length	1024
Batch size	64
Training epochs	10
Learning rate	0.0001
LR scheduler	inverse sqrt
Warmup steps	1000
Optimizer	Adam
Adam- β_1	0.9
Adam- β_2	0.98
Dropout	0.1
Weight decay	0.01
Gradient clipping	1.0

4.8.2 Evaluation set construction

To ensure focused evaluation on syntactic phenomena relevant to our research questions, we selected subsets corresponding to our target phenomena from three evaluation benchmarks: BLiMP

[warstadt2020blimp], Zorro [huebner2021babyberta], and BIG-bench [srivastava2023beyond] as discussed below.

4.8.2.1 Source benchmarks

BLiMP. The Benchmark of Linguistic Minimal Pairs for English [warstadt2020blimp] consists of 67 sub-datasets, each containing 1,000 minimal pairs (67,000 pairs total) isolating specific contrasts in syntax, morphology, or semantics. The data is automatically generated according to linguist-crafted grammar templates, with 96.4% aggregate human agreement with the labels. For subject-verb agreement, we selected the relational-noun distractor paradigm and four plural-agreement paradigms: regular and irregular plurals, each with a verb-number manipulation holding the subject fixed (variant 1) and a subject-number manipulation holding the verb fixed (variant 2). For wh-questions, we selected the object-gap, subject-gap, and long-distance subject-gap wh-question paradigms, together with the four wh-versus-that paradigms crossing gap presence and dependency length. For relative clauses, we selected the relative-clause distractor agreement paradigm.

Zorro. Zorro [huebner2021babyberta] is a targeted grammar test suite designed for small-data, restricted-vocabulary evaluation. It comprises 23 paradigms spanning 13 grammatical phenomena, with 2,000 minimal pairs (4,000 sentences) per paradigm. Its lexical items were selected from whole-word entries in BabyBERTa’s custom vocabulary, making the suite suitable for evaluating models trained on comparatively limited, child-directed linguistic input. For subject-verb agreement, we selected agreement across a prepositional phrase, agreement in questions with an auxiliary, and agreement in simple questions. For wh-questions, we selected the object and subject wh-question paradigms. For relative clauses, we selected agreement across a relative clause.

BIG-bench. The Beyond the Imitation Game Benchmark [srivastava2023beyond] includes several linguistic evaluation tasks. For subject-verb agreement, we selected the simple-English, single-adverb, double-adverb, adverb-conjunction, name-plus-prepositional-phrase, noun-plus-prepositional-phrase, noun-plus-prepositional-phrase-and-adverb, Gulordava-nonce, and frequency-disbalance-nonce subtasks. For relative clauses, we selected the long-nested-inner and long-nested-outer subtasks. No BIG-bench items were used for the wh-question evaluation.

4.8.3 Statistical analyses

Bootstrapped confidence intervals on per-frequency band deltas. For each frequency band, we computed the performance delta (kNN – baseline) and estimated 95% confidence intervals via bootstrap resampling over test items (10,000 resamples). As seen in Table 4.6, in both settings, the intervals for low- and high-frequency items do not overlap, indicating a robust separation in retrieval gains.

Mixed-effects logistic regression. To test whether the effect of retrieval is larger for low-frequency items, we fit a mixed-effects logistic regression predicting correctness from Frequency (high/low) and Model Type (baseline/kNN), with random intercepts for item and for phenomenon.

Table 4.6 – Bootstrapped 95% confidence intervals on the per-frequency band accuracy delta (kNN – baseline), 10,000 resamples over test items ($k = 16$, $\tau = 3$, sequence-level).

Pretraining	Freq	Δ	95% CI
Child-realistic	high	−0.013	[−0.030, 0.004]
	mid	<i>0.040</i>	[0.022, 0.059]
	low	<i>0.140</i>	[0.115, 0.166]
Large-scale	high	<i>0.017</i>	[0.002, 0.033]
	mid	<i>0.023</i>	[0.008, 0.039]
	low	<i>0.153</i>	[0.125, 0.182]

We used treatment coding, with high-frequency items and the baseline model as the reference levels:

$$\text{correct} \sim \text{Frequency} * \text{ModelType} + (1|\text{item}) + (1|\text{phenomenon})$$

Coefficients are reported in Table 4.7. Under this coding, the Model Type coefficient estimates the effect of kNN augmentation within the high-frequency band. This coefficient was not statistically significant ($\beta = -0.04$, $p = 0.62$), indicating no detectable retrieval effect for high-frequency items. Crucially, the Frequency $\times$ Model Type interaction was positive and significant ($\beta = 0.52$, $p < 0.001$), indicating that the retrieval effect (in log odds) was larger for low-frequency than for high-frequency items. Together with the positive accuracy differences and bootstrap confidence intervals for low-frequency items in Table 4.6, this result supports the hypothesis that retrieval disproportionately benefits items for which parametric representations are weak, rather than providing a uniform improvement across frequency bands.

Table 4.7 – Mixed-effects logistic regression coefficients ($\text{correct} \sim \text{Frequency} * \text{ModelType} + (1|\text{item}) + (1|\text{phenomenon})$); $k = 16$, $\tau = 3$, sequence-level).

Term	β	SE	z	p
Frequency (low)	−1.52	<i>0.11</i>	−13.8	< 0.001
ModelType (kNN)	−0.04	<i>0.08</i>	−0.50	<i>0.62</i>
Frequency $\times$ ModelType	<i>0.52</i>	<i>0.10</i>	<i>5.20</i>	< 0.001

4.8.4 Structural match analysis

For each evaluation item and retrieved sequence, we defined a binary structural match indicator equal to 1 if the sequence contained the same target phenomenon—subject-verb agreement (SV), a wh-question (Wh), or a relative clause (RC)—and 0 otherwise. We identified these phenomena using spaCy [honnibal2020spacy] as specified below:

Subject-Verb detection. We applied spaCy’s dependency parser to identify subject-verb relationships. Instances were flagged if they contained `nsubj` (nominal subject) or `nsubjpass` (passive nominal subject) dependencies that link a noun phrase to a finite verb.

Wh-Sentence detection. We used part-of-speech (POS) tagging to identify wh-words (POS tags: `WDT`, `WP`, `WP$`, `WRB`) at clause boundaries. Instances containing interrogative or relative wh-elements were marked for inclusion.

Relative Clause detection. We combined POS tagging with dependency parsing to identify relative clause structures. Specifically, we detected: (1) Relative pronouns (WDT, WP) or relativizers (*that*, *which*, *who*); (2) Clausal complements with `relcl` (relative clause modifier) dependency relations; and (3) Appropriate structural embedding within a noun phrase.

Table 4.8 reports the mean proportion of the 16 retrieved neighbors that contain the target syntactic structure, and Table 4.9 reports two-sample *t*-tests comparing the effect of retrieval with vs. without structural matches, each against the baseline and against each other.

Table 4.8 – Mean proportion of the 16 retrieved neighbors containing the target syntactic structure ($k = 16$, $\tau = 3$, sequence-level retrieval). Structural matches remain particularly sparse for relative clauses in the child-realistic setting.

	SV		Wh		RC	
	high	low	high	low	high	low
Child-realistic	0.74	0.70	0.49	0.62	0.09	0.15
Large-scale	0.95	0.89	0.67	0.76	0.74	0.70

Table 4.9 – *p*-values for structural-retrieval comparisons in the child-realistic and large-scale models ($k = 16$, $\tau = 3$, sequence-level retrieval).

Comparison	Freq	Child-realistic			Large-scale		
		SV	Wh	RC	SV	Wh	RC
Baseline vs. Without	low	0.16	0.38	0.022	0.071	0.49	< 0.001
	high	0.67	0.14	0.43	0.76	0.008	0.28
Baseline vs. With	low	0.006	0.021	< 0.001	< 0.001	0.004	< 0.001
	high	0.36	0.23	0.016	0.44	0.52	< 0.001
With vs. Without	low	0.003	0.046	< 0.001	< 0.001	0.031	< 0.001
	high	0.31	0.078	0.012	0.44	0.001	0.001

The tests support the claim that retrieval benefits are greater when the retrieved examples contain structurally matched examples, and reveal additional nuance: the structural-match advantage is more pronounced in low-frequency bands and for structurally complex phenomena (RCs), consistent with broader trends elsewhere in the manuscript.

4.8.5 Additional Retrieval Configuration Results

This section provides additional results for the retrieval configuration analyses in the main text. All results use the *Full Retrieval* datastore defined in Section 4.4.3; baseline results are shown for reference. Figures 4.6–4.9 vary context-window size ($\tau \in \{1, 3, 10\}$) at $k \in \{1, 1024\}$, whereas Figures 4.10–4.13 vary the number of retrieved neighbors ($k \in \{1, 16, 1024\}$) at $\tau \in \{1, 10\}$. Together with the main analyses, these figures cover the evaluated combinations of neighbor count, context-window size, and retrieval granularity for the displayed high- and low-frequency bands in both pretraining settings. Given many differences between individual configurations are small, we focus here on broad patterns rather than numerical rankings within individual panels.

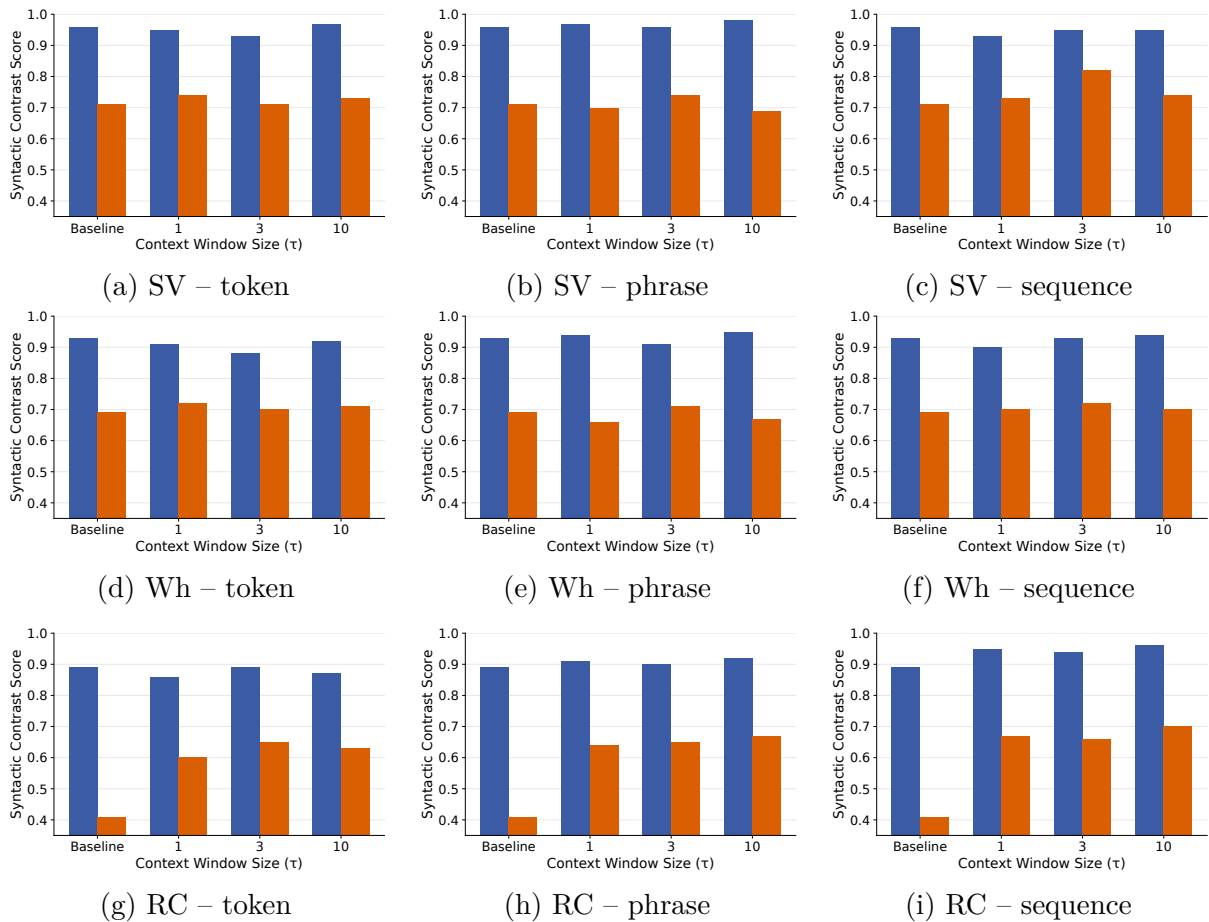


Figure 4.6 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

Effects of context window size. Figures 4.6–4.9 extend Figure 4.4 to $k \in \{1, 1024\}$ and to all retrieval granularities. As in the main analysis, accuracy is non-monotonic with context window size. The direction and magnitude of the effects vary by number of neighbors, retrieval granularity, phenomenon, and frequency band; and are often small. These additional results therefore do not identify a context window size that is uniformly preferred across conditions.

Effects of the number of retrieved neighbors. Figures 4.10–4.13 extend Figure 4.5 to $\tau \in \{1, 10\}$ and to all retrieval granularities. The results are broadly consistent with the pattern reported in the main text: for low-frequency items, the intermediate setting $k = 16$ generally yields the strongest performance among the tested values, whereas effects on high-frequency items are smaller and more mixed. The magnitude and direction of the effects vary across conditions, and the effects are often small. We therefore treat $k = 16$ as a useful representative setting rather than as a universally optimal neighbor count. Comparisons between $\tau = 1$ and $\tau = 10$ similarly provide no evidence for an interesting interaction effect with context window size.

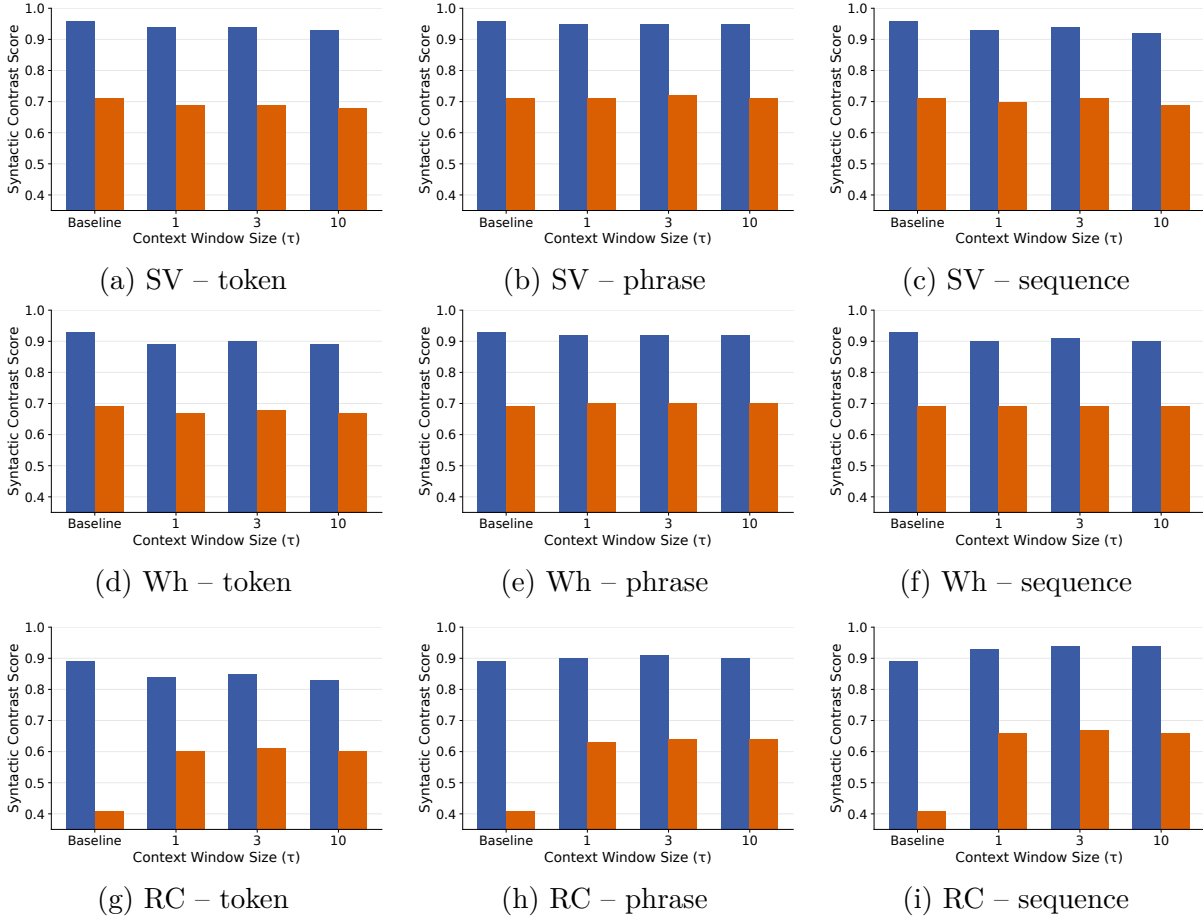


Figure 4.7 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1024$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

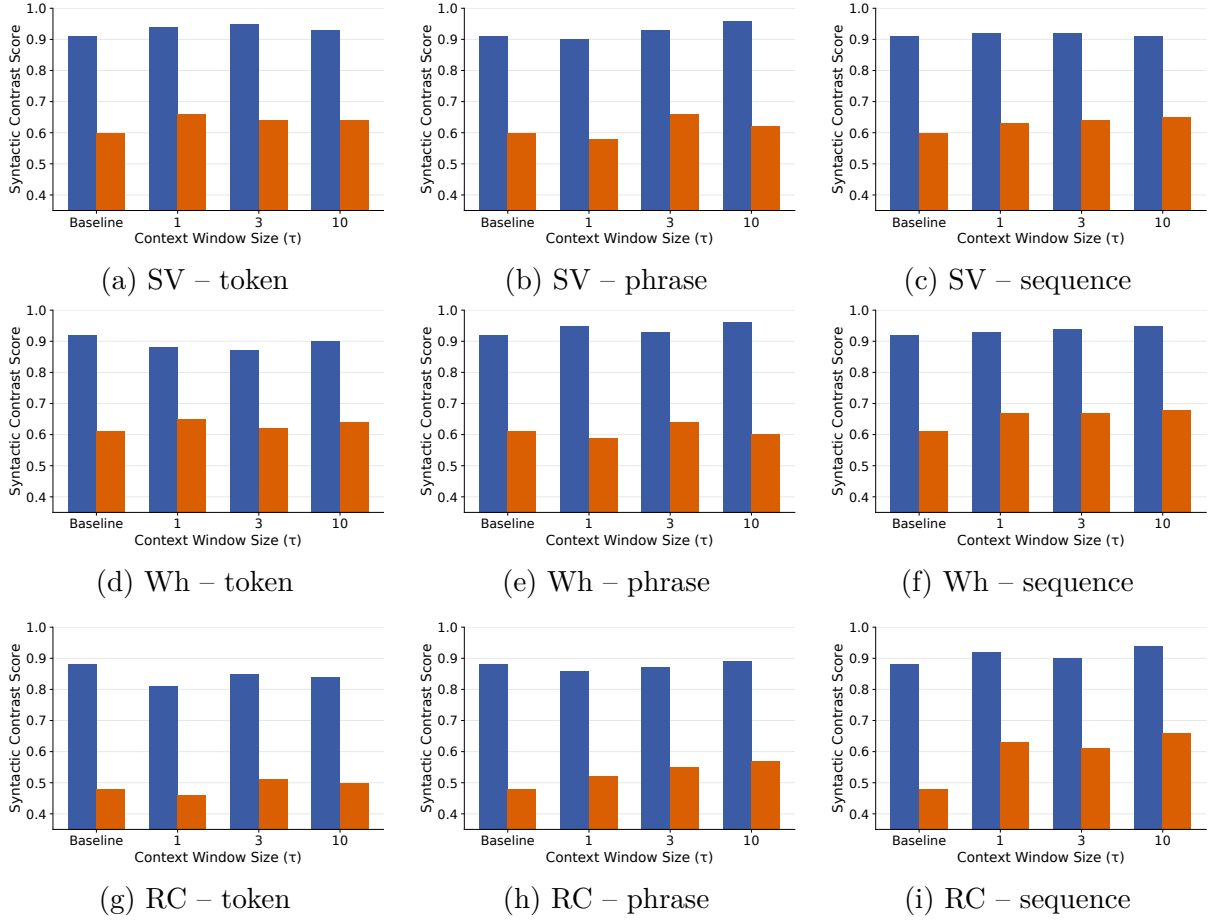


Figure 4.8 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

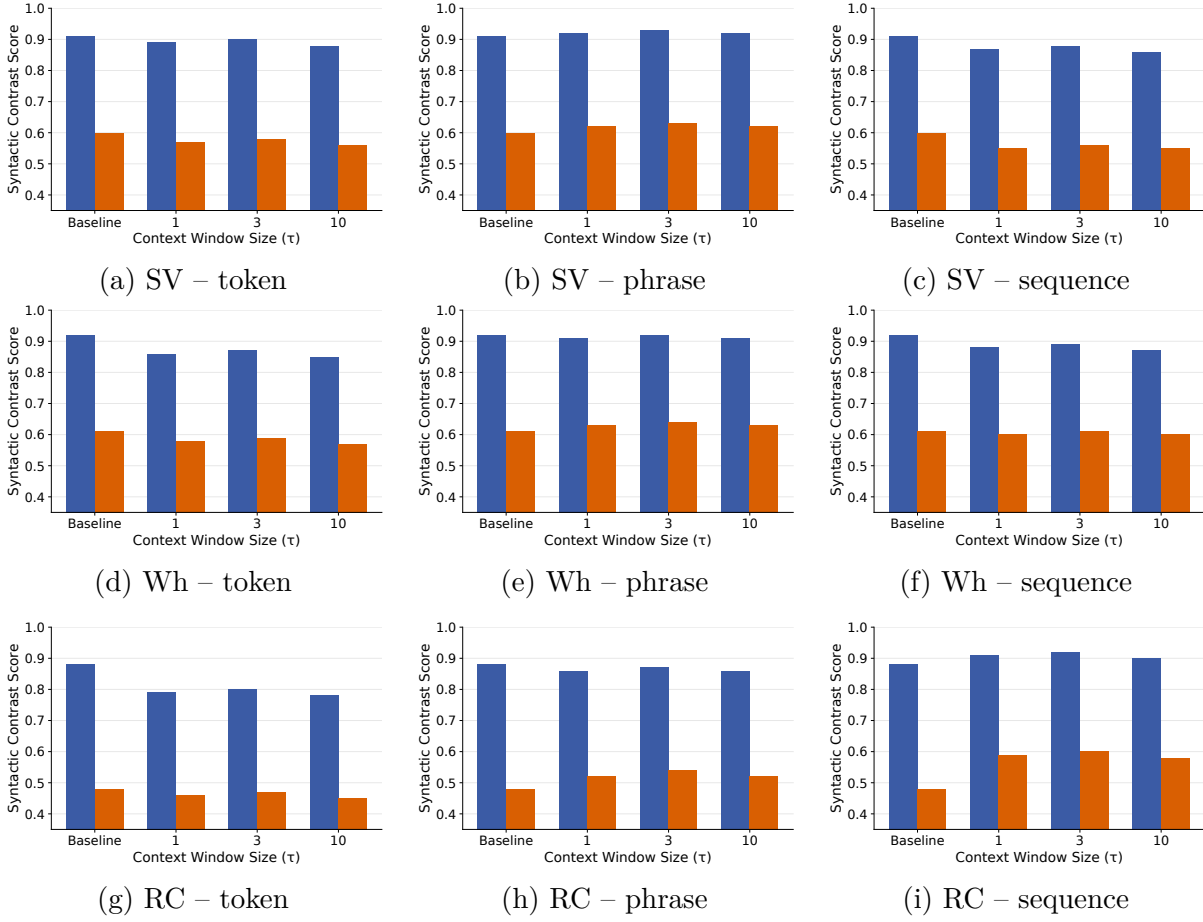


Figure 4.9 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1024$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

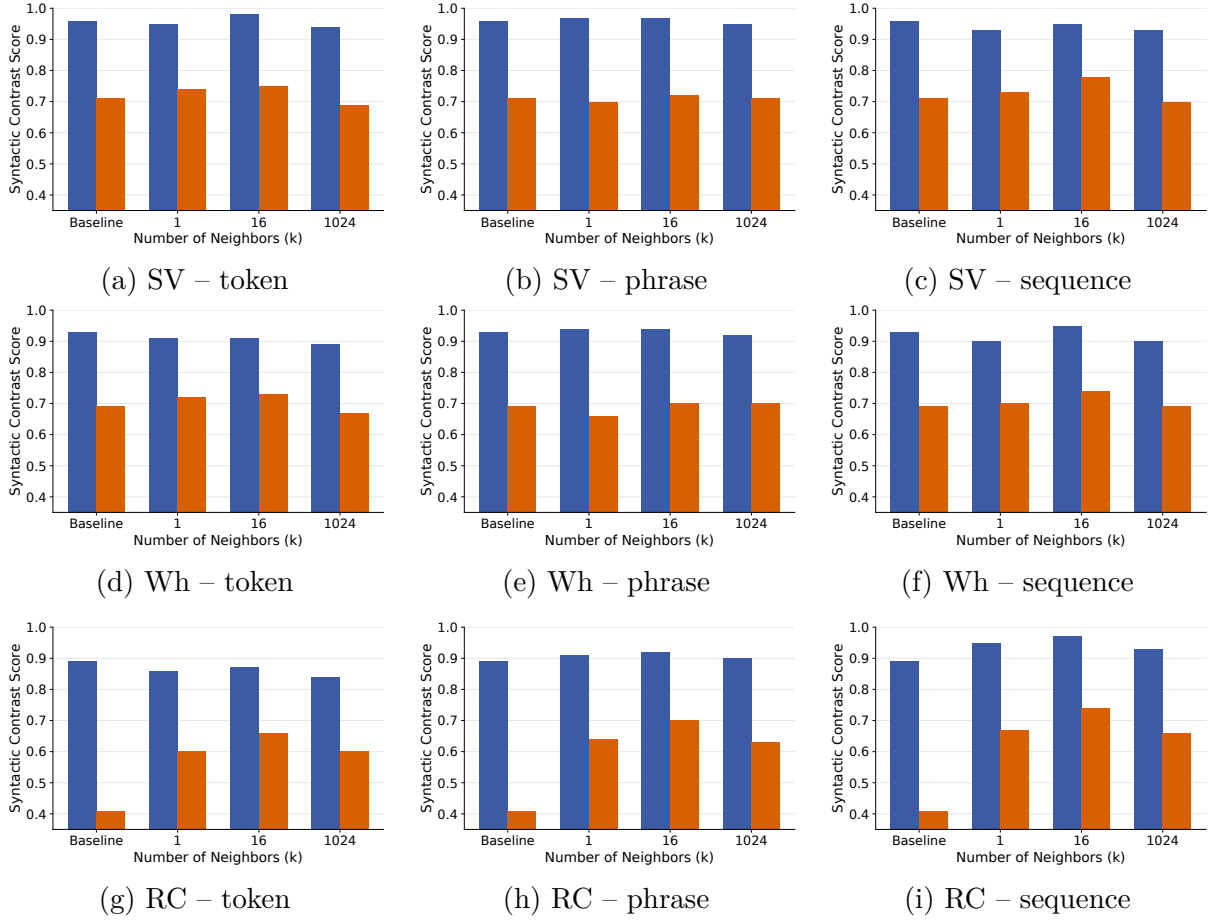


Figure 4.10 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

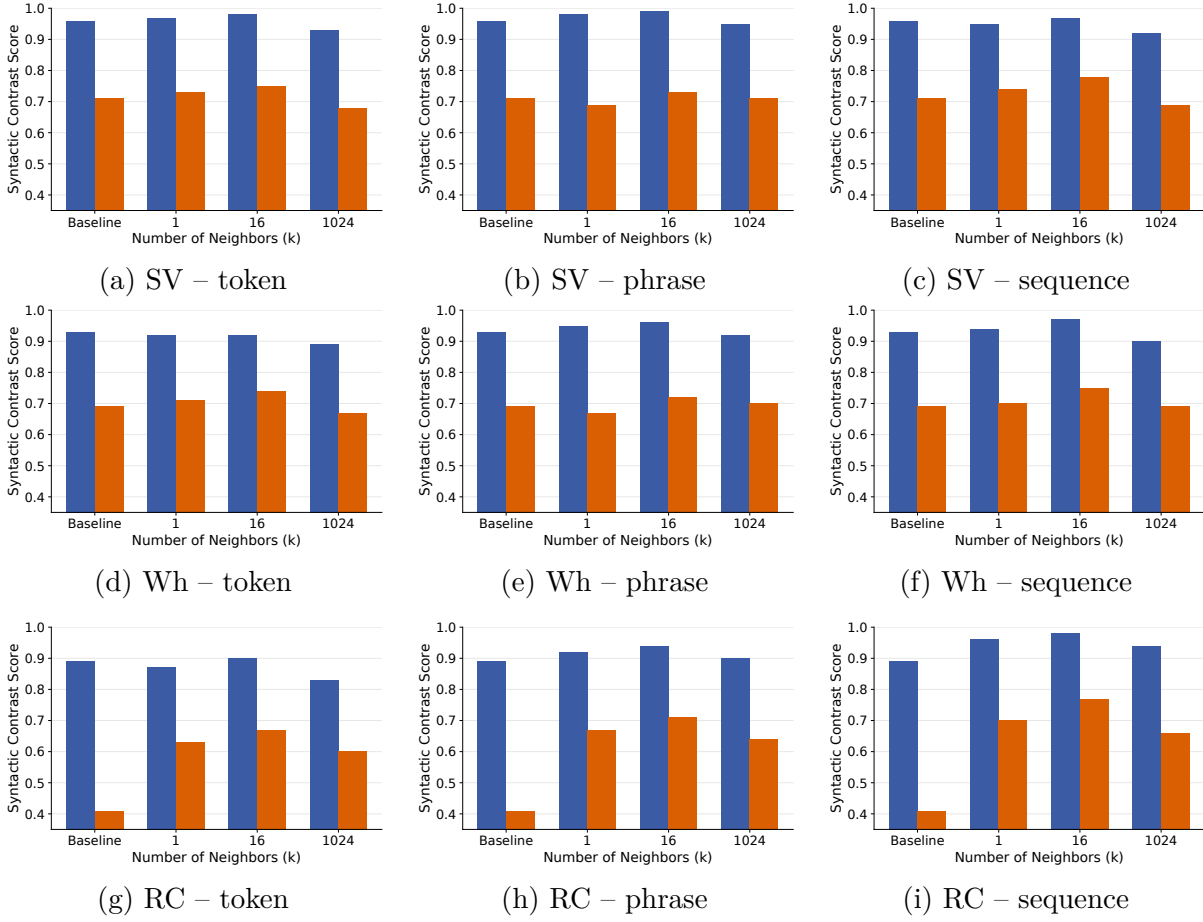


Figure 4.11 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 10$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

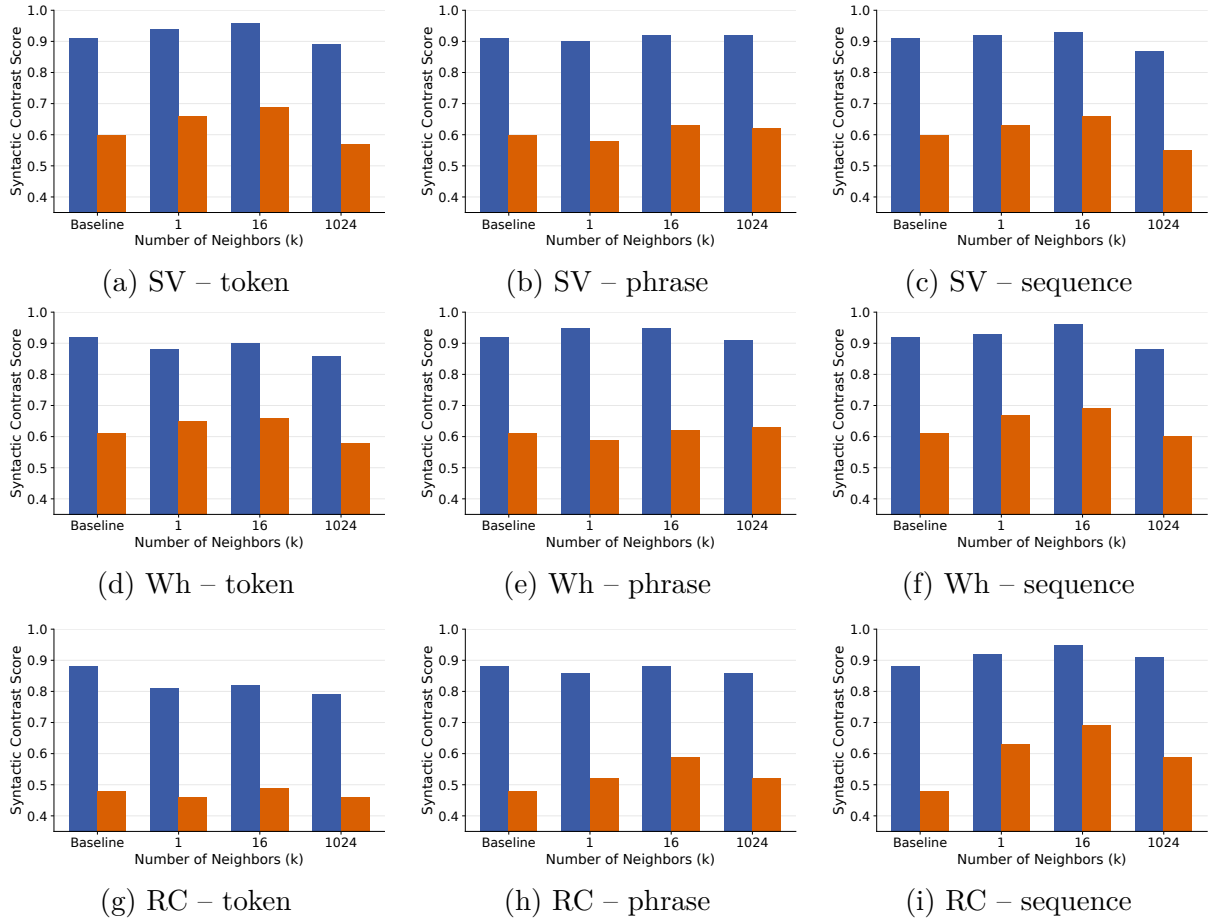


Figure 4.12 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

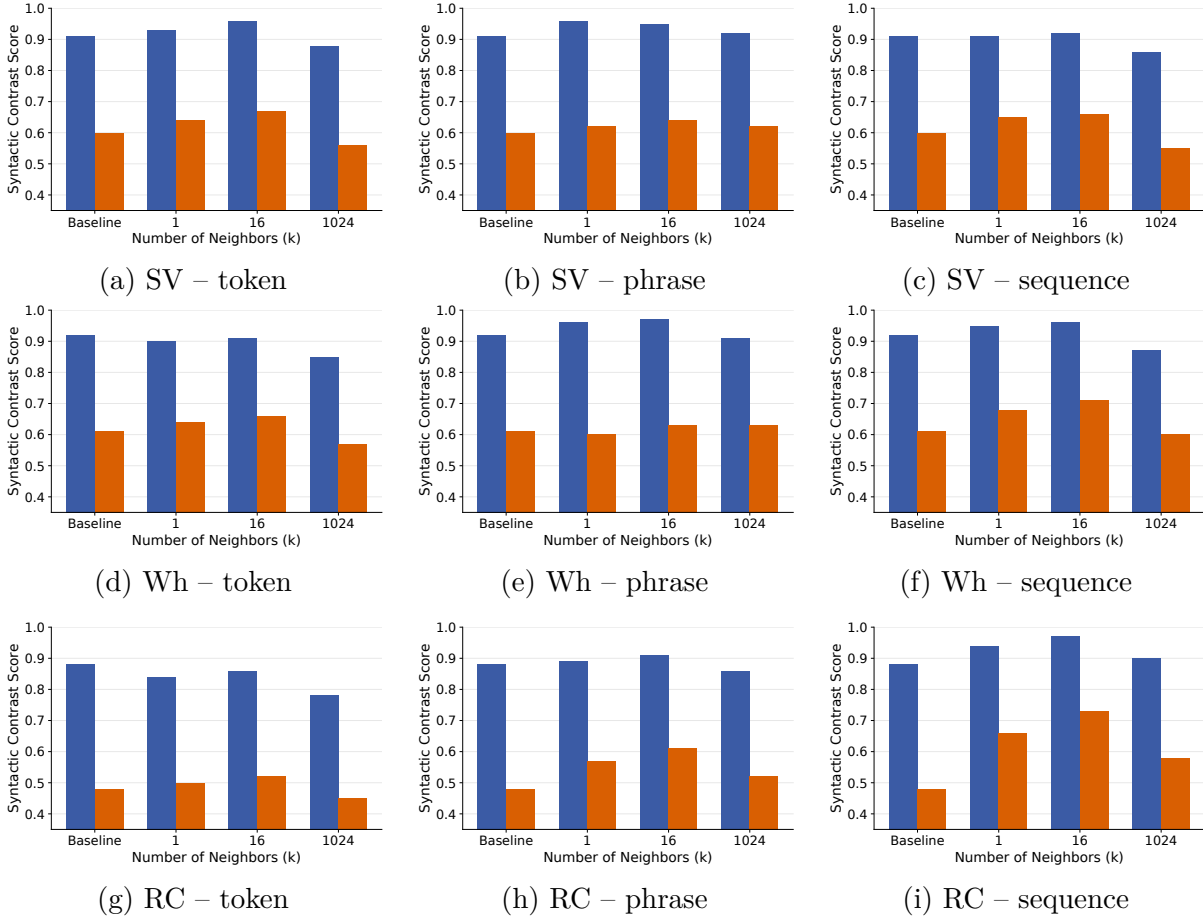


Figure 4.13 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 10$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

Chapter 5

Modeling child-caregiver interactions: benchmarking and reinforcement-learning studies

Objectives

Evaluate language models as simulators of child-caregiver dialogue, then compare how distinct caregiver-derived rewards change generated grammaticality and controlled grammatical preferences in small language models.

5.1 Introduction

5.1.1 Child–Caregiver Interaction in Language Acquisition

Children encounter language in social interactions in which they are not only exposed to caregivers’ utterances, but also contribute utterances of their own. Caregivers respond to these contributions, helping children express their intentions and maintain the conversation. Their responses may include requests for clarification, reformulations, or the reuse of words and structures produced by the child [clark_conversational_2020; fusaroli_caregiver_2023]. Thus, caregiver language provides both linguistic input and responses that are contingent on children’s own behavior [masek_where_2021].

This interactive context raises important questions for language acquisition. Caregiver responses may provide information about whether an utterance was understood and may offer opportunities for children to adjust their subsequent productions [nikolaus2023communicative]. However, the availability of such responses does not, by itself, establish their role in learning. Feedback may concern the child’s intended meaning, intelligibility, or participation rather than grammatical form specifically. Whether these signals provide useful information, and how a learner could exploit them, remain important questions in accounts of grammatical development [demetras_feedback_1986; marcus_negative_1993]. Computational models offer a way to investigate aspects of this interaction and to test hypotheses about its possible contribution to learning.

5.1.2 Research Questions

The present chapter approaches child–caregiver interaction through two complementary questions. The first concerns the simulation of interaction: *To what extent can language models reproduce the linguistic and conversational properties of child–caregiver dialogue?* Addressing this question requires more than assessing whether individual utterances appear child-like or caregiver-like. It also requires examining how the two participants respond to one another and whether the resulting patterns remain comparable to human interaction over multiple turns. Here, language models are used as simulators, and their outputs are evaluated against observations of naturally occurring dialogue.

The second question concerns the learning effects of feedback: *Which forms of caregiver feedback can support grammatical learning in a language model?* In this case, the aim is to isolate particular signals from the broader interaction and examine their effects within a controlled learning framework. Language models are used as learners whose productions can be modified through feedback-based training. This approach allows us to compare signals that occur together in natural conversations, while distinguishing changes in generated language from changes in grammatical preferences on controlled tests.

These questions are related, but they address different aspects of computational modeling. A model may approximate observable properties of children’s speech without having acquired those properties through a child-like learning process. Conversely, a simplified learning model can be used to test the effects of feedback without reproducing a complete child–caregiver conversation.

We therefore consider interaction simulation and feedback-based learning as distinct uses of language models, rather than successive stages of a single system.

5.1.3 Chapter Overview

The chapter brings together two studies addressing these questions. The first develops a benchmark for evaluating language models that simulate children and caregivers, using naturally occurring conversations from CHILDES as a reference [macwhinney2000childes]. It examines properties at the word, utterance, and dialogue levels, compares single-turn responses with extended multi-turn interactions, and assesses how providing conversational examples changes the models’ behavior. Its contribution is to characterize the extent and limits of the models’ ability to reproduce child–caregiver interaction.

The second study investigates learning from caregiver-derived feedback. It uses reward models trained on feedback signals extracted from natural child–caregiver conversations, rather than feedback generated by the simulated interlocutors of the first study. Small language models pretrained on caregiver input are then fine-tuned using different rewards. The study compares the effects of these rewards on the grammaticality of generated utterances and on grammatical preferences measured through minimal-pair benchmarks. This distinction is central to assessing what the feedback changes: more grammatical production does not necessarily establish the acquisition of grammatical knowledge that generalizes. The two studies thus examine, respectively, the empirical adequacy of interaction simulation and the learning effects of selected signals drawn from that interaction.

5.2 Benchmarking LLMs for mimicking child-caregiver language in interaction

5.2.1 Introduction

While LLMs show remarkable capabilities in generating human-like text and engaging in open-ended dialogues and role play in various contexts [feng2024can; yang2024talk2care], their ability to simulate the specificities of child-caregiver interactions has not been systematically investigated. However, these interactions show distinct linguistic and interactive patterns and require dedicated research.

During their linguistic and communicative development, children show non-conventional (i.e., non-adult-like) patterns, such as word omissions, mispronunciations, semantic errors, and non-standard grammatical constructions [bloom1993transition]. They also show non-conventional conversational behaviors, such as incoherence, non-responsiveness, and atypical turn-taking patterns [ninio1996pragmatic]. These behaviors are most apparent in the early years through primary school, although many persist into adolescence [nippold2016later].

Because of their still immature, non-conventional language use, children depend on caregivers to interpret and clarify their communicative intents, thus facilitating communication. Caregivers employ various *scaffolding* strategies, which offer appropriate support tailored to the child’s

current level of cognitive and communicative development. These include the general use of simplified language (a register named child-directed language) as well as interactive strategies such as recasting, repairing, providing follow-up, and offering feedback [berk1995scaffolding; clark2020conversational; snow1977talking; nikolaus2023communicative; soderstrom2007beyond]. This scaffolding is gradually reduced as the child becomes more proficient and ready for independent language use.

Despite growing interest in applying LLMs to specialized interactive scenarios [feng2024can; yang2024talk2care], their ability to simulate child-caregiver interactions remains underexplored. There is, to the best of our knowledge, no systematic examination of whether LLMs can a) properly simulate early child-like utterances with their known non-conventional properties, b) simulate caregiver-like language and its distinctive properties known as child-directed language, and, more importantly, c) beyond mimicking child or caregiver general linguistic properties in *isolation*, simulate child-caregiver language in *interactions*, meaning that we need to simulate the fact that the caregiver’s language is responsive/contingent on the child’s linguistic quirks, providing tailored feedback and scaffolding. This gap is particularly significant given the potential applications of mimicking this scenario in developmental research and its applications [zhang2024mathemyths; seo2024chacha; rasanen2024age; feng2024can].

What the current study is about

As argued above, benchmarking LLMs for effectively mimicking child-caregiver interaction is a complex, multidimensional task. The current work does not claim to provide a complete solution, but rather offers a starting point—an initial exploration into ways to approach some aspect of this task and the insights we learned that can inform future research.

In particular, it is important to note that a comprehensive benchmarking would require evaluating both the semantic content of interactions (*To what extent was the semantic content of the caregiver-like response appropriate to the child-like utterance?*) and linguistic form (*To what extent was this content linguistically framed in an age-appropriate manner?*). The current work focuses primarily on the latter, using data from spontaneous child-caregiver dialogues as a reference (CHILDES dataset)[macwhinney2000childes].

Specifically, we quantify the extent to which two widely used LLMs (GPT-4o and Llama 3) approximate child-caregiver interactions along *structural* metrics at word, utterance, and dialogue levels. These metrics were selected based on insights from previous research [valentini2023automatic; rasanen2024age; french2024aligning]. This previous research focused on a few isolated aspects of child-directed dialogue. Here, we aimed to provide a more synthetic view, capitalizing on insights from these studies.

Furthermore, a major novelty in this work is that we compared two benchmarking approaches: single- and multi-turn testing. In single-turn testing (the most common approach), models receive a child utterance from the CHILDES dataset and generate a caregiver-like response, or conversely, receive a caregiver’s utterance and generate a child-like response. For multi-turn testing, we observe free interactions, over multiple turns, between a "child"-LLM and a "caregiver"-LLM, prompted to communicate like a child and a caregiver, respectively. The resulting dialogues are

then compared, on average, to actual conversations from CHILDES dataset.

Finally, both benchmarking approaches were implemented under zero-shot and few-shot conditions with two objectives: a) for the zero-shot condition, the goal is to characterize the baseline capabilities of LLMs without specific guidance (revealing their initial biases), and b) for the few-shot condition, the goal is to measure LLMs’ ability to improve (relative to the zero-shot baseline) when provided with examples and, in fact, test the sensitivity of the metrics we used to quantify any such improvement.

5.2.2 Related Work and novelty

Research on LLMs’ linguistic and interactive appropriateness to children is still in its early stages. Several studies have explored specific aspects of the interaction, but a comprehensive assessment of LLMs’ capabilities in simulating child-caregiver interactions is still underexplored.

For example, **valentini2023automatic** focused on vocabulary and showed limitations in LLMs’ ability to pick simple words for a young audience, **rasanen2024age** trained a GPT-2 model on caregiver input, and evaluated its capacity to generate language that is similar to caregivers (child-directed language). However, they focused on simulating caregiver data alone, without accounting for the child’s data, and therefore, missing the *interactive* dynamics that can influence the generation. **french2024aligning** studied GPT-3.5 and Llama2’s ability for linguistic alignment to the interlocutor, showing sub-optimal performance in responding appropriately to child-like utterances.

While each of these studies has contributed valuable insights to specific dimensions of the challenge, our work aims to provide a more integrated approach on, at least, two levels. First, we evaluate *both* the caregiver- and the child-like generation. We explicitly study them in interaction, not merely simulating them in isolation. Second, instead of focusing on one metric or linguistic level, we propose metrics at the word, utterance, and dialogue levels, aiming to provide a more comprehensive evaluation framework for LLMs in this specialized communicative context.

5.2.3 Method

5.2.3.1 Data

We used the CHILDES public dataset [**macwhinney2000childest**] for benchmarking, focusing on 2 to 5 years of age.¹

From this dataset, we selected 40 conversations (approximately 300 turns each) evenly distributed across the target age groups—specifically, i.e., 10 conversations at 2, 3, 4, and 5 years.

To prepare the data for analysis and generation, we restructured these conversations into utterance-response pairs. When consecutive turns came from the same speaker (which is often the case with caregivers), we decomposed these into multiple utterance-response pairs and inserted <SILENCE> tokens to mark the positions where the non-speaking interlocutor did not contribute. This preprocessing step preserved the temporal structure of the interactions while

1. This age range was selected because children younger than 2 typically do not engage in extended dialogues, while older age groups had insufficient sample sizes in the dataset to allow for robust analysis.

creating a format suitable for our analytical framework. The resulting benchmarking dataset comprised 6,600 interaction pairs containing a total of 73,300 word tokens. This corpus was further characterized by an asymmetric distribution between participant types, with 26,300 tokens produced by children and 47,000 tokens produced by caregivers.

5.2.3.2 Models

While numerous LLMs are currently available, testing all of them across multiple experimental conditions would be impractical. Nevertheless, to examine generalizability and ensure that our results are not dependent on the idiosyncrasies of a single model architecture, we systematically compared two state-of-the-art LLMs: an open-source model **Llama 3 (8B)** (The chat-optimized version of Meta’s instruction-tuned large language model, [touvron2023llama]) and a proprietary model; **GPT-4o**, (version 2024-08),² the generative pre-trained transformer from OpenAI’s GPT-4 family [achiam2023gpt]. We selected these models (especially the latest version of ChatGPT) as they are some of the most powerful LLMs (at least at the time this research was carried out) and thus are most likely to be capable of adapting effectively to various roles, including those of children and caregivers.

Fine-tuned model In addition to the LLMs described above, which we adapt to child-caregiver interaction via prompting, we also evaluated a smaller—and computationally more manageable—pre-trained conventional model that we directly fine-tuned on child-caregiver data. Specifically, we fine-tuned the distilled encoder-decoder BlenderBot model [roller2020recipes] on conversation data from the CHILDES corpus.³ This approach allowed us to compare prompt-based adaptation of large, general-purpose models with traditional fine-tuning approaches targeting child-caregiver interaction patterns.

5.2.3.3 Benchmarking

Single-turn testing The single-turn testing evaluated LLM’s ability to generate contextually appropriate responses to individual utterances. For each conversation in our benchmarking dataset, we implemented two complementary procedures: a) **Child-to-caregiver direction**: We extracted each child utterance and used it to prompt the LLMs to generate a caregiver-like response. The actual caregiver response from the CHILDES corpus served as the reference against which we evaluated the LLM-generated output. b) **Caregiver-to-child direction**: Conversely, we extracted each caregiver utterance and used it to prompt the LLMs to generate a child-like response. The actual child response from the corpus served as the reference for evaluation. This single-turn setting allows a direct, controlled comparison between the caregiver-LLMs’ response and the caregiver’s or child’s actual response to the *same* utterance/prompt.

Multi-turn testing While single-turn testing enables controlled comparisons with actual responses, it cannot capture the dynamics of extended dialogue interactions. Therefore, we

2. The latest version at the moment we wrote this paper.

3. See appendix for fine-tuning details.

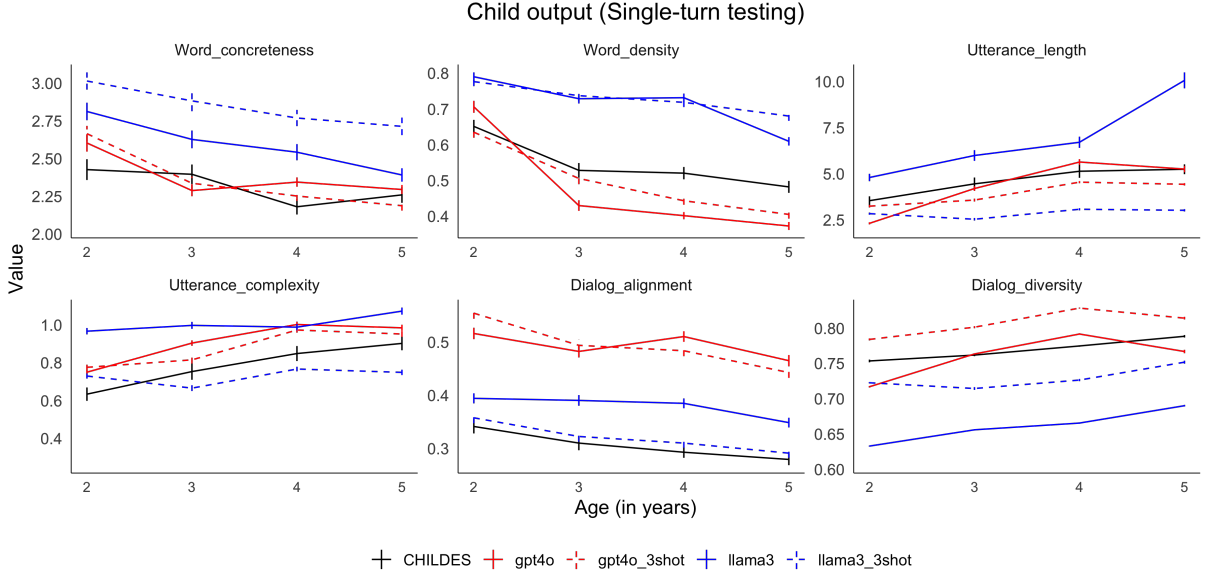


Figure 5.1 – Developmental trajectories of six linguistic/conversational features in children and LLMs. We compare actual human data from the child-caregiver dataset CHILDES vs. LLMs playing a child. Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Points represent averages across all utterances and conversations, and ranges represent 95% confidence intervals.

implemented multi-turn testing protocol to evaluate sustained conversational capabilities. To this end, we simulated complete dialogues using two distinct instances of the same LLM: one prompted to behave as a child (hereafter *child-LLM*) and another prompted to behave as a caregiver (hereafter, *caregiver-LLM*). To initiate these simulated conversations, we used the first utterance from each reference conversation in our dataset as a conversation starter. While the outcome is not as controlled as in the single-turn case, the metrics still allow comparisons to the human reference on average, as will be clear next.

Zero vs. Few-shot settings We tested the initial built-in capabilities and biases of the LLMs in a **zero-shot** setting where LLMs received only the utterance(s) they were expected to respond to, with no additional examples from the CHILDES dataset. In addition, to test our metrics’ ability to capture improvement, we tested the same models again in a **few-shot setting** where the models were given the first three child-caregiver pairs of turns in each conversation, allowing the model to observe examples of both the linguistic structure and interactive dynamics for the same participants.

In both conditions, we maintained consistent and minimal instructional prompts to avoid introducing confounding variables. For example, when prompting for caregiver responses, we used instructions such as: “*You are the parent of a X-year-old English-speaking child. Now, you are having a conversation with your child. Based on the conversation history above, give your response to the child input.*”) (see the prompt templates in Appendix 5.2.7).

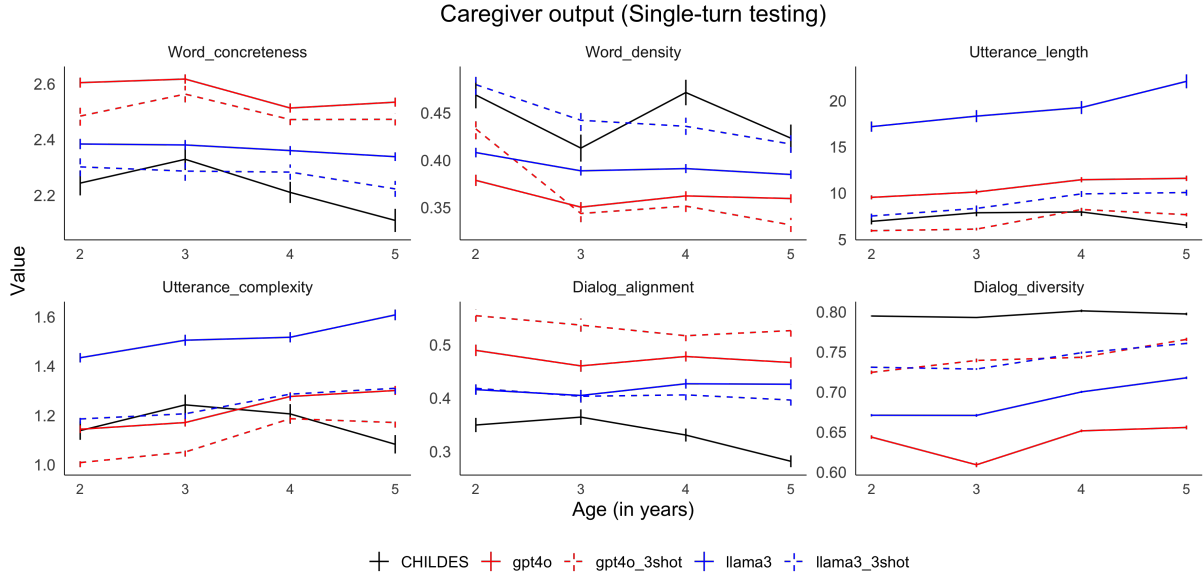


Figure 5.2 – Developmental trajectories of six linguistic/conversational features in caregivers and LLMs. We compare actual human data from the child-caregiver dataset CHILDES vs. LLMs playing a caregiver. Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Points represent averages across all utterances and conversations, and ranges represent 95% confidence intervals.

5.2.3.4 Metrics

We aim to provide a comprehensive benchmarking evaluating key properties at the word-, utterance-, and dialogue levels.

At the *word* level, we followed **dawson2021features** to quantify the *Word concreteness* using human ratings from **brysbaert2014concreteness**. This measure was computed as the average concreteness rating of all content words in each utterance. We also used *Word density*—defined as the proportion of content (vs. function) words in each utterance. This metric reflects the information load and was calculated using the established list of function words by **o2012multi**.

At the **utterance** level, we adopted measures from **rasanen2024age** to capture structural complexity: a) *Utterance length*: the number of words per utterance; b) *Syntactic complexity*: the mean dependency tree depth for each utterance using the Spacy toolkit (v3.7) with its dependency parser based on RoBERTa transformers.⁴ [**liu2008dependency**], where deeper trees indicate more complex structures.

Finally, at the level of **dialogue** dynamics, we distinguish two measures. *Semantic alignment* concerns similarity between interlocutors’ contributions in an exchange pair [**duran2019align**; **french2024align**; **misiek2020development**], represented using sentence embeddings [**reimers-2019-senten**]. *Semantic diversity* concerns variation among a single speaker’s contributions across the conversation. Following **guo2023curious**, this within-speaker measure is the average pairwise cosine distance (1 minus cosine similarity) between utterance embeddings. Higher diversity therefore

4. The dependency parsing implementation provides a structural representation of syntactic relationships between words in an utterance.

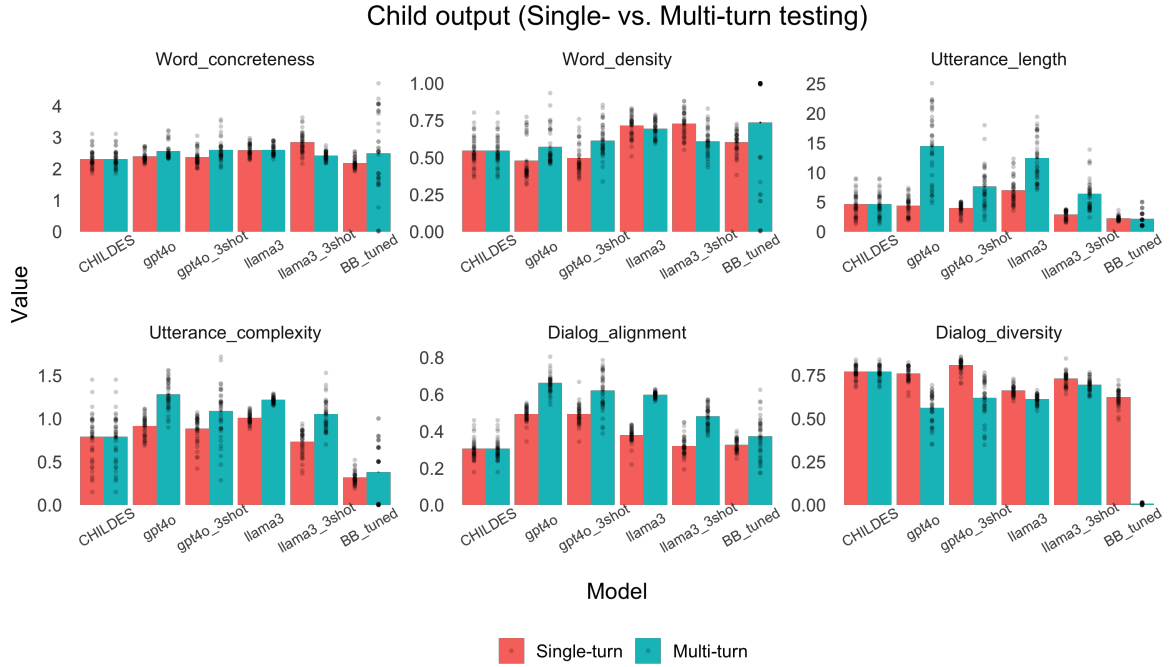


Figure 5.3 – Comparison of linguistic/conversational features across the single-turn and multi-turn testing for children. The bars show the averages across age groups. The points show variability across conversations (40 in total). Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. “BB_tuned” which stands for the BlenderBot model fine-tuned on a subset of CHILDES. Finally, for CHILDES, the data for single-turn and multi-turn is the same.

indicates greater dispersion among that speaker’s contributions, rather than greater alignment with the interlocutor.

5.2.4 Results and Discussion

5.2.4.1 Single-turn testing

Figure 5.1 and Figure 5.2 show the results of single-turn testing for child-LLMs and caregiver-LLMs across age groups.

Child-LLM The age-conditioned child-LLMs recover several trends in the CHILDES reference, including under zero-shot prompting without conversational demonstrations. This evaluates the simulation of age-associated patterns by already pretrained models; it does not show that those models acquired language through the corresponding developmental stages.

Specifically, the models capture decreasing concreteness and lexical density over time (children use more abstract words and function words as they grow older), increasing utterance length and syntactic complexity, and decreasing dialog alignment alongside increasing dialog diversity (as children become able to contribute new information instead of just repeating caregivers’ inputs).

When comparing model performance, GPT-4o more accurately mimicked children than Llama 3, achieving closer alignment with CHILDES reference values across most metrics in the zero-shot condition. The introduction of few-shot examples (three interaction pairs) primarily benefited

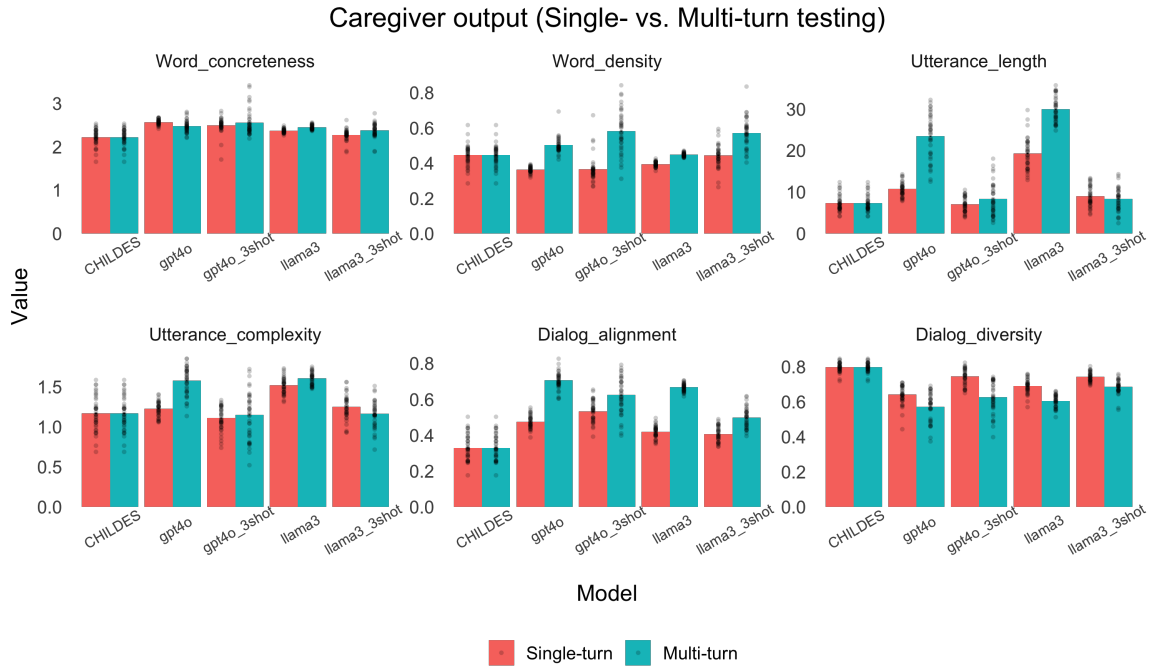


Figure 5.4 – Comparison of linguistic/conversational features across the single-turn and multi-turn testing for caregivers. The bars show the averages across age groups. The points show variability across conversations (40 in total). Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Finally, for CHILDES, the data for single-turn and multi-turn is the same.

Llama 3, improving its alignment with CHILDES across several measures, though not consistently across all metrics. Despite these improvements, GPT-4o maintained better overall performance following few-shot prompting.

Caregiver-LLM In simulating caregiver language, zero-shot GPT-4o more closely approximated CHILDES values for utterance-level measures, while Llama 3 achieved better alignment with word-level and dialogue-level metrics. The few-shot prompting condition improved both models’ performance, with Llama 3 ultimately achieving better overall alignment with CHILDES reference values than GPT-4o after exposure to exemplars. Notably, both models continued to diverge from human data at the dialogue level, exhibiting higher semantic alignment and lower semantic diversity than actual caregivers in the CHILDES corpus. Overall, there were no clear developmental patterns in CHILDES caregivers’ behaviors.

5.2.4.2 Multi-turn testing

Figure 5.3 and 5.4 show the results of multi-turn testing. For easier comparison with single-turn, we show the results side-by-side, averaged over age. Here, we made a caregiver-LLM interact with a child-LLM (using two instances of the same LLM).⁵

5. We also crossed models in child-LLM vs. caregiver-LLM (e.g., child-llama 3 interacting with caregiver-GPT-4o), but observed no noticeable changes.

Child-LLM In zero-shot settings, we observed marked differences between single-turn and multi-turn testing across multiple metrics. Most notably, utterance length, syntactic complexity, and semantic alignment all increased considerably, becoming less comparable to children in CHILDES.

After few-shot prompting, we observed minor to moderate improvements, such as reductions in utterance length⁶ and (slight) decreases in both syntactic complexity and semantic alignment.⁷

Caregiver-LLM Multi-turn testing resulted in a general increase across most metrics compared to single-turn testing (with exceptions in dialog diversity, which decreased, and concreteness, which remained constant). These changes made the multi-turn behavior generally less comparable to human data in CHILDES. Under few-shot prompting, we observed a significant improvement in multi-turn behavior, especially in terms of length and complexity, which became much more comparable to CHILDES. However, we observed only moderate improvement in dialog-level measures.

While LLMs successfully approximated human references for word-level and sentence-level properties under certain conditions, we identified systematic discrepancies in interactive measures for caregiver-LLMs across all experimental configurations. As illustrated in Figure 5.4, in both dialog alignment and diversity, LLMs exhibited higher alignment and lower diversity than caregiver data in CHILDES across LLM type (GPT-4o LLama 3), prompting strategy (zero- vs few-shot) and benchmarking approach (Single- vs. multi-turn).

To verify this qualitative observation statistically, we ran, for each of the two interactive measures (Dialog_alignment and Dialog_diversity), linear regressions comparing models’ output to the CHILDES reference, testing all configurations: 2 LLMs x 2 prompting strategies x 2 benchmarking approach. All 16 comparison models (8 for each measure) revealed highly statistically significant differences between LLM-generated and human caregiver language, suggesting that interactive caregiver properties are potentially more challenging to mimic for LLM.

5.2.5 Conclusion

This paper presents a preliminary exploration into ways we can benchmark LLMs’ ability to simulate child-caregiver interactive dynamics. While previous research has typically focused on measuring the properties of a specific structure/level (e.g., words or utterances) and/or on evaluating language from one part of the dialogue outside the interactive context (mainly focusing on the caregiver), our novel contribution is in three key aspects: a) we evaluate child-caregiver language generation in *interaction*, b) we tested a more comprehensive set of measures covering

6. Note that length could not be reduced further, even when we experimented with explicitly setting an upper bound on children’s utterance in our instructions to the LLM.

7. In addition to using LLMs instructed to play a child, we also used model fine-tuning. To this end, we considered a pre-trained encoder-decoder model- Blenderbot fine-tuned on a subset of caregiver(encoder)-child(decoder) dialogues in CHILDES. We also explored an alternative approach using model fine-tuning rather than prompting. Specifically, we fine-tuned a pre-trained encoder-decoder Blenderbot model on a subset of caregiver (encoder) to child (decoder) dialogues from CHILDES. However, while showing reasonable performance in single-turn testing, this fine-tuned model was erratic and highly repetitive in the multi-turn interactions, also making it unsuitable for evaluating caregiver-LLMs.

the word, utterance, and dialog levels, and c) we compared two benchmarking approaches, evaluating the models in short and extended settings.

We put this benchmarking framework to use, comparing two powerful LLMs: GPT-4o and Llama 3. Indeed, such a comparison was essential to distinguish between findings that are likely generalizable and those that are specific to a particular model. Our evaluation incorporated both single- and multi-turn testing. The former is more controlled, allowing a direct evaluation of the LLMs' response using human references, while the latter, though less controlled, allowed measuring LLMs' behavior in more extended conversations. Interestingly, we found that single-turn evaluations, while insightful, were not totally correlated with the LLMs' behavior in an extended conversation. Indeed, the multi-turn analyses showed an increased divergence of LLMs from human data, particularly in utterance-level properties and discourse dynamics (although both benchmarkings led to qualitatively similar conclusions regarding this linguistic level). This quantitative mismatch highlights the importance of dynamic testing of LLMs, since static testing alone may fail to capture the cumulative effects of sustained interactions—a consideration particularly relevant for applications involving extended communicative exchanges.

Our benchmarking effort aims not only at comparing different LLMs, but also at providing a quantitative tool that can be sensitive to incremental improvement *within* the same models. In particular, comparing zero- and few-shot prompting, the benchmark identified areas where this intervention was more or less successful: Both GPT-4o and Llama 3 matched several caregiver patterns more closely under the few-shot condition, which supplied three interactive examples together with output-length instructions. The comparison demonstrates sensitivity to this combined prompting intervention; it does not isolate the contribution of examples or establish lasting adaptation.

However, the impact was consistently less pronounced for dialogue-level properties, where models (whether in single- or multi-turn settings) continued to exhibit higher alignment and lower diversity compared to human data. This pattern suggests that while surface-level linguistic features (such as word and sentence properties) appear relatively straightforward to adapt to, capturing the interactive nature of child-caregiver communication is more challenging (see also Limitations).

In conclusion, we explored a multi-level benchmarking approach for assessing LLMs' ability to mimic child-caregiver language in interactions, and we showed its usefulness in tracking incremental improvements. We found that single-turn testing of the LLMs, as typical in most benchmarks, was not totally indicative of the LLMs' real behavior in extended conversational contexts, thus emphasizing the need for more dynamic, multi-turn testing in this line of work. Furthermore, few-shot prompting was effective in bringing the LLMs closer to caregivers' data, especially regarding word- and utterance-level properties. It was not as effective on the dialog-level properties. The LLMs exaggerated alignment and showed reduced diversity compared to CHILDES.

5.2.6 Limitations

The supplied few-shot templates combine conversational demonstrations with explicit output-length instructions (Appendix A). Differences from zero-shot prompting therefore reflect this combined change, and cannot isolate the effect of examples on length or complexity.

While our explorative work has led to some initial insights, it also has several limitations and raises questions for future research.

A primary constraint concerns our selection of evaluation metrics. We relied on established measures from previous research to assess LLMs at word, sentence, and dialogue levels. For children’s language production, these metrics successfully captured developmental trajectories, enabling age-specific benchmarking of LLMs (Figure 1). However, when applied to caregiver language, these same metrics failed to reveal clear developmental patterns (Figure 2), thereby limiting the precision of our assessment. This asymmetry highlights the need for developing more refined metrics that can better capture the subtle adaptations in caregiver speech across different stages of child development.

While evaluating LLMs in the single-turn scenario—comparing answers to a reference—is rather straightforward and aligns with the way LLMs are typically benchmarked, a multi-turn approach is also necessary in our context since the ultimate goal behind benchmarking is for these models to be used in an extended interactive context. However, the multi-turn approach is inherently less controlled as it involves two instances of LLMs interacting, each playing the role of an interlocutor. However, since neither instance perfectly models the interlocutor, extended interaction can amplify artifacts, leading the systems to adapt to each other’s quirks rather than approximating genuine child–caregiver interaction. Thus, a multi-turn benchmarking approach can under-estimate the models’ true capabilities to interact with an actual human (child or caregiver). That said, the fact that multi-turn testing was, at least qualitatively, in agreement with single-turn testing (e.g., both of them point to LLMs’ ability to adapt more easily to properties of words and sentences and less easily to properties of the dialog) is a testimony to the multi-turn’s potential as a valid method of evaluation in this context, though, of course, future research is needed to thoroughly investigate this question.

Finally, we reported that few-shot learning improved caregiver-LLM performance in terms of utterance length and syntactic complexity, but had less impact on alignment and diversity. While this finding was, overall, robust across configurations in our setup, it should be taken with a grain of salt, given that we did not systematically test it in a comprehensive set of experiments. Indeed, our main goal in this paper, as we stated above, was not to improve the LLMs’ performance, but to demonstrate, in a simple case, the ability of the benchmarking metrics to identify gaps and track improvement. For example, it is possible that a larger, more systematic exploration of the prompting strategies (which is computationally expensive, and thus, could not be done here in combination with the other experiments we did) could yield improvements across the dialog dimensions as well.

5.2.7 Ethics statement

All data used in this study is already publicly available. This work focuses on model benchmarking and improvement using offline child-caregiver data and internal simulations, aiming to advance fundamental research in this area. We do not consider this testing sufficient for deployment; any future real-life applications should undergo rigorous validation in child-safe environments with appropriate human oversight, such as by teachers or parents.

Appendix A: Prompt templates

We provide prompt templates in different conditions. These are templates for the multi-turn testing. The templates for the single-turn testing are the same, except there is no conversation history.

The zero-shot prompt template for the caregiver [Conversation history ADULT: <Utterance>, CHI: <Utterance>...] You are the parent of a <Month>-month-old English-speaking child. Now, you are having a conversation with your child. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Based on the given conversation history above, give your response to parent input as ADULT. Do not output the speaker label.

The zero-shot prompt template for the child [Conversation history CHI: <Utterance>, ADULT: <Utterance>...] You are a <Month>-month-old English-speaking child. Now, you are having a conversation with your parent. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Based on the given conversation history above, give your response to parent input as CHI. Do not output the speaker label.

The few-shot prompt template for the caregiver [Conversation history ADULT: <Utterance>, CHI: <Utterance>...] You are the parent of a <Month>-month-old English-speaking child. Now, you are having a conversation with your child. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Ensure your response is no longer than 50 words regardless of the prompt. Here are some example interactions: CHI: <Utterance>, ADULT: <Utterance>... Follow the example interactions. Based on the given conversation history above, give your response to parent input as ADULT. Do not output the speaker label.

The few-shot prompt template for the child [Conversation history CHI: <Utterance>, ADULT: <Utterance>...] You are a <Month>-month-old English-speaking child. Now, you are having a conversation with your parent. <SILENCE> indicates silence in the previous turn; <UNINTELLIGIBLE> indicates unintelligible speech. Ensure your response is no longer than 6 words regardless of the prompt. Here are some example interactions: CHI: <Utterance>, ADULT: <Utterance>... Follow the example interactions. Based on the given conversation history above, give your response to parent input as CHI. Do not output the speaker label.

Appendix B: Finetuning details

We fine-tuned the Blenderbot model, a 400M parameter architecture comprising: a retriever for dialogue history; a Seq2Seq generator with 2 encoder layers, 24 decoder layers, 2560-dimensional

embeddings, and 32 attention heads; retrieve-and-refine architectures combining dialogue and knowledge retrieval. The parameters were optimized through preliminary experiments, resulting in a batch size of 16 and a learning rate of 0.0001 maintained over 4 epochs. The fine-tuning data consisted of 2.6M interaction pairs with 15.8M total word tokens (4.4M child tokens and 11.4M adult tokens) restructured from CHILDES dataset [macwhinney2000childes].

Appendix C: Example generations

We present sample generations in Table 5.1. It shows child-caregiver conversations (CHILDES corpus) and GPT-4o generations in a multi-turn setting where the child and caregiver are two instances of GPT-4o. All conversations start from the same child turn (here, all conversations start from a child outputting “silence”). To extract features, we preprocessed all samples by removing non-verbal behavioral annotations (e.g., *giggles*) and lower-casing words to focus exclusively on verbal content.

Table 5.1 – Comparison of real and LLM-generated child-caregiver dialogue exchanges. The LLM generation here is from a multi-turn setting, where two instances of GPT-4o (i.e., a child-LLM and a caregiver-LLM) interact. The table shows three dialogue samples: (1) a real interaction from the CHILDES corpus, (2) a zero-shot generation by GPT-4o given only the initial turn, and (3) a few-shot generation by GPT-4o after being shown three previous examples in the child-caregiver interaction.

Source	Role	Utterance
CHILDES (Human)	C	<i>SILENCE</i>
	A	wanna play catch
	C	let’s play catch
	A	okay almost
	C	yeah
	A	try it again put your hands in like this
	C	almost
	A	you wanna hit it
GPT-4o (Zero-shot)	C	<i>SILENCE</i>
	A	I see you’re still thinking about cherries. It’s okay to feel upset. Would you like a hug while we figure out something else yummy to eat?
	C	<i>SILENCE</i>
	A	I know you’re feeling quiet right now. I’m here when you’re ready. Would you like to hold the banana and see if it feels fun to peel?
	C	Banana! Peel it?
	A	Yes, let’s peel the banana! Can you help me pull the peel down? Then we can eat it together. Yum!
	C	Banana! Peel! Eat!
	A	Yes! Let’s peel the banana together. Peel, peel! Now take a bite—yum! Is it tasty?
GPT-4o (Few-shot)	C	<i>SILENCE</i>
	A	Do you love your big red truck?
	C	Big red truck! My truck!
	A	Is it your big red truck? Where will you drive it?
	C	Drive to park! Big red truck!
	A	Are we driving your big red truck to the park? That sounds so fun!
	C	My truck drive park!
	A	Yes, your big red truck! We can drive it to the park!

Notes: A = Adult, C = Child. All model-generated responses were obtained using default sampling parameters (temperature = 1.0). *SILENCE* marks an empty turn. Human dialogues are from the CHILDES corpus [macwhinney2000chldes].

5.3 Which forms of caregiver feedback support grammar learning?

5.3.1 Introduction

Children do not learn language from exposure to input alone, but also through social interaction in which caregivers respond to their produced utterances [fusaroli_caregiver_2023; masek_where_2021; clark_conversational_2020]. A long-standing question in language acquisition concerns the role of such feedback in shaping grammatical development, especially if caregiver responses can help the learner know when their productions were well- or ill-formed [sokolov_changing_1994].

Although caregivers rarely provide explicit correction of grammatical errors [brown_derivational_1970], researchers have proposed that they nevertheless provide global, noisy feedback from which children can infer whether their utterances were likely grammatical or ungrammatical [hirsh-pasek_brown_1984; demetras_feedback_1986; sokolov_changing_1994]. In this view, the child can be understood as learning in a reinforcement-like fashion, using the valence of the feedback as positive or negative rewards that guide which productions to maintain or adjust in future interactions [whitehurst_what_1988; nikolaus_communicative_2023; schoneberger_three_2010].

The developmental literature has documented several forms of caregiver response that can be interpreted, within a reinforcement-learning framework, as valenced feedback. The most direct examples are *communicative feedback* cues, namely a) acknowledgments (e.g., "okay", "yeah"), which signal communicative success and thus increase the likelihood that the child's sentence was well-formed, and b) clarification requests (e.g., "huh?", "what?"), which signal communicative failure or misunderstanding and thus increase the likelihood that the sentence was ill-formed [clark_conversational_2020; demetras_feedback_1986; newport_mother_1977; saxton_prompt_2005; nikolaus_communicative_2023a; nikolaus_communicative_2023b].

A second relevant class is *semantic contingency*, or the extent to which caregiver responses remain responsive to the child's topic and intended meaning [hoff-ginsberg_topic_1987; che2018assessing; hirsh2015contribution; agrawal_automatic_2024]. In the present model, this degree of contingency defines a reward target. Low contingency may reflect semantic or conversational misalignment, but does not by itself identify a grammatical error in the child's utterance.

There are other forms of caregiver feedback that also provide such valenced signals, even if they have not traditionally been associated with this learning framework. Perhaps most directly relevant to grammar learning is *structural alignment* [fusaroli_caregiver_2023; dale_unraveling_2006; fernandez_quantifying_2014; misiek_development_2020], whereby caregivers respond by reusing the child's syntactic frame while potentially introducing different lexical content (e.g., child: "I eat" (Pronoun-Verb) → caregiver: "Mommy eats too"). Here, the feedback valence is reflected in whether such alignment occurs, signaling support for the child's use of that syntactic frame.

A final type of feedback is *affective feedback*, such as whether caregivers respond with positive affect (e.g., "that's amazing!") versus a more emotionally neutral response [rivero2023relations].

These cues are far more available to children than explicit grammatical corrections, making their role in language learning much more plausible [demetras_feedback_1986; hirsh-pasek_brown_1997]. Yet, they provide only indirect and noisy learning signals. First, they are not always directed at grammar specifically. Second, even in cases when grammar is the main target, the signal is global (i.e., utterance-level) and does not identify the specific error within the utterance. This makes *credit assignment* a central challenge [marcus_negative_1993; pinker_learnability_1989; saxton_contrast_1997], and also makes interactive learning from caregiver feedback an interesting *empirical* case.

Nevertheless, little is known about how these different feedback types contribute to children’s grammatical learning. While corpus studies allow a quantitative account of caregivers’ feedback behavior, *the causal effect* of each type of feedback or reward on learning remains difficult to establish. A common methodological issue facing previous attempts [saxton_negative_2005] is that different types of feedback are provided to the same child, making it difficult—and ethically problematic—to isolate their effect over the developmental timescales on which language is learned.

Recent computational work has begun to revisit these questions by using LMs as controlled learning environments for testing hypotheses about language learning, thereby combining the targeted precision of experimental methods with the ecological validity of corpus studies [misra2024language]. Particularly relevant to our question are studies that used Reinforcement Learning to investigate LMs’ interactive learning strategies [ma_babysit_2024; martins-et-al-2025-upon; zhao-et-al-2023-babystories; padovani2025dialogue; xu2025wanting; nikolaus_modeling_2021].

Recent work has applied this approach to *natural* child-caregiver interactions, using Reward Models trained from caregivers’ actual feedback data [nikolaus_fourthassi_2026]. Language models are then fine-tuned using techniques inspired by Reinforcement Learning from Human Feedback [ouyang_training_2022]. This work showed that caregiver-like clarification requests on model generation can signal negative evidence. Learning from this signal improved grammaticality.

The current study

Existing RLHF-based studies of naturally occurring caregiver feedback have focused primarily on clarification requests, leaving open whether other forms of social feedback have similar, weaker, or even negative effects on grammatical learning. The present study addresses this gap by comparing several theoretically motivated feedback types within the same controlled learning framework.

5.3.2 Methods

Our method uses a two-stage learning setup designed to separate learning from linguistic exposure from learning driven by caregiver-feedback rewards. In a first step, a small causal language model is pretrained on child-directed language (excluding children’s production),

providing an *input-only* baseline that quantifies learning gains from linguistic exposure. In a second step, the model is fine-tuned via RL, using a reward model trained on the target caregiver’s feedback valence.

Pre-training baselines We use a customized version of GPT-2 (2 layers, hidden size 512, 8 attention heads), pre-trained on CHILDES [macwhinney2000chldes], using three pre-training scales (0.1M, 1M, and 10M words, see Appendix 4).

Empirical Reward Models For each feedback type, we trained reward models to capture naturally occurring patterns of caregiver feedback in child–caregiver interactions (using CHILDES dataset). Each reward model is trained to map children’s utterances to a scalar, representing the valence of the target feedback (see Appendix 4). The valence values themselves were annotated prior to reward-model training using a range of automatic annotation tools (see below, and Appendix 3).

Topline Reward Model: we use a grammar-targeted reward as a reference condition for what direct grammatical supervision can achieve in this framework. It is trained on labeled BLiMP and Zorro materials (Appendix 4), rather than on observed caregiver feedback. It is therefore not a formal upper bound or an independent test of developmental sufficiency.

RL fine-tuning For each feedback/reward, the pre-trained models were fine-tuned using the corresponding reward models, using Proximal Policy Optimization [schulman_proximal_2017]. We included several safeguards to mitigate potential language drift (see Appendix 5).

Training seeds and variability To account for stochastic variability, we used, for each training scale, a fully crossed $3 \times 3 \times 3$ design over pretraining seeds, fine-tuning seeds, and generation seeds.

Annotation of the reward signals We considered the four classes of caregiver feedback mentioned in the introduction: communicative feedback, semantic contingency, structural alignment and affective feedback. For each type of feedback, we annotated the CHILDES child-caregiver conversations for the corresponding valence (See Appendix 3).

5.3.2.1 Evaluation

We use two complementary evaluations that capture different aspects of grammatical behavior.

Minimal-pair benchmarks: we use Blimp [warstadt2020blimp] and its CHILDES-adapted version Zorro [huebner_babyberta_2021]. They test whether a model assigns higher probability to a grammatical sentence than to a minimally differing ungrammatical one.

Grammaticality of generated utterances Free generations were scored using Child Conversational Grammaticality [nikolaus_automatic_2024], a model adapted to CHILDES and its conversational context. In addition, we used a measure based on an off-the-shelf Grammatical Error Correction model [rothe_simple_2021]. For each run and generation seed, we sampled up to 10,000 utterances using the Beginning-Of-Sentence (BOS) token as a prompt.

Both evaluation methods are broadly connected to child-language methodologies for evaluating grammatical development, which have included both controlled forced-choice tasks and analyses of children’s spontaneous and elicited productions [ambridge2011child].

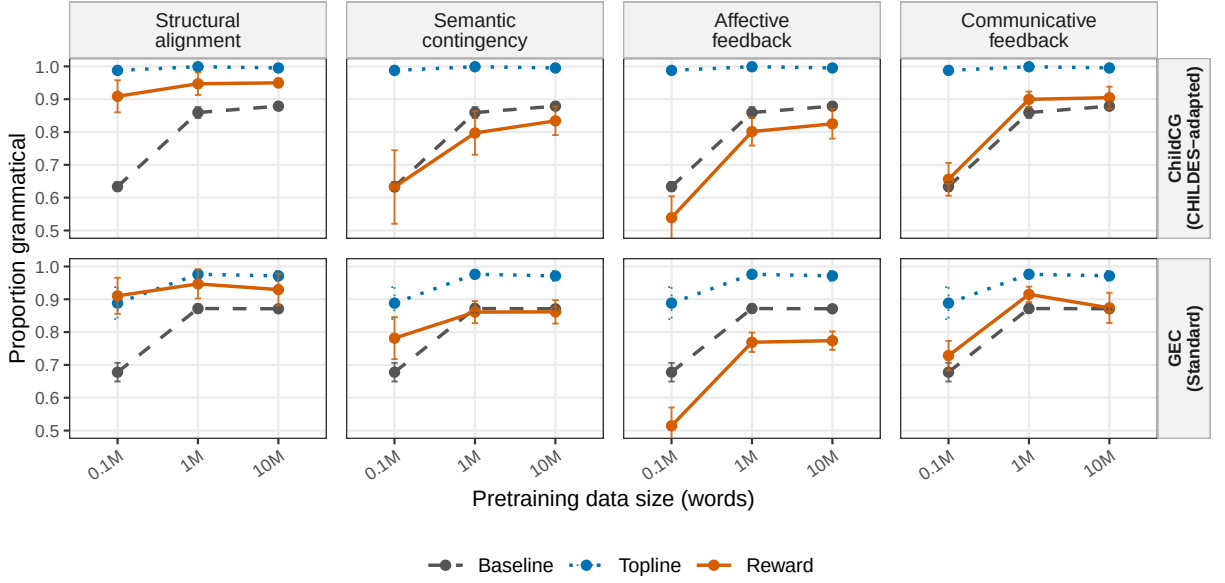


Figure 5.5 – Generated-utterance grammaticality across pretraining scales for each reward. The top row shows results from the CHILDES-adapted ChildCG classifier, while the bottom row shows results using a standard GEC-based metric. Within each panel, baseline and topline trajectories are repeated to make the reward trajectory directly comparable. Means are computed over training seeds units after averaging over generation seeds. Error bars show ± 1 standard deviation across training seeds.

5.3.3 Results

First, we compared the input-only baseline models with the same models fine-tuned using the grammar-targeted reference reward (Figure 5.6 in Appendix). This condition improves minimal-pair performance and generated grammaticality relative to pretraining alone. Because its reward model is trained on BLiMP and Zorro materials, the benchmark gains demonstrate the effect of direct grammar-targeted supervision within this setup; independent generalization would require documented separation of reward-training and evaluation items.

Moving to the empirical reward models, we found no consistent improvement on the minimal-pair benchmarks (Figure 5.7 in Appendix). Across Zorro and BLiMP, none of the reward types reliably improved over the baseline. This echoes a broader pattern in recent studies showing that interactive strategies struggle to improve performance on these benchmarks (see also Limitations). In contrast, the generation results showed measurable effects on grammaticality (Figure 5.5). Across both ChildCG and GEC, the overall trends were largely consistent. Structural alignment produced the strongest and most consistent improvement across scales and metrics. Communicative feedback showed an overall positive effect, especially under ChildCG, but the gains were smaller. In contrast, semantic contingency and affective feedback generally reduced grammaticality below the baseline.

Regarding the effect of pre-training size, the low-resource condition at 0.1M was visibly noisier, with larger variability across runs, excessively high non-word rates (Figure 5.8 in Appendix, bottom row), and overall low grammaticality scores. The effects became much more consistent

Reward family	1M	10M
Structural alignment	+.077***	+.060***
Semantic contingency	−.092***	−.061***
Affective feedback	−.038*	−.031*
Communicative feedback	+.033*	+.024†
Acknowledgement	+.017	+.017
Clarification request	+.050***	+.031*

Table 5.2 – Model-estimated change in grammaticality probability relative to baseline, using the ChildCG metric at the 1M and 10M pretraining scales. † $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

from 1M words onward, suggesting that a minimum amount of pre-training data is needed before reward-specific effects can emerge robustly.

We tested these qualitative patterns statistically using a mixed-effect logistic model. Predictors included reward type, pretraining scale, and their interaction. We additionally added the following predictors as controls a) utterance length, since longer utterances may naturally provide more opportunities for errors, and b) rate of non-words generated, since out-of-dictionary forms may be penalized by grammaticality classifiers for reasons that are not strictly grammatical. Finally, we included a random intercept for generation run to account for the fact that sampled utterances from the same seeds are not independent.

Table 5.2 reports model-estimated changes in grammaticality probability relative to the baseline, using the more robust pretraining scales (1M and 10M). Mirroring the pattern in Figure 5.5, structural alignment showed the clearest improvement. In contrast, semantic contingency and affective feedback both reduced grammaticality. Communicative feedback improved grammaticality at 1M and showed a marginally significant positive effect at 10M. When communicative feedback was split into its two components, only clarification requests showed a significant effect, whereas the acknowledgment-based reward model did not.

5.3.4 Discussion and Conclusions

We investigated which forms of caregiver feedback support LMs’ grammaticality within an RLHF-inspired framework. We isolated and compared major feedback types, providing insight into their specific effects and relative contributions.

The main finding was that structural-alignment rewards systematically improve generated grammaticality. Prior work reports associations between caregiver alignment and grammatical development [fusaroli_caregiver_2023]. The present intervention adds a computational result: optimizing a reward learned from caregiver reuse of structure shifts a model’s production toward utterances judged more grammatical, including after the reported controls for utterance length and non-word rate. This supports a candidate learning route through reinforcement of reusable structural patterns. It does not establish that the same causal pathway operates in children or that the observed production gains constitute broad grammatical generalization. Communicative Feedback also improved grammaticality, though to a lesser extent. The improvement is consistent with prior work, especially on clarification requests [nikolaus_fourtassi_2026;

clark_conversational_2020].

As for Semantic Contingency and Affective Feedback, they *reduced* grammaticality. We did not anticipate this outcome; however, it can be understood in the broader context of development. Based on the supplementary analyses in Figure 5.8, and models’ generation samples in Table 5.3 (Appendix), Affective Feedback increased utterance length, suggesting a role in eliciting longer vocalization [**goldstein2003social**]. Semantic Contingency increased the use of content words and preserved greater lexical diversity, in line with findings in lexical development [**che2018assessing**].

To conclude, different types of feedback push generation in different directions: some improve grammaticality, while others encourage other properties of language learning and use. Future work should adopt a broader framework, investigating how these feedback types may support *complementary* aspects of language learning beyond grammar.

Limitations

A limitation of our study is that the effects of the empirical rewards were observed in model generation, but not in minimal-pair benchmarks. This echoes a broader pattern in recent studies showing that interactive strategies generally struggle to improve performance on minimal-pair benchmarks of grammar such as BLIMP and Zorro [**zhao-etal-2023-babystories**; **padovani2025dialogue**; **nikolaus_fourthassi_2026**; **stopler2025towards**].

This raises an important theoretical question: To what extent does grammaticality in generation reflect grammatical knowledge? This remains a debated question in child language research [**ambridge2011child**; **chater2018language**]; and we do not aim to address this general question here.

In the specific case of the present study, however, it is cautious to interpret the generation-based results more narrowly: they test how reward fine-tuning changes the model’s productive behavior; they do not directly test whether the model has acquired grammatical knowledge that can generalize. Note that even under this conservative interpretation, the generation-based results remain informative for theories of language development. Unlike traditional corpus-level correlations, the rewards here were used as interventions within a computational learning system, allowing us to test whether optimizing for these signals *causally* shifts the model’s production policy in grammar-relevant directions.

Nevertheless, a model that becomes more grammatical in free generation does not necessarily mean it has learned grammatical regularities that generalize beyond the specific utterance types favored during fine-tuning. Minimal-pair benchmarks—though not perfect [**martinez2023evaluating**]*—*are useful precisely because they intend to test this stronger criterion for learning. Our null results on BLiMP/Zorro (Figure 5.7) therefore matter. They suggest that the empirical rewards did not produce grammatical generalization—at least, not of the kind captured by these benchmarks.

That said, while this conclusion may be partly correct, it would be premature to draw it too strongly here without further research. Existing minimal-pair benchmarks do not exhaust the space of possible grammatical generalizations. In particular, even child-adapted

benchmarks such as Zorro primarily target classic grammatical phenomena, but do not directly target some of the errors that are especially common in child language *production* and may therefore also be reproduced in our generated data—including missing subjects, objects, auxiliaries, or determiners, among others [saxton2005prompt; hiller_data-driven_2016; nikolaus_automatic_2024]. This is important because, in an RL fine-tuning setting, errors that are sampled in the model’s productions are the ones that receive direct corrective pressure from the reward signal.

Another concern is that reward fine-tuning may narrow the model’s production distribution, reducing diversity and potentially harming generalization. However, while this may be partly true, some narrowing may still be compatible with grammatical learning, and perhaps even expected if the model moves away from noisy or ill-formed productions. For instance, in our analyses, the grammar topline was relatively the least lexically diverse condition (Figure 5.8, Lexical Entropy, Appendix), yet it produced the clearest gains on the minimal-pair benchmarks (Figure 5.6, Appendix). Thus, reduced lexical diversity did not prevent improvement on these benchmark scores. Because the topline reward used BLiMP and Zorro training materials, this comparison does not independently establish grammatical generalization. The more central question is whether reward-induced changes target grammatical phenomena that can generalize beyond the sampled utterances themselves.

Testing this stronger learning claim, thus, remains an important direction for future work. It requires conducting large scale corpus exploration of the grammatical errors found in both child and model productions, and to use these bottom-up insights to build targeted tests around the relevant phenomena. This would help determine whether the reward-induced gains observed here reflect merely local shifts in surface production patterns or more abstract grammatical generalizations.

Ethical considerations

This work uses existing child-caregiver corpora and does not involve new data collection. Because the data involve children, we treat the corpora as sensitive and report only aggregate model-level results, without attempting to identify individual children, caregivers, or families. The proposed modeling framework is intended as a tool for studying hypotheses about language development, not for evaluating individual children or caregivers.

Appendix 1: Supplementary Analyses

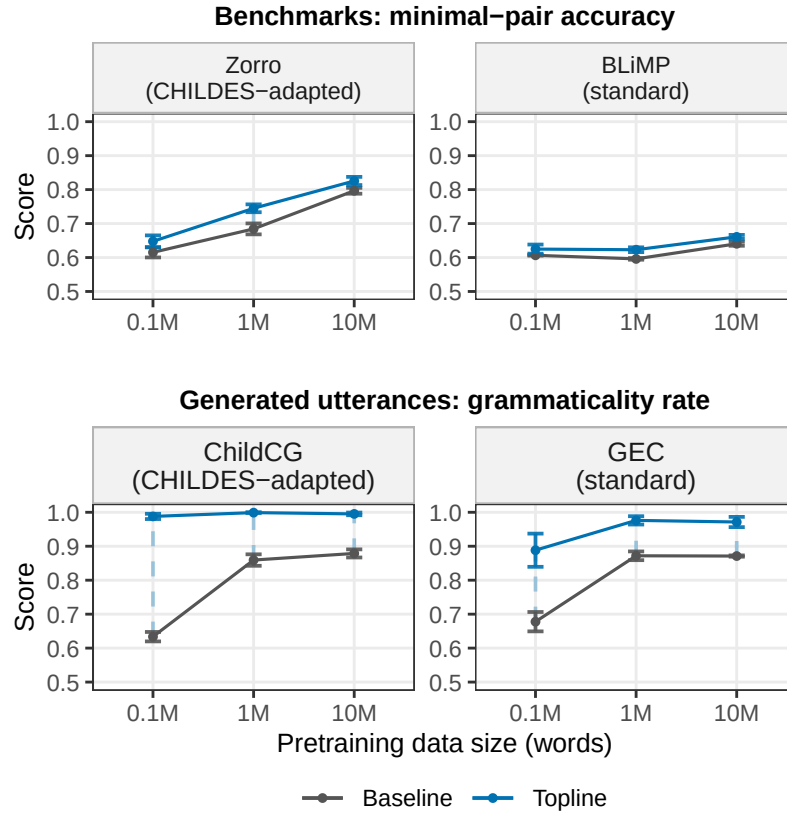


Figure 5.6 – Baseline and Topline performance across pretraining scales. The top row reports minimal-pair benchmark accuracy for Zorro and BLiMP. The bottom row reports grammaticality of generated utterances, using the ChildCG or GEC-based metric. Points show means across trainings seeds (after averaging over generation seeds). Error bars show ± 1 standard deviation across these independent units.

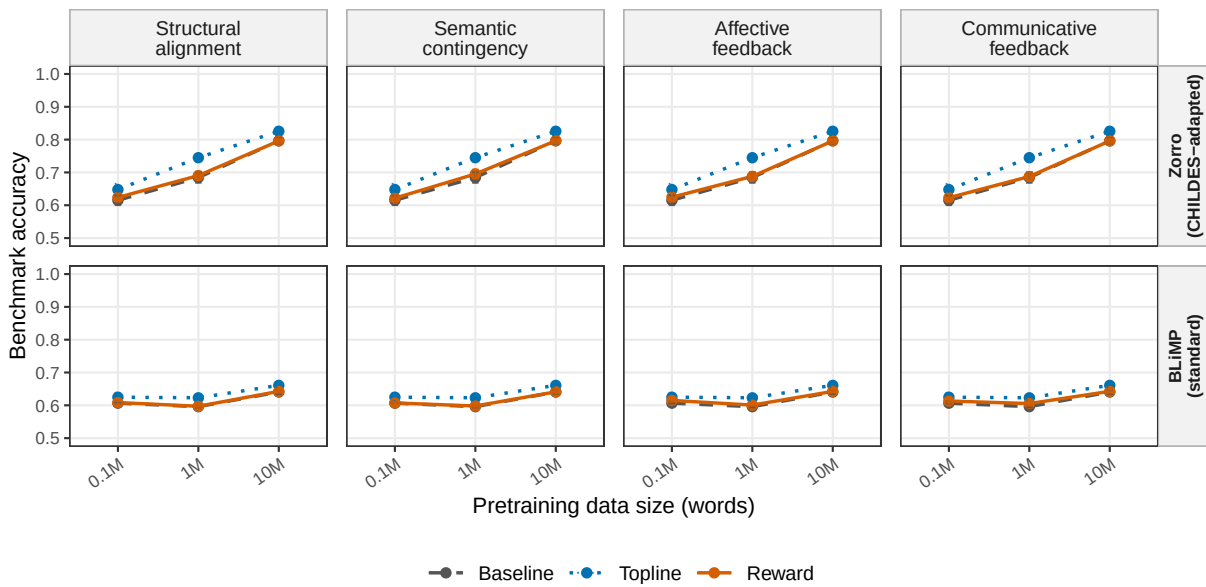


Figure 5.7 – Benchmark performance across pretraining scales for each reward family. Rows show Zorro and BLiMP minimal-pair accuracy; columns show the four empirical reward families. Within each panel, the baseline and topline trajectories are repeated to make the reward trajectory directly comparable at each scale. Means are computed over training seeds, after averaging over generation seeds. Error bars show ± 1 standard deviation across independent units.

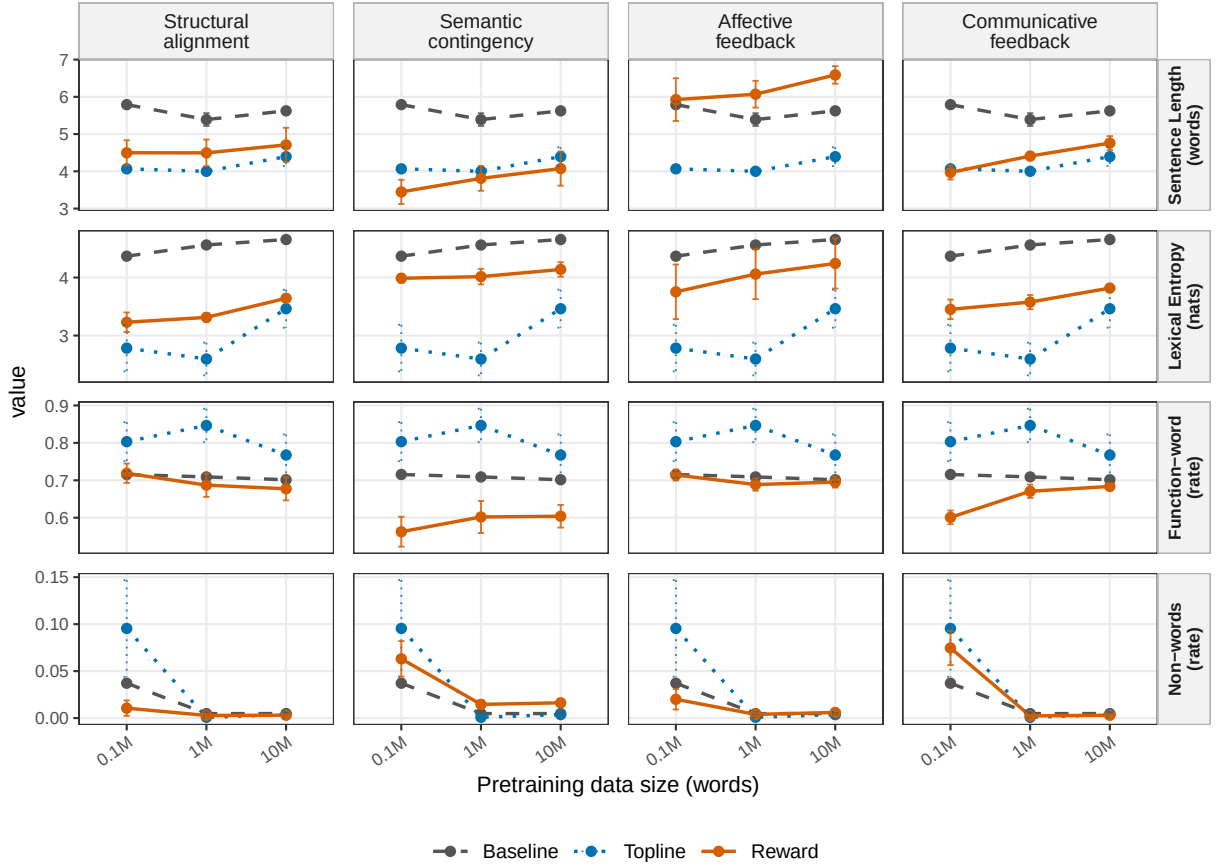


Figure 5.8 – Properties of generated utterances: Sentence length, pooled lexical entropy, function-word density, and rate of non-words. Points show means for each reward across training seeds, with error bars indicating ± 1 standard deviation across these seeds. Within each panel, baseline and topline trajectories are repeated to make the reward trajectory directly comparable at each scale.

Appendix 2: Examples of Model-Generated Utterances

Reward type		ChildCG = 1 (grammatical)	ChildCG = 0 (ungrammatical)
Baseline		<i>you want to look at some more pictures?</i> <i>oh, is that what it looks like?</i> <i>it is a big boy now isn't it?</i> <i>have you not any other little boys?</i>	<i>this piece of cake?</i> <i>because you can see mummy's poorly sick.</i> <i>the dog is saying.</i> <i>he is the name of the little girl.</i>
	Topline	<i>what are you doing?</i> <i>there is a horse.</i> <i>that is a girl.</i> <i>are you going back?</i> <i>that is a chair.</i> <i>what is that one?</i> <i>i will put it there.</i> <i>what did we do with the barn?</i>	<i>why is that side?</i>
Structural Alignment		<i>so that is a duck.</i> <i>he is a turtle.</i> <i>you are a good boy.</i> <i>i am a monkey.</i>	<i>how many trucks in the sky.</i> <i>come down the street.</i> <i>we are a good boy.</i> <i>can't find it.</i>
	Semantic Contingency	<i>yes that is a doggie.</i> <i>well it is a boat.</i> <i>there is an elephant.</i> <i>yes a banana puppy.</i>	<i>where is baby sheep?</i> <i>what is big truck?</i> <i>he a lion.</i> <i>a big flower red.</i>
Affective feedback		<i>and then i will put it in it.</i> <i>i will put it over here if i put it in there.</i> <i>we have to wash her hair first okay?</i> <i>we will shall we put it all together?</i>	<i>if i do it and then we can do it and put it in?</i> <i>now i will have to clean this with your room won't you?</i> <i>want mama do it?</i> <i>because your teeth i am watching this picture aren't they?</i>
	Communicative feedback		
	Acknowledgement	<i>they are called flowers.</i> <i>that is a gold straw.</i> <i>and that is the baby.</i> <i>you have a pink one on there.</i>	<i>have to use your spoon and pretend.</i> <i>you will use them ones with this bits.</i> <i>all day and then the is warm.</i> <i>but two grey stars.</i>
	Clarification request	<i>i don't know.</i> <i>can you see it?</i> <i>can't you?</i> <i>mhm, it is.</i>	<i>look at that isn't it?</i> <i>look it, there is it?</i> <i>see you shouldn't?</i> <i>there doesn't.</i>

Table 5.3 – Examples of utterances generated by RL-fine-tuned models at the 10M-word scale (generation was qualitatively similar at the 1M-word scale). Examples are organized by reward types. Unshaded cells show utterances that were classified as grammatical by ChildCG, while shaded cells show utterances classified as ungrammatical.

Appendix 3: Automatic labeling of rewards

We automatically annotated caregiver feedback signals in child-caregiver conversations and used these annotations to supervise reward model training (see Appendix 4).

Communicative feedback

Clarification requests Clarification requests were identified using a DeBERTa-v3-xsmall classifier fine-tuned on manually annotated caregiver responses in CHILDES, following prior work [nikolaus_fourthassi_2026].

Acknowledgements To identify caregiver acknowledgements, we relied on the feature-based algorithm described in nikolaus_communicative_2023a, which was specifically designed for CHILDES data and combined common keywords (e.g., “okay”, “alright”, and “yeah”) as well as repetition ratios capturing repetition-based acknowledgements (e.g., “It isn’t very nice, is it?” – “It isn’t.”).

Structural Alignment

Following previous research [dale_unraveling_2006; fernandez_quantifying_2014], a child structure was defined as a POS bigram. We used spaCy (en_core_web_sm)⁸ to tag each child utterance and caregiver response for part-of-speech (POS), excluding punctuation and empty tokens. Caregiver alignment was computed as a normalized score of bigram re-use, corresponding to the size of the intersection between the child and caregiver POS-bigram sets, divided by the size of the larger set.

Semantic Contingency

Following previous research [fusaroli_caregiver_2023; foushee2022getting], semantic contingency was computed using sentence embedding similarity. Here, we used all-MiniLM-L6-v2⁹ to encode the child utterance and caregiver response. We computed the cosine similarity between the two embeddings, and clipped the resulting score to the [0, 1] range.

Affective feedback

We annotated affective properties of caregiver responses using roberta-base-go_emotions.¹⁰ The model was applied to adult responses only. For each caregiver utterance, the classifier returned a probability distribution over several labels. We used the probabilities for emotions related to supportiveness, approval, and warmth.

8. https://huggingface.co/spacy/en_core_web_sm

9. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

10. https://huggingface.co/SamLowe/roberta-base-go_emotions

Appendix 4: Baseline and Reward model training

Baseline models

For the input-based baselines, we trained a small GPT-2-style causal language model on caregiver utterances only. The model used 2 hidden layers, 8 attention heads, and a hidden size of 512. We trained a word-level tokenizer on the training data, retaining words that appeared at least twice and capping the vocabulary at 5,000 types. Models were optimized with the standard GPT-2 causal language-modeling objective using AdamW and a batch size of 256. For each data scale, we held out 10% of the data for validation and used validation loss for early stopping and checkpoint selection. To assess the effect of input size, we trained models on three amounts of caregiver language: 0.1M, 1M, and 10M words. Each condition was run with three different random seeds.

Reward models

For each of the four caregiver feedback dimensions above, we train a reward model that maps a child utterance to the corresponding valence of the caregiver response, as annotated using the procedures described in Appendix 3.

All reward models share the same pre-trained model: `microsoft/deberta-v3-xsmall` [he_debertav3_2022], which we fine-tune using linear regression layer and a mean squared error (m.s.e.) loss to predict the target reward for a given utterance.

All reward models are trained on the CHILDES child-caregiver conversation pairs (one row per child utterance with the immediately following caregiver response), the same pre-processed pipeline above. We split the data into 90% training and 10% held-out evaluation with a fixed random seed. We used the AdamW optimizer with an initial learning rate of 1.41×10^{-5} , a batch size of 128, and early stopping based on MSE on the held-out validation set. The best checkpoint is selected on held-out MSE and used as the frozen reward model during PPO fine-tuning.

Topline Reward Model

The topline reward model was obtained using the same procedure, except that instead of learning the mapping between child utterances and the feedback valence, it was trained on the controlled minimal-pair datasets used in Zorro and BLiMP. More specifically, the model was trained to classify sentences from these datasets according to their labels (i.e., grammatical or ungrammatical).

Appendix 5: RL Fine-tuning Details

After pretraining on child-directed CHILDES utterances, we fine-tune each language model with Proximal Policy Optimization [schulman_proximal_2017], implemented using Hugging Face’s TRL library. For each fine-tuning step, we (a) sample utterances from the current language model, (b) compute the corresponding rewards using the reward model, and (c) update the model weights using PPO.

Hyper-parameter	Value
Max PPO total steps	6000
Batch size	1024
Mini-batch size	512
KL controller	adaptive, target $KL^* = 6$
Entropy coef. c_e	1×10^{-3}
LM mixing coef. λ_{LM}	1×10^{-3}
Score clipping	disabled
Sampling	$T=1.0$, top- $p=1.0$, top- $k=0$
Precision	bf16 mixed
Eval frequency	every 100 steps
Log frequency	every 10 steps
Early stopping	no reward gain for 10 log windows

Table 5.4 – PPO fine-tuning hyper-parameters, shared across the four caregiver-feedback rewards, the grammar topline, and all pretraining scales.

To obtain a diverse set of generated utterances, we randomly prompted the model with short utterance prefixes (the first one or two tokens) sampled from the pretraining data, namely the caregiver utterances used to train the input-based baseline model. We additionally included an entropy regularization term (0.001) in the loss.

The models were fine-tuned for a maximum of 6,000 steps, and the best checkpoint was selected based on mean reward. To discourage excessively short or long utterances, we applied rejection sampling by assigning a reward of -1 to generated utterances that were shorter than three tokens or failed to produce an end-of-sequence token within 20 tokens.

To mitigate language drift, a known issue in Reinforcement Learning (RL) fine-tuning, we also added a small language-modeling regularization term (weight = 0.001). We used the TRL default optimizer and learning rate. Other Hyperparameters are shown in Table 5.4. All other PPO hyperparameters were not changed from the default values as implemented in the Huggingface TRL library.

Each PPO run uses one NVIDIA H100 GPU (96 GB) with bf16 mixed precision and completes in approximately 6 GPU-hours.

5.4 Discussion

This chapter examined two aspects of bringing social interaction into computational research on language acquisition: the simulation of child–caregiver dialogue and the learning effects of caregiver-derived feedback. The two studies addressed different questions, using different models and experimental procedures. Their joint contribution lies in identifying what each approach can establish, and what remains necessary for a more integrated account of learning through interaction.

5.4.1 Simulation Fidelity and Learning Effects: Distinct Contributions

The first study showed that language models can approximate some properties of child and caregiver language, but that this approximation depends on the level of analysis and the interaction setting. The few-shot prompting conditions improved several word- and utterance-level properties, especially in caregiver-like generation. However, discrepancies remained at the dialogue level: simulated caregivers showed higher semantic alignment and lower semantic diversity than the human reference. Extended interactions also increased several differences between model-generated and human language. These findings indicate that plausible responses to individual utterances do not, by themselves, establish that a model can reproduce the properties of sustained child-caregiver interaction.

The second study showed that caregiver-derived rewards have different effects on model production. Structural alignment produced the clearest improvements in generated grammaticality. Communicative feedback had smaller effects, with the more consistent benefits associated with clarification requests rather than acknowledgments. Semantic contingency and affective feedback generally reduced grammaticality under the evaluations used, while also changing other properties of generation. Importantly, the empirical rewards did not consistently improve performance on BLiMP or Zorro. The evidence therefore supports changes in productive behavior, without establishing broader grammatical generalization.

These contributions should be distinguished. The first study evaluates models as simulators of interaction; it does not investigate the learning process through which children acquire the simulated behaviors. The second uses models as controlled learners; it does not reproduce a complete conversation with a caregiver. In particular, its rewards are derived from natural conversations, not from the interlocutors evaluated in the first study. The benchmark therefore does not validate the learning environment of the reinforcement-learning study, and the reinforcement-learning results do not demonstrate improved performance on the interaction benchmark.

5.4.2 Implications for Computational Models of Language Acquisition

A first implication concerns the criteria used to evaluate computational models. Across both studies, improvement on one measure did not establish improvement on another. In the interaction benchmark, closer correspondence with human utterance-level properties did not eliminate differences in dialogue organization. In the learning study, increased grammaticality in free generation was not accompanied by consistent gains on controlled grammatical contrasts. Although these discrepancies concern different tasks, they raise a common methodological issue: the adequacy of a model must be assessed with respect to the particular behavior or learning claim under investigation. A single successful outcome is insufficient to establish a general account of language acquisition.

This is especially important when interpreting changes in model production. A model can produce a larger proportion of grammatical utterances by changing which utterance types it favors, without necessarily acquiring grammatical distinctions that transfer to other contexts. At the same time, minimal-pair benchmarks do not exhaust the grammatical regularities that a

learner might acquire. As discussed in the second study, they may not directly target the errors most frequently encountered in the model’s own productions. The disagreement between the two evaluations should therefore neither be ignored nor treated as conclusive evidence that no learning occurred. It identifies a more specific question: which regularities changed, and under what conditions do those changes generalize?

A second implication is that caregiver feedback should not be treated as a single, interchangeable source of support. Research on child language has described several ways in which caregivers respond to children’s contributions [clark_conversational_2020; nikolaus2023communicative]. The learning study makes this distinction experimentally explicit: optimizing different feedback signals leads to different production outcomes. The lower grammaticality observed under semantic and affective rewards does not establish that these forms of feedback are detrimental to children’s development. Rather, it shows that the corresponding computational objectives do not uniformly favor grammatical production. Changes in length or lexical diversity motivate further questions about other dimensions of learning, but do not themselves demonstrate developmental benefits.

The comparison also requires care with similarly named measures. Semantic alignment in the interaction benchmark and structural alignment in the learning study capture different properties and play different experimental roles. The former is used to assess correspondence with human dialogue, whereas the latter supplies a reward for model training. Their results cannot be combined into a general claim that more alignment is either beneficial or harmful. For acquisition research, the relevant question is which property of a caregiver response provides useful information, for which aspect of learning, and under which conditions.

5.4.3 Limitations and Directions toward Integrated Interaction-and-Learning Models

The studies leave several gaps between the present models and children’s language acquisition. In the benchmark, age-conditioned generation reflects the behavior of already trained models asked to simulate children of different ages. It is not a developmental trajectory produced by accumulating child-like experience. In the learning study, controlled input sizes and feedback conditions make experimental comparisons possible, but do not establish that the learning procedure corresponds to children’s learning. In addition, the pretrained components used to predict rewards and evaluate outputs introduce knowledge beyond the learner’s caregiver-input corpus. These distinctions limit how directly the results can be interpreted as accounts of human development.

The measures also provide incomplete descriptions of interaction and learning. The benchmark focuses mainly on linguistic properties and does not comprehensively assess whether a response appropriately addresses a child’s intended meaning, communicative difficulty, or need for repair. Its few-shot prompts also introduce length constraints, so the effects of examples and instructions are not fully separated. Its multi-turn condition involves two imperfect simulators, so observed discrepancies may reflect their combined behavior rather than the limitations of either participant alone. Similarly, the learning study relies on automatic feedback annotations, learned reward predictions, and automatic grammaticality measures. Further validation should examine these

components separately, including whether reward predictions remain reliable on utterances generated by the learner rather than drawn from the original corpus.

A further limitation concerns how feedback is represented. During reinforcement learning, the reward model is fixed and predicts a feedback value from a generated utterance. It does not produce a caregiver response that changes with the evolving conversational context or the learner’s history. This abstraction allows particular signals to be isolated, but leaves out the linguistic content of an online response and the possibility of adjustments by both participants. Natural interaction provides a setting in which caregiver responses and children’s subsequent contributions are related over time [masek_where_2021; fusaroli_caregiver_2023]. Modeling that relation requires more than adding a fixed reward to otherwise independent productions.

An integrated approach could connect a learning model to a caregiver model whose responses are evaluated against natural interaction. The learner would update through repeated exchanges, while evaluation would track both the properties of the interaction and changes in the learner’s language. Such a system would require separate tests of simulation fidelity and learning effects; improvements in one should not be assumed to validate the other. Matched controls could help isolate the contribution of contingent feedback, for example by disrupting the relation between an utterance and its feedback while preserving other properties of the learning setting.

Finally, stronger tests of acquisition would examine whether improvements persist beyond the utterances favored during training. This could involve identifying recurrent errors in child and model productions, constructing targeted contrasts, and testing transfer to new lexical items and combinations. Evaluation materials should be separated from those used to train reward models and optimize the learner. These tests would help distinguish changes in production preferences from the acquisition of regularities that can be applied more broadly. They would also provide a basis for investigating combinations of feedback signals, rather than assuming that the effects of isolated rewards will remain unchanged when they occur together.

5.4.4 Chapter Conclusions

This chapter provides two complementary contributions to computational research on language acquisition. The interaction benchmark establishes a framework for comparing simulated and natural child–caregiver dialogue, and identifies discrepancies that remain despite improvements in individual utterances. The feedback-learning study shows that signals derived from caregiver responses can selectively change model production, while leaving broader grammatical generalization unresolved.

Together, these findings support a differentiated approach to modeling social interaction. Reproducing its observable properties and explaining its learning effects are related but distinct objectives. The present studies make progress on each, without yet combining them into a complete model of language acquisition through interaction.

Chapter 6

General Discussion

Objectives

Synthesize what the empirical studies establish about calibrated child–model comparisons, frequency-sensitive learning, episodic retrieval, and caregiver interaction, while distinguishing behavioral resemblance, learning effects, and claims about developmental mechanisms.

6.1 Revisiting the Central Question

This thesis asks how limited and uneven linguistic experience can support expressive and grammatical behavior, and what language models reveal about the resources involved. It approaches this question through comparisons of expressive vocabulary, an intervention on access to prior instances, a benchmark of interaction, and a comparison of caregiver-derived rewards. These studies require a corresponding methodological question: under which learning conditions and behavioral measures can model results inform theories of language development? Computational models are valuable because they make learning assumptions explicit and allow experience, architecture, and learning signals to be manipulated under controlled conditions [dupoux2018cognitive; warstadt2022what]. Their scientific interpretation is nevertheless conditional. A model can reproduce a behavioral pattern for reasons that differ from those that produce the pattern in children, and it can fail because the modeled input, task, response channel, or learning objective omits resources available to the child. Model performance becomes developmental evidence only when the comparison and the inference drawn from it are specified together.

Chapter 2 developed this argument as a framework for calibrated child–model comparisons. It distinguished the alignment of learning experience from the alignment of observed outcomes, and it emphasized that neither endpoint similarity nor a nominally child-scale training corpus is sufficient on its own. The empirical chapters then used models in several different roles. Chapter 3 treated language models as learners whose expressive outputs could be compared with child vocabulary production under controlled exposure and generation budgets. Chapter 4 used retrieval augmentation as an intervention on a model’s access to prior instances, asking whether that intervention selectively reduced lexical-frequency effects on syntactic contrasts. Chapter 5 first evaluated large language models as simulators of child–caregiver dialogue and then used smaller models as controlled learners to test the effects of caregiver-derived rewards.

These studies provide two complementary forms of evidence. The CORPUS-CDI and dialogue studies are primarily *diagnostic*: they ask where model behavior corresponds to, or diverges from, a human reference under an explicit measurement protocol. The retrieval and feedback studies are primarily *interventional*: they vary a computational resource or learning signal and measure the resulting change. This distinction matters because a diagnostic resemblance does not identify the process that produced it, whereas an intervention can establish an effect inside a model without showing that the same intervention operates in children. The four studies should therefore not be read as stages of one system that first learns vocabulary, then gains episodic memory, and finally enters social interaction. They are linked by a common inferential discipline and by a set of recurring empirical questions, not by shared parameters or a common developmental timeline.

Three conclusions emerge. First, developmental comparison requires calibration of the learner’s experience, the learner and intervention, the observed outcome, and the linking assumptions that connect an outcome to a theoretical construct. Second, frequency is a recurrent boundary condition: the predictive learners studied here make low-frequency lexical evidence less

reliably available, while direct access to stored instances provides only partial and task-specific compensation. Third, social interaction cannot be reduced either to plausible isolated utterances or to a generic feedback signal. Sustained dialogue and different forms of caregiver response place distinct demands on a model, and their effects depend on the outcome used to evaluate them.

The resulting picture is neither a general success nor a general failure of language models as accounts of development. Predictive learning captures substantial regularity; retrieval and feedback can selectively change what a model does; and prompting can elicit some age-associated patterns. The studies also expose specific mismatches: vocabulary-growth profiles differ across learners, retrieval leaves a residual frequency gap, and improvements in generated utterances do not automatically extend to controlled grammatical preferences or sustained dialogue. No single intervention was evaluated across all these outcomes. The contribution of the thesis is thus to identify what each result licenses, which mismatches remain, and how those mismatches can be converted into more precise hypotheses for future modeling.

6.2 Methodological and Computational Contributions

The thesis contributes experimental tools as well as conclusions about model behavior. The first methodological contribution is a production-oriented vocabulary protocol. CORPUS-CDI makes three usually separate choices jointly inspectable: how much language a model encounters, how much it is allowed to produce, and which frequency distribution is represented in the evaluation vocabulary. Combining the resulting score with lexical diversity and non-word rate makes it possible to distinguish broader vocabulary coverage from a change in how widely or accurately a model samples word forms. This supplies a concrete implementation of outcome calibration from [chapter 2](#), while exposing the remaining difference between a count of generated forms and a parent’s report of meaningful word use.

The retrieval study contributes a controlled application and analysis of an existing AI method. It does not introduce k NN language modeling itself. Its contribution is to test whether interpolation with a datastore-derived distribution changes syntactic preferences selectively across lexical-frequency bands, then to vary the stored representations, structural content, granularity, context, and number of neighbors. This design moves beyond an aggregate retrieval score: it identifies conditions in which instance access helps, conditions in which it can hurt, and the information associated with the benefit. The frequency-stratified evaluation is consequently useful both for a computational hypothesis about memory and for diagnosing retrieval-augmented language models.

The interaction benchmark contributes a complementary evaluation design: it tests both interlocutor roles, measures word-, utterance-, and dialogue-level properties, and contrasts responses to human turns with sustained exchanges between model instances. The feedback study then contributes a comparative use of reinforcement learning. It holds the learner family and fine-tuning framework constant while replacing the caregiver-derived reward, and evaluates the resulting changes through two generation measures and controlled minimal pairs. Its novelty lies in the comparison of candidate signals and their differentiated effects, rather than in a new PPO

algorithm. Together these contributions show how established NLP methods can be organized into informative cognitive-science experiments without requiring the four studies to share one artificial learner.

6.3 From Model Performance to Developmental Evidence

The studies collectively extend the two-axis framework of input and outcome alignment into a broader inferential chain:

$$\begin{aligned} \text{experience} &\longrightarrow \text{learner/intervention} \longrightarrow \text{observed outcome} \\ &\longrightarrow \text{linking model} \longrightarrow \text{licensed claim.} \end{aligned}$$

Each link contributes assumptions. Developmental interpretation is strongest when those assumptions are reported and varied independently, and weakest when differences at several links are compressed into a single label such as “child-scale,” “human-like,” or “language competence.”

6.3.1 Calibrating the Learner’s Experience

The amount of input is an important part of a developmental comparison, but it is not a sufficient description of experience. Children encounter speech in recurring activities, with prosody, visible events, interlocutors, and consequences for their own participation. A textual corpus can match the estimated number of words heard by a child while differing in register, modality, temporal organization, and interactional contingency. Conversely, a small naturalistic corpus may preserve child-directed register but sample only a narrow portion of the child’s environment.

This distinction appears differently across the empirical chapters. In [chapter 3](#), child-directed transcripts and audiobook text provide contrasting registers under exposure-scaled training conditions. The design makes nominal corpus size explicit, but successive scales do not automatically represent one continuously developing learner, and repeated optimization over the same material contributes to effective exposure. In [chapter 4](#), the smaller model was trained on approximately 50 million words drawn from CHILDES, conversational transcripts, stories, and subtitles. This is a mixed BabyLM-derived corpus, not 50 million words of child-directed speech alone. The bounded-data and large-scale settings provide evidence that a retrieval effect can be observed under two different regimes; because model size, corpus composition, and training history differ simultaneously, and datastore provenance is incompletely reported, they are replications across settings rather than a controlled scale comparison. In [chapter 5](#), prompted dialogue models inherit broad pretraining, whereas the feedback-learning models are pretrained on a bounded caregiver corpus and receive additional information through learned rewards.

A complete exposure account must therefore distinguish unique linguistic material from total token presentations and from externally supplied information. The ten training epochs reported for the smaller retrieval model, for example, are additional presentations of the bounded corpus, not ten independent samples of a child’s experience. Retrieval datastores, prompt demonstrations,

pretrained tokenizers, reward predictors, and automatic evaluators do not all function as direct input to the target learner, but they affect what the complete experimental system can do. Documenting this *total information budget* requires separating resources that affect learning or inference from evaluators used only to measure the result; an evaluator does not transfer its pretraining to a learner merely by scoring it. It makes clear whether a result concerns learning from child-comparable experience, the use of a resource learned elsewhere, or an engineering system that combines both.

6.3.2 Specifying the Learner and the Intervention

Architecture is most informative when it is treated as an experimental variable rather than assumed to be cognitively plausible by resemblance. The character-level LSTM and Transformer in [chapter 3](#) test whether the observed expressive-production pattern depends on one of two sequence-learning architectures. The comparison reduces dependence on a single model family, but shared failure does not isolate a unique cause: the systems also share a textual response channel, a predictive objective, and a generation-based task model. Their contrast is consequently evidence about robustness across the tested architectures, not proof of a general limitation of statistical learning.

The intervention is more direct in [chapter 4](#). The parametric model and its retrieval-augmented counterpart are evaluated on the same syntactic contrasts, allowing the effect of access to stored instances to be estimated within a common outcome. The full-retrieval keys are contextual representations from the GPT-2 model itself. The interpretation is tied to the datastore, representation, interpolation rule, and retrieval configuration. The result concerns what this form of non-parametric access makes sufficient for the tested decisions; it does not establish that k NN retrieval is a process model of hippocampal memory.

The two studies in [chapter 5](#) intervene at still different stages. Prompting changes the context in which a pretrained simulator expresses its existing capacities. Reinforcement learning changes the model’s production policy in response to a frozen reward. Neither manipulation should be described simply as “more interaction.” One supplies examples at inference time, while the other repeatedly reinforces utterances according to a learned scalar proxy. Naming the stage and content of each intervention is necessary before their effects can be compared.

6.3.3 Aligning Outcomes and Task Models

Children and models rarely expose the same dependent variable directly. Child knowledge is inferred from reports, looking, choices, elicited responses, or spontaneous production. A model exposes probabilities, representations, or generated sequences. A threshold, prompt, decoder, distance function, classifier, or aggregation rule connects that output to the construct under study. This additional component is a *task model*, and its assumptions form part of the empirical claim.

The CORPUS-CDI production threshold in [chapter 3](#), for example, links parent-reported vocabulary and the frequency of word forms in sampled model output. It improves comparability

of observation opportunity, but it does not make the two measures identical. The syntactic minimal pairs in [chapter 4](#) provide a controlled test of probabilistic preference, not a direct measure of spontaneous grammatical production. The dialogue benchmark in [chapter 5](#) aggregates lexical, utterance, and conversational statistics, each of which captures a selected aspect of interaction rather than communicative adequacy as a whole. The feedback study then shows explicitly that two measures intended to bear on grammaticality can disagree: rewards can improve the grammaticality of generated utterances without reliably improving BLiMP or Zorro preferences.

Such disagreements should not be collapsed into an average score. They reveal that an intervention changes some expressions of a capacity but not others. Stronger claims require convergence across outcomes with different measurement biases, or a theory that predicts why a change should be restricted to one outcome. Linking functions should, where possible, be selected on data separate from the final comparison and tested for sensitivity to plausible alternatives. Otherwise, part of the apparent alignment can be introduced by the criterion designed to measure it.

6.3.4 What Success and Failure License

An aligned success supports a bounded sufficiency statement: the specified learner can produce the specified outcome from the specified information under the specified task model. It does not establish that children use the same internal representation or optimization procedure. A controlled contrast can strengthen the conclusion by locating which manipulated resource changes performance. Thus retrieval can be said to cause a change in the evaluated model, and a reward can be said to alter the trained policy, while the developmental relevance of that change remains a separate inference.

Failure is less decisive when several mismatches remain. A text-only learner can fail because it lacks perceptual grounding, because its objective gives little weight to rare events, because its memory cannot preserve particular episodes, or because the evaluation does not measure the knowledge that changed. Such failure does not demonstrate that children require an innate constraint or that the target is unlearnable from experience. It becomes theoretically informative when alternative explanations are converted into controlled comparisons. In this sense, the residual gaps identified throughout the thesis are not merely negative results. They define the next variables that a developmental model must expose.

6.4 Learning from Sparse Evidence

A recurring result across the vocabulary and retrieval studies is that model behaviour becomes less robust when the relevant linguistic evidence is infrequent. This recurrence provides a genuine connection between [chapter 3](#) and [chapter 4](#), but the connection is diagnostic rather than a direct causal chain. The two chapters examine different linguistic domains, models, and outcome measures. [Chapter 3](#) measures how reliably target words appear in free generation under calibrated exposure and production budgets, whereas [chapter 4](#) measures preferences between minimally different sentences in controlled syntactic contrasts. The shared result is

therefore not that both chapters reveal a single latent deficit, nor that retrieval solves the vocabulary-production problem. Rather, they provide converging evidence that frequency is an important boundary condition on what predictive language learners make available under different forms of evaluation.

6.4.1 Frequency Sensitivity in Expressive Production

The contribution of CORPUS-CDI in [chapter 3](#) is to make model generation more comparable to observations of children’s expressive production. Training exposure, generated output quantity, and the frequency distribution of the evaluation vocabulary are brought under explicit control. Within this protocol, the models differ from children not only in their final scores but also in the shapes of their trajectories. Over the evaluated exposure conditions, their generated vocabulary coverage grows more slowly and approximately linearly, and low-frequency target words are especially unlikely to reach the production criterion. Changing the sampling temperature alters the severity of this pattern, but does not yield a setting that simultaneously aligns vocabulary coverage and lexical diversity with the child data.

The discrepancy is not confined to vocabulary coverage, but neither does every model diverge on every individual measure. Lexical diversity and non-word rates vary with architecture and corpus: some CDS-trained conditions approach the child reference in type–token ratio, and the CDS-trained LSTM shows a marked decline in non-word rate. The informative mismatch concerns the joint production profile across exposure conditions. Local agreement or improvement on one measure does not establish simultaneous agreement in reliable expressive coverage, lexical diversity, and word-form accuracy. The content-word analysis adds a further comparison of how output is allocated between content and function words. Together these measures distinguish changes in the number of different forms produced from changes in how reliably evaluation words are available. The non-word contrast additionally involves different response channels: dictionary membership in orthographic output cannot isolate children’s articulatory development.

The frequency-band analysis summarizes fitted curves by the exposure-equivalent point at which they cross an 80% criterion. Some crossings lie beyond the observed training range, so they are extrapolated properties of the fitted curves rather than observed acquisition ages. Their magnitude depends on the curve and the exposure-to-age mapping. The distinction is particularly important because the CDS conditions reuse the 30-month-equivalent checkpoint for later production budgets: a larger generated sample at months 31–36 is not additional training. The supported synthesis is strong frequency sensitivity under this protocol; it does not require interpreting distant fitted crossings as a literal number of months that a learner would need.

This result should be interpreted at the level of the measurement actually used. A CORPUS-CDI score records whether a word is produced sufficiently often in a finite generated sample. It therefore measures *reliable expressive availability under a specified generation protocol*; it is not a direct test of whether the model possesses a lexical representation or understands the word. Failure to cross the count threshold may reflect weak representation, low contextual probability, competition from more frequent alternatives, or the interaction between decoding and the evaluation rule. Conversely, crossing the threshold does not establish child-like semantic

knowledge. The threshold is a linking function between two different response systems—parent-reported child vocabulary and sampled model text—and conclusions must remain conditional on that link.

With this qualification, the chapter nevertheless identifies an informative asymmetry. High-frequency forms occupy a disproportionately large share of model output, whereas many low-frequency content words remain difficult to elicit reliably. This pattern is compatible with the hypothesis that next-token prediction, together with the tested architectures and input distributions, allocates behavioural probability unevenly across the lexical long tail. It does not, by itself, establish cross-entropy learning as the unique cause. Corpus composition, representational choices, model capacity, repeated exposure, and decoding all contribute to the observed learner–output system. The appropriate conclusion is consequently that the tested predictive learners do not reproduce children’s relative robustness to sparse lexical evidence under the CORPUS-CDI production measure.

6.4.2 Retrieval as Selective Compensation

Chapter 4 asks a narrower intervention question: when parametric predictions are frequency-sensitive, can explicit access to stored instances selectively improve performance? The answer is conditionally positive. Adding k NN retrieval produces larger gains for low-frequency than for high-frequency items across the tested syntactic phenomena and model settings. The inferential analyses support the selectivity of this effect: retrieval is not merely a uniform increase in accuracy. At the same time, low-frequency items remain less accurate than high-frequency items after augmentation. Retrieval narrows the gap without eliminating it, and some high-frequency conditions show no benefit or modest degradation.

The size of the effect can be seen directly in the reported accuracies. For the smaller model, low-frequency subject–verb agreement improves from 0.60 to 0.76 while high-frequency agreement changes from 0.91 to 0.90. Low-frequency relative-clause accuracy rises from 0.48 to 0.65, while high-frequency accuracy changes from 0.88 to 0.89 (Table 4.2). These within-model contrasts make clear that the narrower gap includes a substantial improvement for the difficult items, rather than arising only from degraded performance at the high-frequency end. The item-bootstrap analyses support the greater average gain for low-frequency items (Table 4.6). They do not make absolute frequency bands comparable across the two differently trained models.

The content and configuration analyses further constrain this result. Semantic similarity alone is generally insufficient to reproduce the syntactic gains, while retrieval sets containing structurally relevant instances tend to be more useful. In the low-frequency conditions, similarity-based selection without a structural match does not outperform random selection from sets containing a structural match (Figure 4.2). The relevant match is operationalized by detection of the target phenomenon, rather than identity of a complete syntactic derivation. However, this does not demonstrate separable, independently necessary structural and semantic components of human episodic memory. Contextual representations may retain several kinds of information at once, and apparently “semantic-only” representations may not remove structure completely. Moreover, the best retrieval granularity, context size, and number of neighbours vary across

phenomena. The observed benefit is therefore produced by a particular interaction among the base model, datastore, query representation, retrieval rule, and target construction.

Accordingly, the k NN system should be understood as a computational intervention that isolates one useful property of an episodic system: access to particular stored instances without additional gradient updates. It does not instantiate the full set of functions attributed to human hippocampal memory, such as selective consolidation, adaptive encoding, online updating, or task-sensitive retrieval control. The results show that instance-based access is sufficient to improve some low-frequency syntactic decisions in the tested models. They do not show that the same mechanism explains children’s frequency robustness or that the hippocampus is the source of that robustness.

Taken together, the two chapters motivate a plural rather than a replacement account. Parametric prediction captures substantial distributional structure but remains frequency-sensitive; non-parametric access can compensate selectively when useful instances are available and retrievable. This is a principled connection between the chapters, but it remains a hypothesis about complementary computational resources. A stronger test would apply retrieval to the same expressive-production outcome used in [chapter 3](#), or compare parametric and retrieval-based learners on matched production and minimal-pair tasks. Such work would also need to count the datastore as part of the learner’s total information budget, following the calibration principles developed in [chapter 2](#). Retrieval does not provide information for free: its developmental interpretation depends on what is stored, when it becomes available, and how access is controlled.

6.5 Competence Depends on How It Is Measured

The studies in this thesis do not reveal a single quantity that can be called “language competence.” Instead, they show that an apparent success or failure depends on the behavioural outcome, the task used to elicit it, and the linking assumptions connecting that outcome to the construct of interest. This dependence is not merely methodological noise. It determines which developmental claims the evidence can support.

6.5.1 Endpoints and Trajectories Answer Different Questions

An endpoint comparison asks whether a model eventually reaches a specified level of performance. A trajectory comparison asks whether changes with increasing experience resemble the timing, ordering, and shape observed in children. These questions can dissociate. Two learners may reach similar endpoints through very different paths, while learners with similar early trajectories may later diverge. The vocabulary study makes this distinction especially clear: a model’s eventual production of more CDI words does not compensate for a growth curve, frequency dependence, or lexical allocation that differs from the child trajectory.

Developmental labels must therefore be attached only to manipulations that warrant them. The exposure-scaled conditions in [chapter 3](#) approximate increasing quantities of linguistic experience, but they are not equivalent to observing one child longitudinally, particularly when separate training sets or models contribute to successive points. Likewise, the age-conditioned

prompts in the dialogue study of [chapter 5](#) test whether a pretrained model can *simulate* age-associated behaviour. They do not constitute evidence that the model learned through those developmental stages. Reinforcement learning in the same chapter is a genuine intervention on model behaviour, but its before–after contrast is still not a child developmental trajectory. Keeping these comparisons distinct prevents endpoint resemblance, prompted role performance, and learning-induced change from being treated as interchangeable evidence.

6.5.2 Free Generation and Controlled Preferences Expose Different Capacities

Free generation is valuable because model and child outputs can be analysed using related production measures. It is also jointly determined by representation, contextual preference, decoding, output length, and sampling variability. In [chapter 3](#), a target word that rarely appears may be weakly represented, or it may simply lose repeatedly to more probable alternatives. In the feedback study of [chapter 5](#), improvements in generated grammaticality demonstrate a change in the production policy, but do not necessarily entail a corresponding change in abstract grammatical preferences.

Minimal-pair evaluation reduces some of this ambiguity. The contrasts in [chapter 4](#) hold much of the lexical and propositional content constant and ask whether the model assigns greater probability to the grammatical member. This design is more sensitive to a particular distinction, but it is also narrower and less naturalistic. Success does not establish that the same knowledge will guide spontaneous production or interaction. Conversely, weak free generation does not imply failure on a controlled preference test.

The divergence between generation-based grammaticality and BLiMP/Zorro-style judgments after feedback learning in [chapter 5](#) is therefore theoretically informative. It suggests that an intervention can make sampled utterances more acceptable without producing measurable transfer to controlled syntactic contrasts. Across the thesis, productive coverage, lexical diversity, sentence preference, conversational form, and generated grammaticality should consequently be treated as complementary outcomes rather than interchangeable estimates of a single human-likeness score.

6.5.3 Uncertainty Includes Sampling and Measurement Choice

Quantitative uncertainty is broader than an error bar around a mean. It includes variation across corpus samples, model initialisations, generated samples, test items, linguistic phenomena, conversations, and human participants. The relevant unit of analysis differs across studies: generated tokens are nested within model runs, test sentences within paradigms, and utterances within conversations. Confidence intervals and hierarchical analyses should respect these dependencies rather than treating every observation as independent.

A second form of uncertainty arises from analytic choices. In [chapter 3](#), estimated vocabulary trajectories depend on the production threshold, decoding temperature, curve family, frequency-band definition, and any extrapolation beyond observed exposure. Sensitivity analyses are therefore part of the substantive result, not an optional robustness appendix. In [chapter 4](#), item-level bootstrap intervals and mixed-effects modelling support the claim that retrieval benefits are

larger for low-frequency items, while uncertainty across phenomena and retrieval configurations limits claims of universality. Similar care is required in [chapter 5](#), where automatic evaluators and dialogue metrics introduce their own measurement error and construct assumptions.

The appropriate response is triangulation. A developmental claim becomes stronger when it survives plausible linking functions and appears across measures whose biases differ. When measures disagree, the disagreement should be preserved and explained rather than averaged away. This principle follows directly from the calibration framework in [chapter 2](#): evidence about a learner is always mediated by a task model. Reporting that mediation explicitly allows the thesis to distinguish robust conclusions from conclusions that remain conditional on a threshold, prompt, decoder, retrieval configuration, or evaluator.

6.6 From Static Input to Social Interaction

The studies in this thesis broaden the learning environment under consideration, from distributions in static linguistic input, through access to stored instances, to signals derived from social interaction. This progression should not be interpreted as a single model acquiring successively more child-like components. The studies use different models, tasks, and outcome measures. Rather, they examine complementary ways in which information may become available to a learner. [Chapter 3](#) evaluates what predictive learning produces under calibrated exposure and production opportunities; [chapter 4](#) asks whether direct access to prior instances can compensate for weaknesses in parametric knowledge; and [chapter 5](#) investigates both the simulation of interaction and the use of caregiver-derived signals during learning.

6.6.1 Interaction as an Object of Simulation

The dialogue benchmark in [chapter 5](#) first treated language models as *simulators*, asking whether they could reproduce observable properties of child–caregiver conversations. Its multi-level design proved important. Depending on the model and prompting condition, generated language approximated aspects of the CHILDES reference at the word and utterance levels, including lexical density, utterance length, and syntactic complexity. The few-shot prompting condition also brought several measures closer to the human reference, particularly for caregiver-like generation. Because the supplied prompt templates include length constraints as well as examples, these results concern the prompting condition as implemented; they do not isolate examples as the sole cause of changes in utterance length. However, this improvement did not extend uniformly to the dialogue level. Under the study’s operationalizations, simulated caregivers remained more semantically aligned and less semantically diverse than human caregivers, and discrepancies often increased when two model instances interacted over multiple turns.

These findings show why simulation fidelity cannot be inferred from locally plausible responses. A model may generate an appropriate-looking reply to a human utterance while producing a different interactional regime when its own outputs shape the subsequent context. In multi-turn simulation, small biases can accumulate as each model conditions its next response on the other’s output, producing patterns that are not apparent in single-turn evaluation. This is adaptation

through the conversation context, without online parameter learning. Because both interlocutors are simulated, the resulting divergence cannot be assigned to one role alone or directly predict performance with a human interlocutor. Interaction is therefore not merely a longer sequence of independently adequate utterances; it is an evolving relation between interlocutors whose behavior is mutually conditioning.

Crucially, simulation fidelity is not evidence of acquisition. The age-conditioned models were already pretrained and were prompted to enact children or caregivers of particular ages. Apparent age-related differences consequently reveal knowledge that the pretrained model can express under a role instruction, not a developmental trajectory caused by accumulating child-like experience. Likewise, the few-shot contrast demonstrates sensitivity to prompt context and instructions, not lasting learning. The benchmark is informative about the behavioral adequacy of a simulator, but it does not establish how children acquire the behaviors being simulated.

6.6.2 From Caregiver Responses to Reward Signals

The reinforcement-learning study in [chapter 5](#) addressed a different question by treating models as controlled learners. Small causal language models were first trained on caregiver utterances and then fine-tuned using rewards derived from naturally occurring child-caregiver pairs. The comparison distinguished communicative feedback, structural alignment, semantic contingency, and affective feedback. These signals produced different outcomes. Structural alignment yielded the clearest gains in the grammaticality of generated utterances; communicative feedback produced smaller and less consistent gains, with clarification requests more informative than acknowledgements. Semantic-contingency and affective rewards did not improve grammaticality under the measures used and sometimes reduced it, while also changing utterance length, lexical entropy, and function-word density.

This finding is stronger than an unadjusted difference in sample averages. In the reported mixed-effects analysis, the reward contrasts control for utterance length and non-word rate and include a generation-run intercept. Structural alignment remains positively associated with ChildCG grammaticality, with estimated changes relative to baseline of +0.077 and +0.060 at the 1M- and 10M-word scales, respectively ([Table 5.2](#)). The positive direction is also present under the GEC-based generation measure. Thus the result is not explained away by these two measured covariates, although it remains dependent on automatic assessments of grammaticality. The noisier 0.1M condition further shows that the learner’s initial language distribution matters for how effectively the reward can guide production.

This manipulation provides a conditional causal result within the computational system: changing the reward changes the learner’s production policy. It does not, however, instantiate complete online interaction. Each frozen reward model maps a generated child utterance to a scalar value learned from correlations between child utterances and subsequent caregiver behavior. During fine-tuning, it neither generates the caregiver’s response nor conditions its feedback on an evolving dialogue, shared context, or the learner’s history. The learner therefore receives a prediction of the feedback that an utterance tends to elicit, rather than a situated response from an adaptive interlocutor. This abstraction is useful because it isolates candidate signals, but

it tests the utility of fixed reward proxies, not the full developmental effect of reciprocal social interaction.

The distinction also prevents the two studies in [chapter 5](#) from being treated as a single pipeline. The simulated interlocutors from the benchmark did not supply the rewards used in fine-tuning, and the reward-trained learners were not evaluated on the interaction benchmark. Better simulation would not by itself validate a learning signal, just as a reward-induced production change would not establish that the resulting model sustains more human-like dialogue.

6.6.3 Production Gains and Grammatical Generalization

The divergence between free generation and controlled grammatical evaluation is one of the most informative results of the feedback study. Caregiver-derived rewards, especially structural alignment, could increase the proportion of generated utterances classified as grammatical without consistently improving preferences on BLiMP or Zorro. One possible explanation is a redistribution of probability toward familiar or stable utterance types, without corresponding improvement on the grammatical contrasts tested in the benchmarks. This remains an explanation to test, not a demonstrated account of the structural-alignment effect; its persistence after adjustment for length and non-word rate argues against attributing it simply to shorter outputs. Changes in generation therefore support a claim about productive behavior, but not yet a claim about general grammatical knowledge.

The converse qualification is also necessary. Minimal-pair benchmarks sample particular constructions and may not target the errors that occur frequently in the learner’s own productions. Their null results do not prove that no grammar-relevant change occurred. Instead, disagreement between the measures localizes the unresolved question: which regularities changed, and how far do they transfer? As argued in [chapter 2](#), a general acquisition claim requires convergence across measures whose linking assumptions are independently justified.

The grammar-targeted topline provides a further comparison: it improves both generated grammaticality and the reported minimal-pair scores. However, its reward model was itself trained on grammaticality labels from BLiMP and Zorro, and the present description does not establish an independent evaluation split. It therefore shows that explicitly grammar-supervised reward optimization can improve the reported outcomes, but cannot serve as an independently validated upper bound on grammatical generalization or prove that the empirical rewards fail solely because they are indirect. This distinction leaves the caregiver-reward contrast intact while limiting what the topline adds to its interpretation.

6.7 Theoretical Implications for Language Acquisition

Taken together, the findings support a plural and functionally differentiated account of the information that may contribute to language learning. Predictive exposure, episodic access, and interaction-derived feedback alter a learner at different stages and in different ways. The thesis does not show that these mechanisms form a single developmental architecture, but it clarifies hypotheses that an integrated account would need to distinguish.

6.7.1 Complementary Sources of Support

Predictive learning provides gradual sensitivity to distributional regularities, but the results of [chapter 3](#) suggest that, under the production measure used there, this sensitivity remains strongly shaped by lexical frequency. [Chapter 4](#) provides a limited computational demonstration that inference-time access to stored instances can selectively improve performance when baseline decisions are less accurate, narrowing the frequency gap on syntactic contrasts without eliminating it. The result is compatible with the general idea that consolidated statistical knowledge and instance-based access can make complementary contributions [[mcclelland1995there](#); [kumaran2016learning](#)]. Because the vocabulary and retrieval studies use different models and outcomes, however, retrieval cannot be claimed to explain or repair the expressive-vocabulary trajectory observed in [chapter 3](#).

Feedback introduces a third form of support: it changes which productions are reinforced and thereby updates the learner’s parameters, whereas the retrieval intervention changes access to stored examples at inference time. The structural findings in [chapters 4 et 5](#) offer a cautious point of contact. Retrieved instances containing useful structural information supported performance on syntactic contrasts, while a reward associated with caregiver reuse of POS-bigram structure improved generated grammaticality. These manipulations are not equivalent, and neither identifies the mechanism used by children. Together they identify two distinct routes through which structure-related information can improve model behavior: selecting useful instances at inference and reinforcing a production policy during fine-tuning. This motivates a testable hypothesis that the way a learner accesses and uses structural evidence matters alongside the amount of exposure. Whether children exploit analogous information remains an empirical question.

6.7.2 Feedback Is Not a Unitary Developmental Resource

The reward results argue against treating “caregiver feedback” as a homogeneous source of evidence. A clarification request, structural reuse, semantic continuation, and positive affect may all be contingent on a child’s contribution, yet they need not provide information about the same target. Structural alignment may preserve a reusable linguistic frame; clarification requests may indicate communicative difficulty without locating its source; semantic contingency may support common ground; and affect may regulate engagement rather than grammatical form. Their value should consequently be assessed relative to a specified developmental outcome.

For the same reason, the weaker grammaticality observed under semantic and affective rewards does not imply that such responses are harmful to children. Grammar was the principal outcome of the study, whereas these signals may serve semantic, pragmatic, motivational, or relational functions that were not measured. The theoretically relevant question is therefore not whether feedback is globally beneficial, but which signal can support which change, for which learner, under which interactional conditions.

6.7.3 From Computational Effects to Developmental Claims

Across these studies, computational interventions establish forms of conditional sufficiency rather than mechanistic identity. Retrieval shows that external instance access *can* reduce a model’s frequency sensitivity; reward fine-tuning shows that caregiver-correlated scalar signals *can* reshape production. Neither result shows that children use the same representation, optimization procedure, or learning dynamics. In particular, natural interaction supplies temporally situated linguistic responses, multimodal context, joint attention, and mutual adaptation that are absent from a frozen reward function.

The broader theoretical contribution is therefore a disciplined decomposition of the acquisition problem. Distributional exposure determines what regularities are available for gradual learning; memory determines which particular experiences can be re-accessed; and interaction can make some learner behaviors more consequential than others. Their effects must be tested separately before they are combined. Consistent with the calibration framework in [chapter 2](#), claims about development should remain tied to the manipulated information, the learner in which it was manipulated, and the outcome through which its effect was observed.

6.8 Thesis-wide Limitations

The main limits concern the available experience, the validity of the measurements, and the populations to which the results can generalize. The distinctions between model roles and developmental scales discussed above also prevent treating the four studies as a cumulative experiment on a single learner.

6.8.1 Textual Input and Missing Grounding

All studies represent language as text. Children encounter speech embedded in perceptual and social events, with acoustic variability, prosody, gaze, gesture, and visible referents [[Tomasello1995](#); [Csibra2009](#)]. Transcription removes much of this information while supplying discrete orthographic units. In [chapter 3](#), character input retains spaces and punctuation, so it does not recreate segmentation from continuous speech. In [chapter 4](#), stored text preserves linguistic instances without the perceptual or social content of an experienced episode. In [chapter 5](#), dialogue is represented by written turns, and feedback learning further reduces the caregiver response to a scalar prediction. These simplifications support tractable interventions, but they do not establish what adding any particular missing modality would change. That contribution requires a matched intervention.

6.8.2 Automatic Measures and Validation Data

The thesis makes its task models explicit, but several remain incompletely validated. The CORPUS-CDI threshold is chosen through correspondence with the human vocabulary trajectory, making independent calibration important for subsequent comparisons. Dictionary-based non-word judgments can penalize legitimate child forms and do not isolate articulation. The dialogue

measures quantify linguistic form and embedding-based relations, without fully assessing whether a caregiver addresses a child’s intended meaning or repairs a misunderstanding. Their exact definitions, normalization, and uncertainty matter to the interpretation of alignment and diversity.

In the feedback study, automatic annotation, reward prediction, and generated-utterance evaluation introduce separate sources of error. A predictor fitted to corpus utterances may become unreliable on the learner’s evolving output distribution. The agreement of ChildCG and GEC on the principal reward pattern is useful converging evidence, but neither substitutes for a targeted human assessment of the affected errors. Likewise, a held-out reward-validation set does not by itself establish separation from the final acquisition test; the grammar topline requires the specific data-provenance check described above.

6.8.3 Information Budgets and Population Generalizability

Unique corpus size, repeated exposure, inference-time memory, and pretrained reward components contribute different information to the experimental system. Their provenance is necessary for claims about learning from child-comparable experience alone. In particular, the smaller retrieval condition combines several text genres and repeated epochs; broad pretraining supports the dialogue simulators and parts of the feedback pipeline. These choices are compatible with controlled computational comparisons, but should not be compressed into a single label of developmental realism.

The studies also concentrate on English and rely heavily on CHILDES as input or reference [macwhinney2000childes]. Recorded families, activities, age ranges, and transcription conventions sample only part of children’s experience. Aggregate curves can conceal variation across children and communities [WeislederFernald2013; bergelson2023everyday]. Conclusions about lexical frequency, word classes, and caregiver response may depend on English morphology and word order as well as the sampled interactional practices. Their scope is the populations and corpora analyzed here until those dependencies are tested directly.

6.9 Toward Integrated Developmental Models

The next step is to test the unresolved links between the present results. Integration will be informative when it connects outcomes and resources while keeping their individual contributions identifiable.

6.9.1 Longitudinal and Multimodal Experience

To distinguish differences in vocabulary-growth dynamics from artifacts of independent exposure conditions, a learner could update continuously from a temporally ordered stream. The same learner could be evaluated at predefined checkpoints while its unique exposure, repeated exposure, optimization history, and memory state are recorded separately. Matched text, speech, and grounded conditions could then test whether particular perceptual or referential information changes rare-word availability and the content/function-word balance at the same exposure

budget. Such a model would still require an explicit account of what was available to the learner’s attention; adding synchronized modalities should not be equated automatically with reproducing a child’s effective experience.

6.9.2 Adaptive Episodic Memory

The residual frequency gap and the structural-match analyses motivate memories that are updated online and selected according to more than embedding similarity. Candidate mechanisms include preferential storage of surprising or informative events, retrieval conditioned on structural demands, adaptive context and neighbour selection, interference, forgetting, and gradual consolidation into parametric knowledge. These mechanisms should be compared through controlled ablations, keeping the stored material and target tasks comparable. Applying the intervention to both CORPUS-CDI generation and syntactic preferences would directly test whether the observed compensation extends across the two frequency-related findings. Their value would lie in explaining when access to prior instances improves transfer, not in treating a successful retrieval system as direct evidence for a particular neural implementation.

6.9.3 A Contingent Caregiver

An interactive learner could produce an utterance, receive a context-sensitive caregiver response, and update before the next exchange. The caregiver would need to condition on dialogue history and the learner’s changing behavior, rather than returning a fixed scalar prediction. Its responses should first be validated against natural interaction at utterance and dialogue levels. Yoked, delayed, or shuffled-feedback controls could then isolate contingency while preserving the amount and surface form of input. A factorial comparison of structural and semantic rewards would also test whether their isolated effects combine, trade off, or depend on the learner’s current production distribution; the present single-reward comparisons cannot decide this. Because two learned agents can amplify one another’s errors, simulation fidelity and learning effects would still require separate evaluation.

6.9.4 Held-out, Multi-outcome Evaluation

Evaluation should follow the learner longitudinally across expressive vocabulary, word–referent mapping, controlled syntactic contrasts, spontaneous production, and sustained interaction. Calibration materials should be separated from final tests, and reward models or evaluators should not be trained on the same items used to support a generalization claim. Targeted contrasts should first be built around recurring errors in both child and model production, such as omissions, and then tested with held-out lexical items and combinations. Additional transfer tests can hold out constructions, speakers, and interaction contexts. Reporting endpoints, curve shapes, item-level errors, and uncertainty together would show whether an intervention produces a narrow production shift or a regularity that generalizes across outcomes.

6.9.5 Cross-linguistic and Individual-level Validation

An integrated account should also be tested across languages and across individual learning histories. The goal is not to make every model reproduce an average child curve, but to ask whether variation in a learner’s own experience predicts variation in its subsequent behavior under comparable measures. Multilingual and community-diverse corpora would test whether conclusions about frequency, memory, and feedback survive changes in morphology, word order, interactional practice, and exposure distribution. Wherever privacy and sample size permit, hierarchical analyses should preserve child-, caregiver-, conversation-, item-, and model-level variation rather than collapsing them into one population mean.

6.10 General Conclusion

This thesis asked how limited and uneven linguistic experience is transformed into expressive and grammatical behavior, and how models can clarify the contributions of distributional exposure, access to prior instances, and caregiver-derived feedback. The findings show that the amount of exposure is only part of the explanation: the accessibility of particular instances, the content of a learning signal, and the task used to observe behavior also matter. Developmental interpretation requires the learner’s experience, the learner and intervention, the observed outcome, and the linking assumptions to be specified together. Similar performance under one task is not sufficient to establish a shared mechanism, while failure under a mismatched comparison is not sufficient to establish that an outcome is unlearnable.

Within these boundaries, the empirical chapters identify a coherent pattern. The expressive-vocabulary study reveals developmental and frequency-related discrepancies under a production-based comparison. The retrieval study shows that access to stored instances can selectively improve performance on low-frequency syntactic items, although it does not eliminate the frequency gap. The interaction studies show that plausible individual utterances do not guarantee realistic sustained dialogue, and that caregiver-derived rewards have differentiated effects: structural alignment most consistently improves generated grammaticality, without establishing broader grammatical transfer.

The resulting contribution is a set of tools and findings about how linguistic evidence becomes available in behavior: a calibrated expressive-vocabulary measure, an analysis of frequency-selective retrieval, a benchmark of sustained dialogue, and a comparison of caregiver-derived learning signals. Predictive learning, instance access, and feedback make distinct, task-dependent contributions whose effects must be measured separately. Language models are most informative here not as artificial children, but as controlled systems in which experience, mechanisms, and evaluation can be varied. Used with calibrated comparisons and explicit inferential limits, they can help transform broad claims about language acquisition into testable questions about which learning resources are sufficient for which developmental outcomes.

RÉSUMÉ

L'acquisition du vocabulaire expressif et de la grammaire à partir d'une expérience linguistique limitée, où les formes sont inégalement fréquentes, constitue une question centrale en sciences cognitives. Les modèles permettent de tester des hypothèses d'apprentissage ; leur portée développementale dépend de la comparaison des expériences et des comportements. Cette thèse associe une calibration des comparaisons enfants-modèles à quatre études computationnelles sur le vocabulaire, la syntaxe avec récupération d'exemples, la simulation du dialogue et l'apprentissage par récompenses dérivées des réactions des adultes. Avec des modèles et des entraînements distincts, ces études examinent les contributions de l'exposition, de l'accès aux exemples mémorisés et du feedback.

Le protocole CORPUS-CDI ajuste l'exposition estimée, la quantité de productions générées et le vocabulaire évalué pour comparer le développement expressif des enfants à celui de LSTM et de Transformers au niveau des caractères. Les modèles testés ne reproduisent pas conjointement les trajectoires enfantines de diversité lexicale, d'exactitude des formes lexicales et de croissance du vocabulaire ; les mots peu fréquents restent particulièrement difficiles à produire. Une étude distincte augmente des modèles paramétriques par récupération des k plus proches voisins. Cette intervention améliore leur sensibilité aux contrastes syntaxiques contenant des mots peu fréquents et réduit partiellement l'écart lié à la fréquence lexicale, avec une contribution importante de l'information structurelle. Un benchmark d'interactions enfant-adulte montre ensuite que certaines propriétés des énoncés et tendances liées à l'âge peuvent être approchées, tandis que les conversations prolongées révèlent des écarts plus marqués. Les adultes simulés restent trop alignés sur leurs interlocuteurs et insuffisamment diversifiés. Dans une étude indépendante d'apprentissage par renforcement, les récompenses produisent des effets différenciés : l'alignement structurel améliore le plus régulièrement la grammaticalité des énoncés générés, sans amélioration cohérente aux tests de paires minimales BLiMP et Zorro.

Ces bénéfices sélectifs coexistent avec des écarts persistants entre trajectoires et tâches. La thèse apporte des benchmarks de production et d'interaction, des interventions computationnelles contrôlées et un cadre reliant leurs résultats à des conclusions développementales circonscrites. Elle motive des théories distinguant exposition, mémoire et interaction, dont les effets doivent être testés par plusieurs mesures de l'usage du langage.

MOTS CLÉS

Acquisition du langage ; modélisation computationnelle ; vocabulaire expressif ; modèles de langage augmentés par récupération ; interaction enfant-adulte ; apprentissage par renforcement

ABSTRACT

How children develop expressive vocabulary and grammatical behavior from limited, unevenly distributed linguistic experience is a central question in cognitive science. Language models make learning assumptions experimentally tractable, but their developmental relevance depends on how experience and behavior are compared. This thesis combines a framework for calibrating child-model comparisons with four computational studies of expressive vocabulary, retrieval-supported syntax, dialogue simulation, and learning from caregiver-derived rewards. The studies use distinct learners and training regimes to examine what predictive exposure, access to stored instances, and feedback contribute to specific linguistic outcomes.

The CORPUS-CDI benchmark aligns estimated training exposure, generated production quantity, and evaluation vocabulary to compare children's expressive development with character-level LSTMs and Transformers. The tested models do not jointly reproduce children's trajectories of lexical diversity, word-form accuracy, and vocabulary growth; low-frequency vocabulary remains particularly difficult to produce. A separate study uses k -nearest-neighbor retrieval to supplement parametric language models. Retrieval improves sensitivity to syntactic contrasts involving low-frequency words and partially narrows the lexical frequency gap, with structural information playing an important role. Turning to interaction, a benchmark comparing simulated child-caregiver exchanges with naturalistic dialogue shows that selected utterance-level and age-related patterns can be approximated, while sustained conversations reveal greater divergence. Simulated caregivers remain overly aligned with their interlocutors and insufficiently diverse. In an independent reinforcement-learning study, caregiver-derived rewards have differentiated effects: structural alignment most consistently improves the grammaticality of generated utterances, without consistent improvements on BLiMP and Zorro minimal-pair tests.

Together, these findings identify selective benefits of retrieval and feedback alongside persistent mismatches across developmental trajectories and evaluation tasks. The thesis contributes production and interaction benchmarks, controlled computational interventions, and a framework connecting model outcomes to bounded developmental claims. It motivates theories that distinguish the contributions of exposure, memory, and interaction, and test their effects across multiple measures of language use.

KEYWORDS

Language acquisition; computational modeling; expressive vocabulary; retrieval-augmented language models; child-caregiver interaction; reinforcement learning